

Vasoactive Molecules and the Kidney

Philip A. Marsden

CHAPTER OUTLINE

RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM, 288
ENDOTHELIN, 298

NATRIURETIC PEPTIDES, 304
KALLIKREIN-KININ SYSTEM, 313

KEY POINTS

- In addition to the classical endocrine renin-angiotensin system (RAS), which primarily functions through systemically circulating angiotensin II, a distinct local RAS operates within the kidney, acting in paracrine, autocrine, and potentially intracrine modes.
- While much of our knowledge of the RAS centers on the effects of angiotensin II, various other angiotensin-related proteins, enzymes, and receptors also play significant roles in kidney physiology and disease.
- For decades, RAS-blocking therapies have been used to attenuate decline in kidney function, while agents that either inhibit (e.g., endothelin receptor antagonists) or enhance (e.g., neprilysin inhibitors combined with angiotensin receptor blockade) the effects of other vasoactive molecules are currently being explored.
- Blockade of endothelin type A receptors, either alone or in combination with type B receptor blockade, has been studied in the treatment of IgA nephropathy and diabetic nephropathy.
- Inhibition of neprilysin prevents the degradation of the natriuretic peptides, specifically atrial natriuretic peptide and brain natriuretic peptide, but its effects can be limited by an unopposed RAS. Concurrent angiotensin receptor blockade and neprilysin inhibition improves outcomes in patients with heart failure, although its effects on renal outcomes are currently unknown. Sacubitril/valsartan, an angiotensin receptor–neprilysin inhibitor combination, lowers blood pressure more effectively than traditional RAAS blockade and enhances heart failure outcomes in patients with CKD.
- The development of therapies that affect other vasoactive molecules (e.g., the kallikrein–kinin system, urotensin II, guanylin, uroguanylin, and adrenomedullin) has been limited.

Vasoactive peptides, arising from both the systemic circulation and from local tissue-based generation, play important roles in kidney physiology in not only the regulation of renal blood flow (RBF) but also electrolyte exchange, acid-base balance, and diuresis. More recent interest has focused on the role of these peptide systems in kidney development and the pathogenesis of organ injury. In this chapter, we review the physiology of vasoactive peptides that exert potent effects on the kidneys and therapies that modulate the actions of these molecules to ameliorate kidney damage.

RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM

In their groundbreaking 1898 report, *Niere und Kreislauf*, Robert Tigerstedt, and Per Bergman, working at the Karolinska Institute in Sweden, documented the sustained vasopressor effects of crude kidney extracts. Even though the extract was crude, Tigerstedt named the unidentified active compound “renin,” reflecting its renal origin. More than 125 years later, our understanding of the

renin-angiotensin system (RAS) continues to evolve. A major step forward occurred in 1967 when Prof. Arakawa in Japan first isolated human angiotensin II and defined its structure. The ongoing pursuit of a comprehensive understanding of the RAS is driven by not only scientific curiosity but also a recognition of the critical therapeutic value of its inhibition. This stems from the now well-established kidney-protective benefits of RAS inhibitors, which have demonstrated efficacy in reducing proteinuria and preserving kidney function.¹ The ongoing exploration of the RAS aims to refine existing therapies and uncover new strategies to enhance kidney protection and cardiovascular health. The kidney-protective effects of RAS inhibition were first observed with angiotensin-converting enzyme inhibition, as described by Anderson, Meyer, Rennke, and Brenner in 1985 using a rodent model of progressive kidney disease.²

Clinical Relevance

The renin-angiotensin system plays a fundamental role in blood pressure, plasma volume, electrolytes, and acid-base homeostasis. Beyond its function in kidney physiology, however, angiotensin II, the primary effector molecule of the renin-angiotensin system, raises intraglomerular pressure, induces proteinuria, and stimulates the production of extracellular matrix that leads to glomerulosclerosis and interstitial fibrosis. Accordingly, blockade of angiotensin II synthesis by angiotensin-converting enzyme inhibition or antagonizing its action at the angiotensin I receptor with an angiotensin receptor blocker is at the cornerstone of strategies that attenuate progressive kidney function decline in most forms of chronic kidney disease.³

RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM: A CLASSICAL VIEW

The classical view of the RAS focuses on the endocrine aspects of this peptidergic system. Angiotensinogen synthesized by the liver enters the circulation, where it is

cleaved to form angiotensin I by renin, a peptidase that is secreted from the juxtaglomerular apparatus (JGA) of the kidney. The terminal two amino acids of angiotensin I are then removed to form angiotensin II, as it traverses through the circulation, exposed to angiotensin-converting enzyme (ACE), a peptidase robustly expressed on endothelial cells, especially in the pulmonary vasculature. Angiotensin II, an octapeptide hormone, is the principal effector molecule of the RAS. Angiotensin II binds to the type 1 receptor (AT₁R), resulting in vasoconstriction, sodium retention, thirst, and aldosterone secretion. This traditional view of the RAS is still valid but has been considerably augmented in recent years by both the discovery of new enzymes, peptides, and receptors and an appreciation that the RAS has an independently functioning local tissue-based component that acts through paracrine, autocrine, and possibly intracrine mechanisms (Fig. 11.1).

ANGIOTENSINOGEN

Angiotensinogen is primarily, although by no means exclusively, synthesized in the liver, particularly the pericentral zone of the hepatic lobules.⁴ In humans, it is coded by a single gene, composed of five exons and four introns, that spans about 13 kb of genomic sequence on chromosome 1 (1q42-q43). It is translated to a 453 amino acid globular glycoprotein with a molecular weight between 45 and 65 kDa, depending on the extent of its glycosylation, that then undergoes posttranslational cleavage of a 24- or 33-amino acid signal peptide,⁵ giving rise to the mature circulating form of angiotensinogen.⁶ New approaches to target the RAS are either direct antisense oligonucleotides that inhibit angiotensinogen RNA translation or small interfering RNA (siRNA) that function via the RNA interference pathway in the liver.⁷ Since all angiotensins stem from angiotensinogen, lowering angiotensinogen has the potential to circumvent the RAS escape phenomenon.

Structurally, angiotensinogen bears substantial homology to the serpin superfamily of protease inhibitors and, like many members of its family, behaves as an acute-phase reactant in the inflammatory setting,⁸ reflecting the

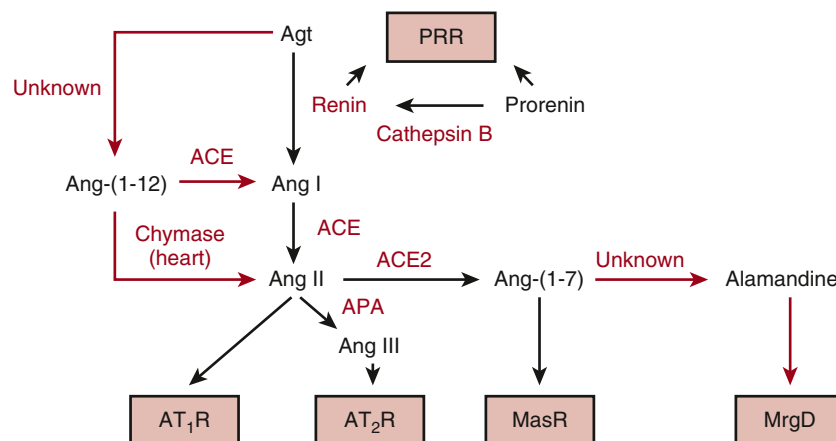


Fig. 11.1 Schematic depiction of the renin-angiotensin system components and selected actions. The enzymes of the system are shown in red. Newly described enzymatic pathways are shown as red arrows. Receptors are shown in the boxes. ACE, Angiotensin-converting enzyme; Agt, angiotensinogen; Ang, angiotensin; APA, aminopeptidase A; AT₁R, angiotensin type-1 receptor; AT₂R, angiotensin type-2 receptor; MasR, Mas receptor; MrgD, Mas-related G protein-coupled receptor; PRR, (pro)renin receptor. (Modified from Carey RM. Newly discovered components and actions of the renin-angiotensin system. *Hypertension*. 2013;62: 818–822.)

presence of an acute-phase response element that binds the transcription factor, NF- κ B (nuclear factor kappa-light-chain-enhancer of activated B cells).⁹

RENIN

Like angiotensinogen, the gene-encoding renin is also located on the long arm of chromosome 1 (1q32) and contains 10 exons and 9 introns, similar to other aspartyl proteases.¹⁰ Unlike humans and rats that have only a single renin gene, the mouse has two genes, *Ren1* and *Ren2*, expressed primarily in the submandibular gland and kidney, respectively.

Following its synthesis as a 406-amino acid preprohormone, the 23-amino acid leader sequence of preprorenin is cleaved in the rough endoplasmic reticulum, giving rise to prorenin (also called inactive renin and “big” renin), which may be then rapidly secreted directly from the Golgi apparatus or from protogranules.⁵ Alternatively, and virtually exclusively in the JGA, prorenin may be packaged into mature, dense granules that instead of being immediately secreted, undergo further processing to the active enzyme, renin (active renin). Contrasting with the more constitutive secretion of prorenin, the release of renin-containing granules is tightly regulated.¹⁰

Mature, active renin is a variably glycosylated 340-amino acid 37–40-kDa aspartyl protease that is active at neutral pH, and in contrast to the more promiscuous activities of most other proteases in this class, it has only a single known substrate, cleaving the decapeptide angiotensin I from the amino terminal of angiotensinogen. Although the kidney produces both renin and prorenin, a range of extrarenal tissues including the adrenals, gonads, and placenta produce prorenin and contribute to its presence in plasma. However, as evidenced by the near total absence of active renin in anephric patients, the kidney, and the JGA in particular, appears to be the only source of circulating renin in humans.

Factors that chronically stimulate renin secretion, such as a low-sodium diet and ACE inhibition, lead to an increase in the number of renin-secreting cells rather than

an increase in cell size or the number of granules that each JGA cell contains. This expansion of the renin-secreting mass occurs proximally by metaplastic transformation of smooth muscle cells within the walls of the afferent arteriole. Although sometimes mentioned, ectopic renin expression within the extraglomerular mesangium appears to be an uncommon event.¹¹ The mechanisms that inhibit renin expression in nonexpressing cell types remain poorly understood.

Prorenin activation

Prorenin is maintained as an inactive zymogen through the occupation of its catalytic cleft by its prosegment. Removing this prosegment by either proteolytic or nonproteolytic means yields active renin, a term that denotes its enzymatic activity rather than its amino acid sequence (Fig. 11.2).

Within the dense core secretory granules of the JGA, acidification by vacuolar adenosine triphosphatases (ATPases) provides the optimal pH for the prosegment-cleaving enzymes (proconvertase 1 and cathepsin B) and may also assist the pH-dependent, nonenzymatic activation of prorenin as well.^{11–13} Although various peptidases such as trypsin, plasmin, and kallikrein can also cleave the prosegment of prorenin in vitro, these do not appear to contribute to the generation of renin in the in vivo setting. Although traditionally viewed as occurring only in the JGA, cell culture-based studies suggest that proteolytic activation of renin can also occur in cardiac and vascular smooth muscle cells by as yet unidentified serine proteases.^{14–16} The significance of these findings in the intact organism, however, remains to be established.

In addition to proteolytic cleavage of its prosegment, prorenin can also be reversibly activated nonenzymatically by a conformational change such that the prosegment no longer occupies the enzymatic cleft. Under usual circumstances, <2% of prorenin is in this open active conformation. This process can, however, be induced by acid (pH 4.0)^{17,18} and to a lesser extent by cold.¹⁹ More recently, the putative (pro) renin receptor (PRR; discussed later) has also been shown to nonproteolytically activate prorenin.²⁰

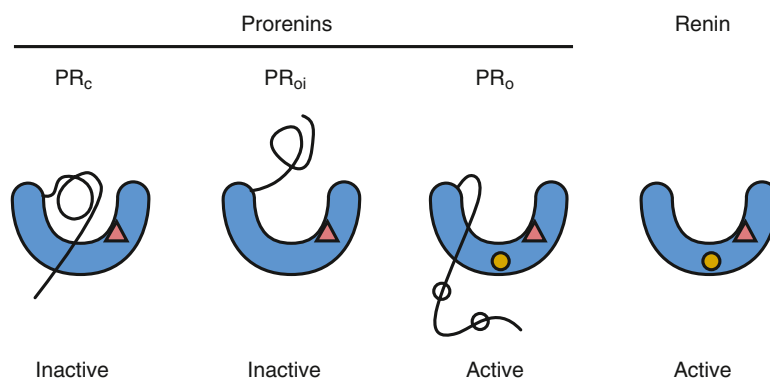


Fig. 11.2 The conformational changes and the expression of immunoreactive epitopes associated with the activation of prorenin are depicted. The main body of the molecule (blue), the substrate-binding cleft, and the prosegment (black line) are shown. The closed triangle represents the epitope of the main body expressed by PR_c (prorenin in the inactive closed conformation), PR_{oi} (prorenin in the inactive intermediary open conformation), PR_o (prorenin in the active open conformation), and renin. The closed circle (yellow) represents the epitope of the main body, expressed by PR_o and renin, but not by PR_c and PR_{oi}. The open circles represent epitopes of the prosegment expressed by PR_c and PR_{oi}. (Modified from Schalekamp MA, Derkx FH, Deinum J, et al. Newly developed renin and prorenin assays and the clinical evaluation of renin inhibitors. *J Hypertension*. 2008;26:928–937.)

Regulation of renin secretion

Mechanical, neurologic, and chemical factors regulate the activity of the RAS by modulating renin secretion.

Renal baroreceptor. Renin-expressing juxtaglomerular (JG) cells possess an intrinsic pressure-sensing mechanism(s) that regulates renin synthesis and release in response to changes in perfusion pressure. The existence of a renal baroreceptor mechanism was first conceptualized by Skinner and colleagues²¹ to explain how renin secretion increases when afferent arteriolar perfusion pressure falls. Studies in conscious dogs show that changes in renal perfusion pressure have only a small effect on renin secretion until a threshold of about 90 mm Hg is reached, below which renin secretion abruptly increases, doubling with every 2 to 3 mm Hg fall in pressure.²² Accordingly, reduction in pressure below this level profoundly stimulates renin secretion, thereby acutely activating the RAS and resulting in a range of angiotensin II-dependent phenomena that collectively serve to restore systemic pressure. Despite the importance of the baroreceptor function, several decades of research have not identified precisely how the pressure signal is transduced into renin release, though postulated mediators include Piezo1, a mechanosensitive ion channel, endothelins (ETs), and prostaglandins.²³

Neural control. The JGA is endowed with a rich network of noradrenergic nerve endings and their $\beta 1$ receptors. Stimulation of the renal sympathetic nerve activity leads to renin secretion that is independent of changes in RBF, glomerular filtration rate (GFR), or Na^+ reabsorption. Moreover, this effect can be blocked surgically (denervation) and pharmacologically, by the administration of β -adrenoreceptor blockers.²² The role of cholinergic, dopaminergic, and adrenergic activation is controversial, though these agents have also been shown to modulate renin release under certain circumstances.

Tubule control. Chronic diminution in luminal NaCl delivery to the macula densa is a potent stimulus for renin secretion, reflecting a coordinate interaction between a range of mediators including adenosine, nitric oxide (NO), and prostaglandins that affect renin release and its transcription.²⁴ This mechanism is thought to account for the chronically high plasma renin activity (PRA) in persons ingesting a low-salt diet.²⁵

Metabolic control. The tricarboxylic acid (TCA) cycle provides a final common pathway by which carbohydrates, fatty acids, and amino acids converge in the process of adenosine triphosphate (ATP) generation by aerobic electron transfer. Although the TCA cycle operates within mitochondria, its intermediates can be detected within the extracellular space, increasing in abundance when local energy supply and demand are mismatched or when cells are exposed to hypoxia, toxins, or injury.²⁶ Succinate, for instance, has been shown to stimulate renin release and its intravenous administration leads to hypertension, though the mechanisms underlying this effect have only recently been unraveled. In 2004, He and colleagues²⁷ reported that α keto-glutarate and succinate are ligands for the previously orphaned G protein-coupled receptors (GPCRs), GPR99 and GPR91, respectively, and that succinate-induced hypertension is abolished in GPR91-deficient mice. Indeed, in follow-up studies from this group, GPR91 was localized to the apical plasma membrane of macula densa cells where succinate stimulation was shown to activate p38 and Erk 1/2 mitogen-activated protein (MAP) kinases (MAPKs), inducing cyclooxygenase-2 (COX-2)-dependent synthesis of prostaglandin E₂, a well-established paracrine mediator of renin release.²⁸ Moreover, the ability of tubule succinate to induce JG renin secretion suggests that this phenomenon is likely an important determinant of JGA function in both physiologic and pathophysiologic settings. In diabetic rats, for instance, elevated succinate has been detected in both plasma and urine.²⁸

Vitamin D receptor. The vitamin D receptor (VDR) is a negative regulator of the RAS such that VDR-null mice display a marked increase in renin expression and angiotensin II production in conjunction with hypertension and cardiac hypertrophy.²⁹ Importantly, these effects occurred independently of calcium and parathyroid hormone, both of which have been reported to also modulate renin expression. The molecular basis for the interaction between vitamin D and renin expression has also been, at least partly, unraveled in a series of studies by Yuan and colleagues.³⁰ Under usual circumstances, activation of the cyclic adenosine monophosphate (cAMP)–protein kinase A pathway by the sympathetic nervous system or macula densa leads to phosphorylation of a cAMP response element (CRE) binding protein (CREB) and recruitment of CREB-binding protein/p300 to the CRE in the promoter region of the renin gene. VDR-bound 1,25 (OH)₂D₃, however, blocks binding of CREB to the CRE DNA *cis*-element, leading to a reduction in renin gene transcription (Fig. 11.3).

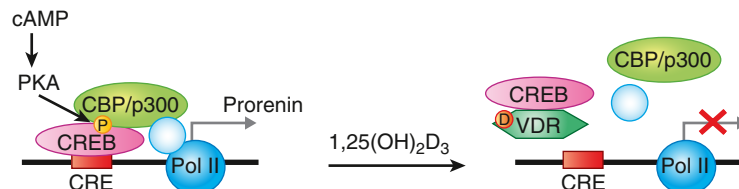


Fig. 11.3 Model of 1,25(OH)₂D₃-induced transrepression of renin gene expression. The cAMP–PKA pathway activates CREB by phosphorylation, leading to recruitment of CBP/p300. In the presence of 1,25(OH)₂D₃, liganded VDR interacts with CREB and blocks its binding to CRE, leading to reduction of renin gene transcription. *cAMP*, Cyclic AMP; *CBP*, CREB-binding protein; *CRE*, cAMP response element; *CREB*, CRE-binding protein; *D*, 1,25(OH)₂D₃; *P*, phosphorylation; *PKA*, protein kinase A; *Pol II*, RNA polymerase II; *VDR*, vitamin D receptor. (Modified from Yuan W, Pan W, Kong J, et al. 1,25-Dihydroxyvitamin D₃ suppresses renin gene transcription by blocking the activity of the cyclic AMP response element in the renin gene promoter. *J Biol Chem*. 2007;282:29821–29830.)

From a clinical perspective, this interaction between vitamin D and renin may explain the well-documented inverse relationship among plasma vitamin D₃, blood pressure, and PRA. While several studies examining the effects of vitamin D supplementation as a kidney-protective or antihypertensive measure have been undertaken, their findings have been mixed and long-term, randomized controlled trials with clinically meaningful endpoints have not been performed.

Other local factors. In addition to the factors discussed earlier, a large range of locally produced biologically active molecules have also been shown to alter renin secretion. These include peptides (ANP, kinins, vasoactive intestinal polypeptide, ET, calcitonin gene-related peptide [CGRP]); amines (dopamine and histamine); and arachidonic acid derivatives.²²

Plasma Prorenin and Renin

Under usual circumstances, the plasma concentration of prorenin is approximately 10 times greater than renin. In some patients with diabetes, however, plasma prorenin is disproportionately increased, where it predicts the development of diabetic nephropathy (including microalbuminuria) and retinopathy.^{31,32}

In addition to its role in the research setting, measurement of plasma renin is an important clinical assay, providing important information, for example, when evaluating patients with possible hyperaldosteronism, assessing volume status, and in predicting the response to, or monitoring drug adherence to, an ACE inhibitor or angiotensin receptor blocker (ARB). In broad terms, plasma renin is determined by either activity or immunologic assay methods.³³ The most commonly used method involves the measurement of PRA. With this method, the rate at which angiotensin I is produced from plasma angiotensinogen is assayed. To prevent angiotensin I's degradation or its conversion to angiotensin II, inhibitors of angiotensinase and ACE are added to the assay. Accordingly, PRA is dependent on renin and endogenous angiotensinogen concentrations and will also overestimate the extent of inhibition by renin inhibitors due to the displacement of protein-bound drugs by the peptidase inhibitors. The latter scenario may be diminished by using an antibody capture method in which antiangiotensin I antibody, instead of peptidase inhibitors, is used to protect angiotensin I from further catabolism.³⁴

The nomenclature of renin assays can be quite confusing in that plasma renin concentration (PRC) may be measured by both activity and immunologic assays. With the activity method (PRCa), exogenous angiotensinogen is added to the assay, thereby avoiding the influence that endogenous levels of the substrate might have. However, PRCa may also be affected by the presence of renin inhibitors, though like PRA, it may take advantage of antibody capture methodology.³³ In the immunologic assay for renin (PRCi), the concentrations of renin and prorenin in its active, open conformation are assessed so that, like PRA and PRCa, the PRCi assay is also time and temperature dependent because lower temperatures will increase the proportion of prorenin in its active conformation. Moreover, renin inhibitors, by binding to the active site of prorenin in its open

conformation, prevent the refolding of the prosegment and may therefore lead to an overestimation of PRCi.^{33,34}

ANGIOTENSIN-CONVERTING ENZYME

ACE is a zinc-containing dipeptidyl carboxypeptidase that cleaves the terminal histidyl-leucine from angiotensin I to form the octapeptide angiotensin II. In contrast to the single-substrate specificity of renin, ACE is not specific, cleaving the two terminal acids from peptides with the C-terminal sequence R₁-R₂-R₃-OH, where R₁ is the protected (noncleaved) amino acid, R₂ is any nonproline L-amino acid, and R₃ is any nondicarboxylic (cysteine, ornithine, lysine, arginine) L-amino acid with a free carboxyl terminal.⁵ Importantly, therefore, ACE also catalyzes the inactivation of bradykinin. Although encoded by a single ACE gene, two distinct tissue-specific messenger RNAs (mRNAs) are transcribed, each with different initiation and alternative splice sites.³⁵ The somatic form, present in almost all tissues, is a 1306-amino acid, 140–160-kDa glycoprotein with two active sites, whereas the 90- to 100-kDa testicular or germinal form is found exclusively in postmeiotic male germ cells and contains a single active site and appears to be involved in spermatogenesis.^{5,36,37} The somatic form of ACE is widely distributed with activity present in not only tissues but also most biological fluids. In the human kidney, ACE is present to the greatest extent within proximal and distal tubules; however, both its magnitude of expression and its site-specific distribution may be altered by disease.³⁸

ANGIOTENSIN TYPE 1 RECEPTOR

The AT₁R mediates most of the known physiologic effects of angiotensin II. The gene for this widely distributed 359-amino acid, 40-kDa, seven-transmembrane GPCR is located on chromosome 3 in humans.³⁹

Within the kidney, AT₁Rs are widely expressed. In the glomerulus, they are found in both afferent and efferent arterioles, as well as in the mesangium, in the endothelium, and in podocytes.⁴⁰ Consistent with angiotensin II's role in Na⁺ reabsorption, AT₁Rs are highly abundant on the brush borders of proximal tubule epithelial cells.⁴¹ Prominent expression has also been found in renal medullary interstitial cells, located between the renal tubules and vasa recta, where angiotensin II is purported to have a potential role in the regulation of medullary blood flow.⁴²

Angiotensin II binding to AT₁R initiates cell signaling by several different pathways that have been mostly studied in vascular smooth muscle cells.⁴³ These include G protein-mediated pathways and the activation of tyrosine kinases, NADH/NADPH oxidases, and serine/threonine kinases.³⁹

G protein-mediated signaling

In the classical G protein-mediated pathway, AT₁R ligand binding leads to activation of phospholipases C, D, and A₂. Phospholipase C rapidly hydrolyses phosphatidylinositol bisphosphate to inositol trisphosphate and diacylglycerol (DAG), initiating calcium release from intracellular stores and protein kinase C (PKC) activation, respectively. Phospholipase D similarly generates DAG and activates PKC, whereas phospholipase A₂ (PLA₂) leads to the formation of various vasoactive and proinflammatory arachidonic acid derivatives.

Reactive oxygen species

Although previously regarded as toxic waste products, emerging evidence indicates that reactive oxygen species may also act as second messengers, activating other cell signaling cascades such as p38 MAPK and a number of transcription pathways implicated in the pathogenesis of inflammatory and degenerative disease.³⁹ Although the mechanisms by which the AT₁R stimulates NADH/NADPH are not well understood, angiotensin II binding to this receptor results in the generation of both superoxide and hydrogen peroxide.

Tyrosine kinases

Angiotensin II binding to the AT₁R “transactivates” a number of nonreceptor tyrosine kinases (Src, Pyk2, focal adhesion kinase [FAK], and Janus kinase [JAK]), as well as the growth factor receptor tyrosine kinases for epidermal growth factor (EGF)^{44,45} and platelet-derived growth factor (PDGF).^{46,47} By binding to the AT₁R, angiotensin II initiates the translocation of tumor necrosis factor- α (TNF- α)-converting enzyme (TACE, ADAM17) to the cell surface. TACE then cleaves TNF- α from its membrane-associated precursor (pro-TNF- α), allowing it to bind to the EGF receptor (EGFR) on the cell surface. This ligand-receptor interaction then induces EGFR autophosphorylation and activates its downstream signaling pathways that include Akt, Erk-1/2, and mammalian target of rapamycin (Fig. 11.4). The *in vivo* relevance of this transactivation pathway has been confirmed. Using mice that express a dominant negative form of EGFR, Lautrette and colleagues⁴⁸ showed that despite similar blood pressures, mutant mice infused with angiotensin II had less proteinuria and renal fibrosis than their wild-type counterparts. Consistent with these findings and the pivotal role of the RAS in diabetic nephropathy, studies using a specific EGFR tyrosine kinase

inhibitor (i.e., PKI 166) have also shown a reduction in early structural injury in a rat model of diabetic nephropathy.⁴⁹

Like EGFR, the transactivation of the PDGF receptor (PDGFR) by AT₁R is also complex, involving the adaptor protein Shc.^{46,47} In addition to studies that have explored the angiotensin-PDGFR interaction in cell culture or organ baths,⁵⁰ a further report has shown that despite continued hypertension, inhibition of the PDGFR kinase *in vivo* can also dramatically attenuate angiotensin II-induced vascular remodeling.⁵¹

Angiotensin type 1 receptor internalization

In addition to the conventional ligand-receptor-mediated pathways, a range of other signaling mechanisms that involve the AT₁R have also been described. These include the discovery of receptor-interacting proteins, heterologous receptor dimerization, and ligand-independent activation (Fig. 11.5).⁵² These additional insights, although adding greater complexity to our understanding of the RAS, also provide the potential for new therapeutic targets in disease prevention and management.

Following ligand binding and the initiation of signal transduction, AT₁Rs are rapidly internalized, followed by either lysosomal degradation or recycling back to the plasma membrane. Several mechanisms account for AT₁R internalization including interaction with caveolae, phosphorylation of its carboxyl terminal by G-protein receptor kinases,³⁹ and association with the newly described AT₁R interacting proteins.⁵² To date, two such interacting proteins, AT₁ receptor-associated protein (ATRAP)⁵³ and AT₁ receptor-associated protein-1 (ARAP1),⁵⁴ have been described. ATRAP interacts with the C terminal of AT₁R, downregulating cell surface AT₁R expression and attenuating angiotensin II-mediated effects.⁵² ARAP1, though somewhat similar to ATRAP, promotes AT₁R

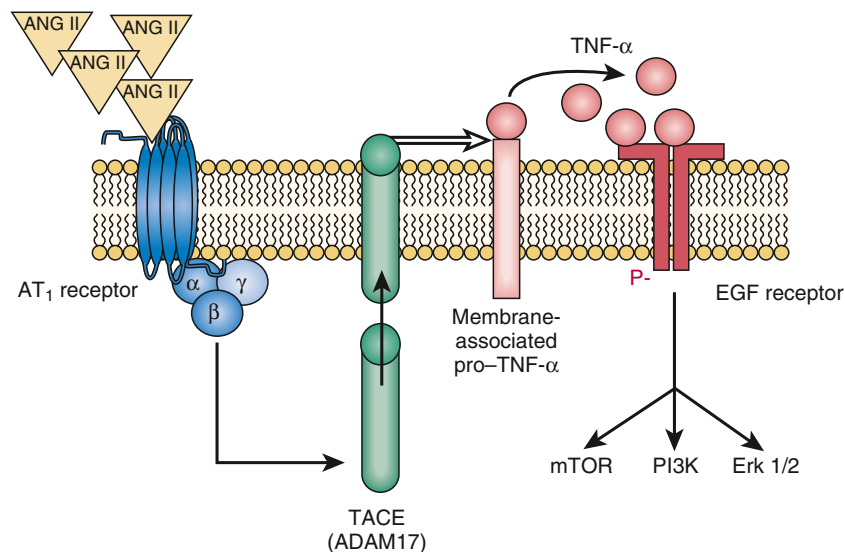


Fig. 11.4 Angiotensin II (ANG II) binds to its angiotensin II type 1 (AT₁) receptor, a G protein-coupled receptor lacking intrinsic tyrosine kinase activity. Through as-yet-undescribed mechanisms, this interaction leads to the translocation of the metalloprotease tumor necrosis factor- α (TNF- α)-converting enzyme (TACE) from the cytosol to the cell surface, where it cleaves TNF- α from its membrane-associated pro-molecule, allowing it to bind and activate the epidermal growth factor (EGF) receptor. *Erk 1/2*, Extracellular signal-regulated kinases 1 and 2; *mTOR*, mammalian target of rapamycin; *PI3K*, phosphatidylinositol-3-kinase. (Modified from Wolf G. “As time goes by”: angiotensin II-mediated transactivation of the EGF receptor comes of age. *Nephrol Dial Transplant*. 2005;20:2050–2053.)

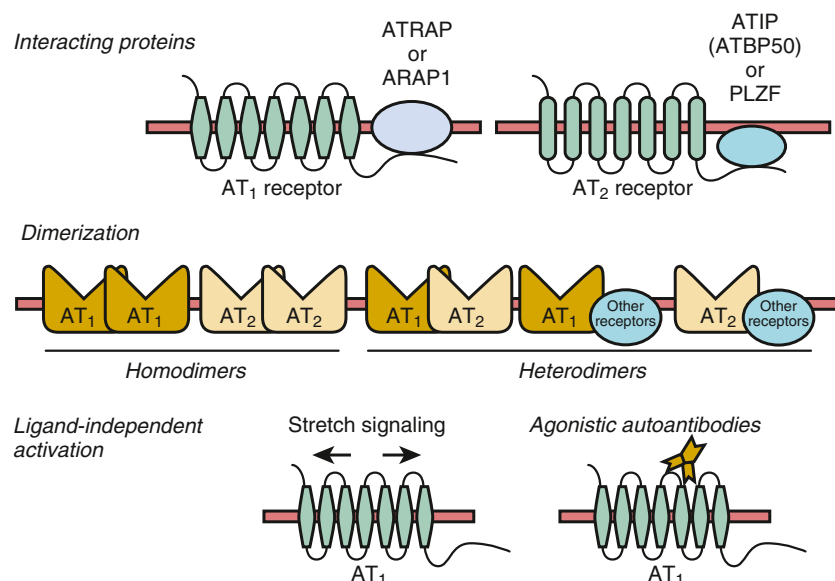


Fig. 11.5 Regulation of angiotensin receptors. *ARAP1*, AT_1 receptor-associated protein 1; *ATBP50*, AT_2 receptor-binding protein of 50 kDa; *ATIP*, angiotensin type 2 receptor-interacting protein; AT_1 receptor, angiotensin type 1 receptor; AT_2 receptor, angiotensin type 2 receptor; *ATRAP*, AT_1 receptor-associated protein; *PLZF*, promyelocytic leukemia zinc finger. (Modified from Mogi M, Iwai M, Horiuchi M. New insights into the regulation of angiotensin receptors. *Curr Opin Nephrol Hypertens*. 2009;18:138–143.)

recycling to the plasma membrane such that its kidney-specific overexpression induces hypertension and renal hypertrophy.⁵⁴

Angiotensin type 1 receptor dimerization

In addition to their ability to induce cell signaling in their monomeric state, GPCRs like AT_1R may also associate to form both homodimers and heterodimers.⁵⁵ Beyond its constitutive homodimerization,⁵⁶ AT_1R may dimerize with angiotensin type 2 receptor (AT_2R) and also form heterooligomers with receptors for bradykinin (B_2), epinephrine (β_2), dopamine ($D_{1,3,5}$), ET (B), Mas, and EGF that modulate their function.⁵⁷⁻⁶⁰

Ligand-independent angiotensin type 1 receptor activation

Without involvement of angiotensin II, cell stretch induces a conformational switch that initiates AT_1R 's intracellular signaling pathways.^{61,62} As might be expected from an understanding of this mechanism, an AT_1R blocker, acting as an inverse agonist, will abrogate these effects, as described in both cardiac⁶² and mesangial cells.⁶³ A similar means of ligand-independent activation has also been shown to result from the binding of agonist antibodies to AT_1R in some women with preeclampsia⁶⁴ and in certain cases of renal allograft rejection.⁶⁵

PHYSIOLOGIC EFFECTS OF ANGIOTENSIN II IN THE KIDNEY

The traditional actions of angiotensin II relate primarily to its effects on vascular tone and fluid balance that are mediated by its actions on the vasculature, heart, kidney, brain, and adrenal glands by the AT_1R . In vascular smooth muscle, stimulation of AT_1Rs by angiotensin II induces cell contraction and consequent vasoconstriction. In the adrenal cortex, this ligand-receptor interaction stimulates aldosterone release, thereby promoting sodium

reabsorption in the distal nephron. Moreover, angiotensin II will directly enhance sodium retention by the proximal tubule and in the brain, it will stimulate thirst and salt craving. Additional effects include sympathoadrenal stimulation and the augmentation of cardiac contractility. Together, these effects serve to maintain extracellular fluid volume and systemic blood pressure. Given the central role that the kidney has in the regulation of these key aspects of mammalian homeostasis, it is not surprising that angiotensin II should have profound effects on renal physiology.

Renal hemodynamic actions of angiotensin II

The effects of exogenously administered angiotensin II are dose dependent. At low doses, angiotensin II infusion increases renal vascular resistance (RVR) and lowers RBF without affecting GFR so that the filtration fraction is increased. At higher doses of angiotensin II, RVR is further increased, leading to an augmented reduction in RBF and fall in GFR.⁶⁶ However, because GFR is reduced to a lesser extent than renal plasma flow, the filtration fraction remains elevated. Such studies are consistent with the view that limited stimulation of the RAS would mostly serve to enhance tubule sodium reabsorption, as is seen, for instance, in societies unaccustomed to contemporary diets.²⁵ Greater activation of the RAS, by contrast, as might be found in the setting of severe volume depletion, would result in angiotensin II-dependent reduction in RBF that would aid in sustaining systemic blood pressure while further stimulating sodium reabsorption.

Kidney micropuncture has been used extensively to explore the intrarenal sites of angiotensin II's effects on vascular resistance. These studies demonstrate that although angiotensin II increases both afferent and efferent arteriolar resistance, intraglomerular capillary pressure (P_{GC}) is consistently elevated due to a greater increase in efferent than afferent arteriolar resistance⁶⁷ and the ultrafiltration coefficient (K_f) is reduced.⁶⁶ Moreover, as predicted by

mathematic modeling, the glomerular hypertension induced by angiotensin II does not lead to acute proteinuria because the structural barriers to macromolecular passage remain intact.⁶⁶ Chronic angiotensin II infusion with sustained intraglomerular hypertension, by contrast, leads to glomerular capillary damage and substantial proteinuria.

Tubule transport

Sodium. Consistent with its importance in the regulation of volume status, angiotensin II has profound effects on renal Na^+ handling. The proximal tubule is responsible for the reabsorption of approximately two-thirds of the sodium from the glomerular filtrate, and binding sites for angiotensin II are particularly abundant in the proximal tubule with immunohistochemical localization of the AT_1R to both apical and basolateral surfaces.⁶⁸ At picomolar concentrations, angiotensin II stimulates the luminal Na^+/H^+ exchanger, the basolateral $\text{Na}^+/\text{HCO}_3^-$ cotransporter, and the Na^+-K^+ ATPase. However, at concentrations $>10^{-9}$ M, angiotensin II inhibits the very same transporters. The mechanisms underlying this dose-dependent effect of angiotensin II on Na^+ transport, which seem to also occur in the loop of Henle,⁶⁸ are incompletely understood. In the distal tubule, the effects of angiotensin II on Na^+ transport are site dependent. In the early distal tubule, for instance, angiotensin II stimulates apical Na^+/H^+ exchange while in the late distal tubule it stimulates the amiloride-sensitive sodium channel.⁶⁸

Acid-base regulation. The kidney has a key role in the maintenance of physiologic pH by regulating the secretion/reabsorption of acids and bases. As for Na^+ , angiotensin II also has substantial effects on acid-base transport in the proximal tubule, distal tubule, and collecting duct. Interest has focused, in particular, on its actions in the collecting duct, where angiotensin II stimulates not only Na^+/H^+ exchangers and $\text{Na}^+/\text{HCO}_3^-$ cotransporters but also the vacuolar H^+ -ATPase in intercalated A-cells via its AT_1R receptor.⁶⁹ Moreover, elegant and detailed electron microscopic studies have helped to unravel the mechanisms by which angiotensin II exerts its effects at this site, revealing translocation of the H^+ -ATPase from the cytoplasm to the apical surface in response to ligand stimulation.⁷⁰

EXPANDED RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM: ENZYMES, ANGIOTENSIN PEPTIDES, AND RECEPTORS

ANGIOTENSIN TYPE 2 RECEPTOR

In humans, the AT_2R is a 363-amino acid protein that maps to the X chromosome and is highly homologous to its rat and mouse counterparts.⁷¹ Like AT_1R , AT_2R is also a seven-transmembrane GPCR, though it shares only 34% homology.

Despite substantial research, the actions of AT_2R are still not well understood and remain somewhat controversial.⁷² In general, however, the actions of AT_2R stimulation oppose those of AT_1R . For instance, whereas AT_1R vasoconstricts and promotes Na^+ retention, AT_2R stimulation leads to vasodilation⁷³ and natriuresis,⁷⁴ consistent with its abundance on the epithelium of the proximal tubule.⁷⁵ The vasodilatory effects of AT_2R stimulation are mediated by increasing NO synthesis and cyclic guanosine monophosphate (cGMP)

by bradykinin-dependent and bradykinin-independent mechanisms.⁷⁶ Its natriuretic effects, however, seem to be dependent on angiotensin II's conversion to angiotensin III by aminopeptidase N.⁷⁷

Like the AT_1R , the activity of AT_2R may also be modulated by oligomerization, in association with various interacting proteins and ligand-independent effects.⁷⁶

(PRO)RENIN RECEPTOR

In 2002 an apparently novel, 350-amino acid, single-transmembrane protein that binds both renin and prorenin with high affinity was identified.²⁰ Ligand binding to this protein was shown to induce a fourfold increase in the catalytic cleavage of angiotensinogen, as well as stimulating intracellular signaling with activation of MAPKs extracellular signal-regulated kinases 1 and 2 (ERK1/2),²⁰ leading to it being named the (pro)renin receptor ([P]RR). The designation (pro)renin refers to its ability to interact with both renin and prorenin.

Given its localization to the mesangium in initial studies, its actions in augmenting local angiotensin II production, and its ability to increase mesangial transforming growth factor- β (TGF- β) production,⁷⁸ the (P)RR was understandably implicated in the pathogenesis of kidney disease.⁷⁹ However, despite the appeal, it has been difficult to reconcile this view of the (P)RR with a number of other experimental findings, regarding both its potentially pathogenetic role and pattern of distribution within the kidney and its homology to other proteins. For instance, given the purported pathogenetic role of the (P)RR, the increased abundance of renin that follows the use of ACE inhibitors and ARBs would be expected to be adverse, yet these classes of drugs have been repeatedly shown to be kidney protective. Second, although the (P)RR was initially localized to the glomerular mesangium, more recent detailed studies have shown that (P)RR is primarily expressed in the collecting duct.⁸⁰ Third, although initially reported as having no homology with any known membrane protein,²⁰ database interrogation shows that the (P)RR is identical to two other proteins: CAPER (endoplasmic reticulum-localized type 1 transmembrane adaptor precursor) and ATP6ap2 (ATPase, H^+ transporting, lysosomal accessory protein 2),⁸¹⁻⁸⁵ a protein that associates with the vacuolar H^+ -ATPase.⁸⁶ Indeed, the predominant expression of the (P)RR at the apex of acid-secreting cells in the collecting duct, in conjunction with its colocalization and homology with an accessory subunit of the vacuolar H^+ -ATPase, suggests that the (P)RR may function primarily in urinary acidification.⁸⁰ However, the vacuolar H^+ -ATPase is not restricted to the kidney but is widely distributed in the plasma membrane and the membranes of organelles in several tissues where it functions, not only in urinary acidification but also in endocytosis, conversion of proinsulin to insulin, and osteoclast bone resorption.⁸⁷ Whereas the prevailing data indicate that (P)RR is an accessory subunit of the vacuolar H^+ -ATPase that also binds renin and prorenin, the precise functions of the prorenin-binding and renin-binding subunit remain to be unraveled in the kidney and elsewhere.

ANGIOTENSIN-CONVERTING ENZYME 2

In 2000, two groups independently reported the existence of the first ACE homolog, ACE2, an apparently novel zinc

metalloprotease but with considerable homology (40% identity and 61% similarity) to ACE.^{88,89} The gene-encoding ACE2, located on the X chromosome (Xp22), contains 18 exons, several of which bear considerable similarity to the first 17 exons of human *ACE*. Its transcript is 3.4 kb, generating an 805-amino acid peptide that is most highly expressed in kidney, heart, and testis but is also present in plasma and urine.^{90,91} In contrast to ACE, ACE2 functions as a carboxypeptidase, removing the terminal phenylalanine from angiotensin II to yield the vasodilatory heptapeptide, angiotensin 1–7 (Ang [1–7]). ACE2 may also indirectly lead to the formation of Ang (1–7) by cleaving the C-terminal leucine from angiotensin I, thereby generating angiotensin 1–9 (Ang [1–9]), which may then give rise to Ang (1–7) under the influence of ACE or neutral endopeptidase (NEP).⁹¹ Thus ACE2 contributes to both angiotensin II degradation and Ang (1–7) synthesis. Accordingly, ACE and ACE2 were initially viewed as having opposing actions with regard to vascular tone and tissue injury. However, emerging data suggest that the situation is far from clear. For instance, while lentivirus-induced overexpression of ACE2 in the heart exerted a protective influence following experimental myocardial infarction,⁹² ACE2 overexpression led to cardiac dysfunction and fibrosis, despite lowering systemic blood pressure.⁹³

In the kidney, ACE2 colocalizes with ACE and angiotensin receptors in the proximal tubule, while in the glomerulus it is predominantly expressed within podocytes and to a lesser extent in mesangial cells, contrasting the endothelial predilection of ACE at that site.⁹⁴ Numerous studies have explored changes in ACE2 expression in human kidney disease, as well as in a range of animal models, reporting both increased and decreased levels.⁹⁵ As such, it is uncertain whether increased ACE2 might be detrimental or a beneficial response to injury. With this in mind, the findings of intervention studies are of particular importance.

In experimental diabetic nephropathy, for instance, pharmacologic ACE2 inhibition with MLN-4760 led to worsening albuminuria and glomerular injury;⁹⁴ similar findings were reported in ACE2 knockout mice that were crossed with the Akita model of type 1 diabetes.⁹⁶ As might be expected from these findings, augmenting ACE2 activity by the infusion of human recombinant protein (hrACE2) was shown to attenuate diabetic kidney injury in the Akita mouse. In this study, hrACE2 improved kidney structure and function and also showed that the protective effects were likely due to reduction in angiotensin II and an increase in Ang (1–7) signaling.⁹⁷ In addition to its role in the RAS, ACE2 has been shown to be the receptor for the severe adult respiratory syndrome (SARS) coronavirus.⁹⁸ The relevance of some of these findings to the human setting will hopefully be clarified when results of the recently initiated clinical trials of a recombinant human ACE2 peptide (GSK2586881) become available.

OTHER ANGIOTENSIN PEPTIDES

Angiotensin III or angiotensin (2–8)

Formed by the actions of aminopeptidase A, the heptapeptide angiotensin III (angiotensin 2–8), like angiotensin II (angiotensin 1–8), exerts its effects by binding to the AT₁R and AT₂R.⁹⁹ Initially, angiotensin III was thought

to have a predominant role in regulating vasopressin release.¹⁰⁰ However, more recent studies indicate that while angiotensin III is equipotent to angiotensin II with regard to its effects on blood pressure, aldosterone secretion, and renal function, its metabolic clearance rate is approximately five times as rapid.¹⁰¹

Angiotensin IV or angiotensin (3–8)

Angiotensin IV is generated from angiotensin III by the actions of aminopeptidase M. Although some of its actions are mediated by the AT₁R, the majority of angiotensin IV's biological effects are thought to result from its binding to insulin-regulated aminopeptidase.¹⁰² Previously viewed as inactive, there has been considerable interest in angiotensin IV with regard to its actions in the central nervous system (CNS), where it enhances learning and memory, possesses anticonvulsant properties, and protects the brain from ischemic injury.¹⁰²

In addition to its CNS effects, angiotensin IV has also been implicated in atherogenesis, principally related to its ability to activate NF-κB and upregulate several proinflammatory factors that include monocyte chemoattractant protein-1 (MCP-1), intercellular adhesion molecule-1, interleukin-6 (IL-6), and TNF-α, as well as enhancing the synthesis of the prothrombotic factor plasminogen activator inhibitor-1.^{103,104} In the kidney, angiotensin IV is reported to have variable effects on blood flow and natriuresis.¹⁰²

Angiotensin (1–7)

The ostensibly vasodilatory and antitrophic angiotensin 1–7 may be formed by the actions of several endopeptidases that include removal of the terminal tripeptide of angiotensin I by NEP, cleavage of the C-terminal phenylalanine of angiotensin II by ACE2, and the excision of the dipeptidyl group from the C terminal of angiotensin 1–9 by ACE. Evolving evidence indicates that the actions of this heptapeptide are mediated by its binding to the orphan GPCR *masR*.¹⁰⁵ Angiotensin 1–7 induces vasodilation by a number of mechanisms that include the amplification of bradykinin's effects, stimulating cGMP synthesis, and inhibiting the release of norepinephrine.¹⁰⁶ Additionally, angiotensin 1–7 inhibits vascular smooth muscle proliferation and prevents neointima formation following balloon injury of the carotid arteries.¹⁰⁷ Contrasting these findings, however, is a report that exogenous angiotensin (1–7), rather than ameliorating diabetic nephropathy, as might have been predicted on the basis of the prevailing paradigm, actually accelerated the progression of the disease.⁹⁶

Angiotensin (2–10)

In addition to angiotensin II and the other C-terminal cleavage products discussed earlier, angiotensin I (1–10) may also give rise to a number of other potentially biologically active peptides that result from removal of amino acids from its N terminus. Of these, angiotensin 2–10, produced by the actions of aminopeptidase A, has been found to modulate the pressor activity of angiotensin II in rodents.¹⁰⁸

Angiotensin (1–12)

Angiotensin (1–12) is a dodecapeptide, first isolated in rat intestine but also found to be present in the kidney and heart, that is cleaved from angiotensinogen by a heretofore

unidentified nonrenin enzyme.¹⁰⁹ Notably, in the kidney, angiotensin (1–12), akin to other components of the intrarenal RAS, is primarily localized to the proximal tubule epithelium.¹¹⁰ Although its biological activity is incompletely understood, its main mode of action is thought to be mediated by its ability to serve as a precursor to angiotensin II by the site-specific and possibly species-specific actions of ACE and chymase.¹¹¹ Other pathways may, however, also contribute to the overall effects of angiotensin (1–12), which in the rat kidney may also include the formation of angiotensin (1–7) and angiotensin (1–4) by neprilysin.¹¹⁰

INTRARENAL RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM

In the traditional view of the RAS, angiotensin II functions as a hormone that, in classical endocrine fashion, circulates systemically to act at sites distant from those where it was formed. However, since the cloning of its components, it has become increasingly clear that there is an additional local, tissue-based RAS that functions quasiindependently from its systemic counterpart, acting in paracrine, autocrine, and possibly even intracrine modes.¹¹² This is most clearly seen in the kidney where pioneering work of Navar and colleagues^{113–115} has shown that the kidney possesses all the necessary molecular machinery to synthesize angiotensin II and other bioactive angiotensin peptides. Moreover, their concentrations in glomerular filtrate, tubule fluid, and interstitium are between 10- and 1000-fold higher than in plasma.^{41,116}

Within the kidney, renin-expressing cells have traditionally been considered to be terminally differentiated and confined to the JGA. However, in a series of elegant studies using a fate-mapping cre-loxP system, Sequeira López and colleagues¹¹⁷ showed that renin-expressing cells are precursors to a range of other cell types in the kidney, including those of the arteriolar media, mesangium, Bowman capsule, and proximal tubule. While normally quiescent, these cells may undergo metaplastic transformation to synthesize renin when homeostasis is challenged.¹¹⁷ Such threats include those related to volume depletion and also tissue injury. For instance, in the setting of single nephron hyperfiltration and consequent progressive dysfunction that follows renal mass reduction, Gilbert and colleagues¹¹⁸ noted the *de novo* expression of renin mRNA and angiotensin II peptide in tubule epithelial cells.

In addition to resident kidney cells, infiltrating mast cells may also contribute to activation of the local RAS in disease. Traditionally associated with allergic reactions and host responses to parasite infestation, mast cells have been increasingly recognized for their role in inflammation, immunomodulation, and chronic disease. In the kidney, interstitial mast cell infiltration accompanies most forms of chronic kidney disease (CKD), where their abundance correlates with the extent of tubulointerstitial fibrosis and declining GFR, though not proteinuria.¹¹⁹ Notably, mast cells have been shown to synthesize renin,¹²⁰ such that their degranulation will release large quantities of both renin and chymase, accelerating angiotensin II formation in the local environment.

INTRACRINE RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM

Peptide hormones traditionally bind to their cognate receptors on the plasma membrane and produce their

effects through the generation of secondary intermediates. However, emerging evidence suggests that certain peptides may also act directly within the cell's interior, having arrived there by either internalization or intracellular synthesis. For instance, angiotensin II has been localized within the cytoplasm and nucleus, but its introduction into the cytoplasm was shown more than a decade ago to have major effects on intracellular calcium currents.¹²¹ Uptake of angiotensin II from the extracellular space likely contributes to its intracellular activity; however, newer studies have focused predominantly on its endogenous synthesis. Consistent with the potential role for intracellular angiotensin II, transgenic mice that express an enhanced cyan fluorescent protein–angiotensin II fusion protein that lacks a secretory signal so that it is retained intracellularly develop hypertension with microthrombi in glomerular capillaries and small vessels.¹²² To date, numerous canonical and noncanonical pathways in the cytoplasm, nucleus, and mitochondria have been implicated in the intracrine RAS,^{123,124} providing a new forefront for the role of the RAS in physiology, pathophysiology, and therapeutics.

RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM IN KIDNEY PATHOPHYSIOLOGY

GLOMERULAR HEMODYNAMIC EFFECTS

In a critical series of experiments in the 1980s, Brenner and colleagues¹²⁵ studied the hemodynamic effects of renal mass ablation in 5/6 nephrectomized rats, a now well-established model of progressive kidney disease. In the setting of nephron loss, those glomeruli that remain undergo compensatory enlargement with increased single-nephron GFR (SNGFR) and elevations in intraglomerular pressure (P_{GC}), argued to initiate a vicious cycle of glomerulosclerosis and progressive loss of function. That this phenomenon might be related to angiotensin II was suggested by previous work in which angiotensin II infusion was demonstrated to also result in elevated P_{GC} .¹²⁶ Together, these studies suggested that intraglomerular hypertension, as a consequence of angiotensin II's action, was a pivotal factor underlying the inexorable progression of kidney disease and that strategies to reduce P_{GC} would lead to its amelioration. Indeed, in proof-of-concept studies, blockade of angiotensin II formation with the ACE inhibitor, enalapril, was shown to dilate the glomerular efferent arteriole, reduce P_{GC} , and slow disease progression in 5/6 nephrectomized rats.¹²⁷ By contrast, combination therapy with hydralazine, reserpine, and hydrochlorothiazide, though equally effective in lowering systemic blood pressure, failed to ameliorate intraglomerular hypertension and disease progression.¹²⁷ These studies were soon followed by similar ones in other disease models, particularly diabetes, which like the 5/6 nephrectomized rats is also characterized by increased SNGFR and elevated P_{GC} .^{128,129} For a more detailed discussion of glomerular hemodynamic changes after 5/6 nephrectomy, see [Chapter 50](#).

FIBROSIS

Considerable research has also focused on many of the nonhemodynamic effects of angiotensin II. For instance, in addition to their effects on P_{GC} , ACE inhibitors and ARBs are also highly effective in reducing interstitial fibrosis and tubule

atrophy, each close correlates of progressive kidney dysfunction. Underlying these effects is the ability of angiotensin II to potentially induce expression of the profibrotic and proapoptotic growth factor and TGF- β in a range of kidney cell types.^{130,131} Consistent with these *in vitro* studies, TGF- β overexpression is seen in both the glomerular and tubulointerstitial compartments in 5/6 nephrectomized rats and diabetic rats, where studies also showed that both ACE inhibitors and ARBs were effective at reducing TGF- β and disease progression.^{132,133} Similarly, in human diabetic nephropathy, the ACE inhibitor perindopril was found to reduce TGF- β mRNA in a sequential renal biopsy study¹³⁴ and losartan was shown to lower urinary TGF- β excretion.¹³⁵ For further discussion of the nonhemodynamic effects of angiotensin II, see [Chapter 50](#).

PROTEINURIA

The development of proteinuria is both a cardinal manifestation of glomerular injury and a pathogenetic factor in the progression of renal dysfunction. While P_{GC} remains an important factor in determining the transglomerular passage of albumin, more recent work has focused on the potential contribution of the podocyte. Indeed, podocyte injury is a cardinal manifestation of proteinuric renal disease, where foot process effacement has been shown to be prevented by both ACE inhibition and angiotensin receptor blockade.¹³⁶ In consideration of its crucial role in the development and function of the glomerular filtration barrier, other studies have focused on the podocyte slit pore membrane protein nephrin. Of note, podocytes express the AT₁R and respond to the addition of angiotensin II to the cell culture medium by dramatically decreasing their expression of nephrin.¹³⁷ Consistent with these findings, the reduction in nephrin expression in patients with diabetic nephropathy was shown to be ameliorated by ACE inhibitor treatment for 2 years.¹³⁸ For further discussion of the pathophysiology of proteinuria in CKD, see [Chapters 29 and 50](#).

INFLAMMATION AND IMMUNITY

Inflammatory cell infiltration is a long-recognized feature of CKD that is attenuated in rodent models by agents that block the RAS.¹³⁹ In the *in vitro* setting, angiotensin II activates NF- κ B by both AT₁R- and AT₂R-dependent pathways, stimulating the expression of a number of potent chemokines such as MCP-1 and regulated on activation, normal T cell expressed and secreted (RANTES), as well as cytokines, like IL-6.¹⁴⁰ In addition to angiotensin II, angiotensin (1–7), acting via the Mas receptor, activates NF- κ B, inducing proinflammatory effects in the kidney under both basal and disease settings.¹⁴¹

In addition to macrophages, mast cells, and other components of the innate immune system, the adaptive immune system also appears to be involved in the pathogenesis of angiotensin II–mediated organ injury. Of note, suppression of the adaptive immune system prevents the development of angiotensin II–dependent hypertension in experimental models¹⁴² and adoptive transfer of CD4⁺CD25⁺ regulatory T cells is able to ameliorate angiotensin II–dependent injury.¹⁴³

THE DIABETES PARADOX

Despite the fact that patients with long-standing diabetes characteristically have low plasma renin,^{144–146} suggesting

that the RAS is not activated by the disease, agents that block the RAS are the mainstay of therapy in diabetic nephropathy.¹⁴⁷ Compounding this apparent paradox, although PRA is normal or low in diabetes, plasma prorenin is characteristically elevated. This dichotomy suggests differences in cell-specific responses to diabetes since the JGA is the primary source of renin secretion while prorenin is secreted by a much wider range of cell types. Peti-Peterdi and colleagues¹⁴⁸ have proposed a model for the (pro)renin paradox of diabetes. Although early diabetes would lead to augmented succinate and enhanced JGA renin release, elevated angiotensin II levels would thereafter suppress JGA renin secretion. Contrasting this negative feedback at the JGA, angiotensin II has been shown to have the opposite effect in the tubule, with diabetes causing a 3.5-fold increase in collecting duct renin that could be reduced by AT₁R blockade.¹⁴⁹ These observations also support a role for the intrarenal component of the RAS in CKD progression, discussed earlier.

ENDOTHELIN

Clinical Relevance

The actions of the endothelins (ETs) are mediated by two receptors: ET types A (ET-A) and B (ET-B). ET receptor antagonists (particularly ET-A receptor antagonists) are being investigated for their efficacy in slowing progression of chronic kidney diseases, especially diabetic kidney disease. However, whereas ET receptor antagonists lower proteinuria, their clinical development has been hampered by the adverse effect of fluid retention and a narrow therapeutic window.

ETs are potent vasoconstrictors that, although expressed primarily in the vascular endothelium, are also notably present within the renal medulla. The biologic effects of the ET system are mediated by two receptors: ET types A (ET-A) and B (ET-B). In the kidneys, these receptors contribute to the regulation of RBF, salt and water balance, and acid-base homeostasis, as well as potentially mediating tissue inflammation and fibrosis. An important therapeutic role for ET receptor antagonism first emerged in the treatment of pulmonary hypertension;^{150–153} ET receptor antagonists have been granted regulatory authority approval for this indication in the United States and Europe.¹⁵⁴ ET receptor blockade as a therapeutic strategy has been investigated in a range of renal diseases. Clinical trials have demonstrated the antiproteinuric and antihypertensive properties of ET receptor antagonists, which have a relatively narrow therapeutic window for the treatment of CKD. Blockade of endothelin type A receptors, either alone or in combination with type B receptor blockade, has been studied in the treatment of IgA nephropathy and diabetic nephropathy.

STRUCTURE, SYNTHESIS, AND SECRETION OF THE ENDOTHELINS

ETs consist of three 21-amino acid isoforms that are structurally and pharmacologically distinct: ET-1, ET-2, and ET-3. The dominant isoform in the cardiovascular system is

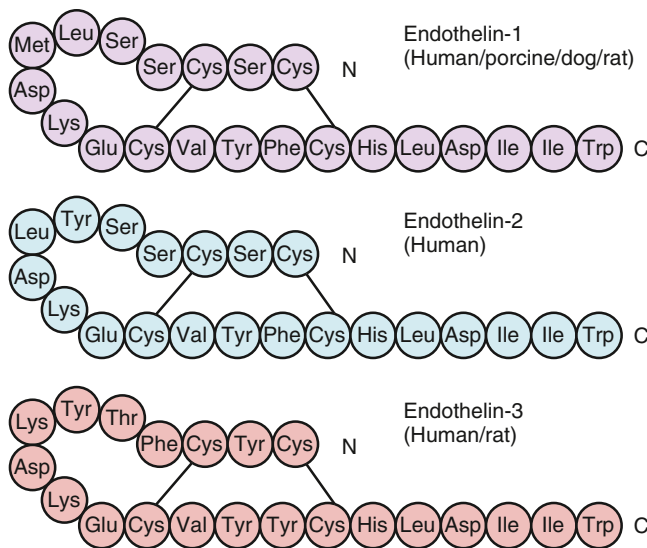


Fig. 11.6 Molecular structure of the three endothelin isoforms. (Modified from Schiffrin EL. Vascular endothelin in hypertension. *Vascul Pharmacol.* 2005;43:19–29.)

ET-1. Differences in the amino acid sequence among the isopeptides are minor. All three isoforms share a common structure with a typical hairpin-loop configuration that results from two disulfide bonds at the amino terminus and a hydrophobic carboxy terminus that contains an aromatic indol side chain at Trp₂₁ (Fig. 11.6). Both the carboxy terminus and the two disulfide bonds are responsible for the biologic activity of the peptide. ETs are synthesized from preprohormones by posttranslational proteolytic cleavage mediated by furin and other enzymes. Dibasic pair-specific processing endopeptidases, which recognize Arg-Arg or Lys-Arg paired amino acids, cleave preproETs, reducing their size from approximately 203 to 39 amino acids. Subsequent proteolytic cleavage of the largely biologically inactive big ETs is mediated by endothelin-converting enzymes (ECEs), the key enzymes in the ET biosynthetic pathway. ECE1 and ECE2 are type II membrane-bound metalloproteases whose amino acid sequence is significantly homologous to that of neprilysin (NEP 24.11).

Secretion of ET-1 is dependent on de novo protein synthesis, which is constitutive. However, a range of stimuli may also increase ET synthesis through both transcriptional and posttranscriptional regulation (Table 11.1). Once it is synthesized, ET-1 is secreted by endothelial cells into the basolateral compartment, toward the adjacent smooth muscle cells. Because of its abluminal secretion, plasma levels of ET-1 do not necessarily reflect its production.¹⁵⁵

Within the kidneys, ET-1 expression is most abundant in the inner medulla. In fact, this region possesses the highest concentration of ET-1 of any tissue bed.¹⁵⁶ In addition to their presence in the inner medullary collecting ducts (IMCDs), ETs have also been described in glomerular endothelial cells,¹⁵⁷ glomerular epithelial cells,¹⁵⁸ mesangial cells,¹⁵⁹ vasa recta,¹⁶⁰ and tubule epithelial cells.¹⁶¹ The kidneys also synthesize ET-2 and ET-3, although at much lower levels than they do ET-1.¹⁶² As with ET-1, ECE1 mRNA is also more abundant in the renal medulla than in the cortex under normal

Table 11.1 Endothelin Gene and Protein Expression

Stimulation

Vasoactive Peptides

Angiotensin II
Bradykinin
Vasopressin
Endothelin-1
Epinephrine
Insulin
Glucocorticoids
Prolactin
Inflammatory Mediators
Endotoxin
Interleukin-1
Tumor necrosis factor- α
Interferon- β

Growth Factors

Epidermal growth factor
Insulin-like growth factor
Transforming growth factor- β
Coagulation
Thromboxane A₂
Tissue plasminogen activator
Other
Calcium
Hypoxia
Shear stress
Phorbol esters
Oxidized low-density lipoproteins

Inhibition

Atrial natriuretic peptide
Brain natriuretic peptide
Bradykinin
Heparin
Prostacyclin
Protein kinase A activators
Nitric oxide
Angiotensin-converting enzyme inhibitors

conditions. However, in disease states such as chronic heart failure, ECE1 mRNA is upregulated primarily within the cortex.¹⁶³ In human kidneys, ECE1 has been localized to endothelial cells and tubule epithelial cells in the cortex and medulla.¹⁶⁴

ENDOTHELIN RECEPTORS

ETs bind to two seven-transmembrane domain GPCRs, ET-A and ET-B. Within the vasculature, ET-A receptors are found on smooth muscle cells, where they mediate vasoconstriction. Although ET-B receptors localized on vascular smooth muscle cells can also mediate vasoconstriction, they are mainly expressed on endothelial cells, where their activation results in vasodilation through the production of NO and prostacyclin.¹⁶⁵ In addition to their role in mediating vascular tone, ET-B receptors also act as clearance receptors for ET-1,¹⁶⁶ particularly in the lung, where ET-B receptor binding accounts for approximately 80% of clearance.¹⁶⁷ Because of its natriuretic and vasodilatory actions, the ET-B receptor is generally considered to confer predominantly kidney-protective effects.

In the kidneys, expression of both ET-A and ET-B receptors is most prominent within the IMCDs, although binding of ET-1 also occurs in smooth muscle cells, endothelial cells, renomedullary interstitial cells, thin descending limbs, and medullary thick ascending limbs.¹⁶⁰ ET-A receptors are localized to several renovascular structures, including vascular smooth muscle cells, arcuate arteries, and pericytes of descending vasa recta, as well as glomeruli. ET-B receptors, although prominently represented within the medullary collecting system, have also been demonstrated in proximal convoluted tubules, collecting ducts of the inner cortex, medullary thick ascending limbs, and podocytes.¹⁶⁸

PHYSIOLOGIC ACTIONS OF ENDOTHELIN IN THE KIDNEY

The ETs have several effects on normal renal function including regulation of RBF, sodium and water balance, and acid-base homeostasis. Although ET-1 has hemodynamic effects in almost all vessels, the sensitivity of different vascular beds varies. The renal vasculature, along with the mesenteric vessels, is the most sensitive: Vasoconstriction occurs at picomolar concentrations of ET-1,^{169,170} increasing RVR and decreasing RBF. However, long-lasting vasoconstriction that is mediated by the ET-A receptor may be preceded by a transient ET-B receptor-mediated vasodilation.¹⁷¹ Because of the site-specific distribution of ET receptors, ET-1 may exert different vasoconstrictive and vasodilatory effects in different regions of the kidneys. For example, by inducing NO release from adjacent tubule epithelial cells, ET-1 may actually increase blood flow in the renal medulla, where ET-B receptors predominate.¹⁷²

In addition to its effects on RBF, the ET system also plays a direct role in renal sodium and water handling.¹⁷³ In the renal medulla, ET is regulated by sodium intake and exerts its natriuretic and diuretic effects through the ET-B receptor.¹⁷⁴⁻¹⁷⁶ In addition to natriuretic and diuretic effects, the ET-B receptor may also contribute to acid-base homeostasis by stimulating proximal tubule sodium/proton exchanger isoform 3 (NHE3).¹⁷⁷ Although the role of ET-B receptor activation in urinary sodium excretion has been appreciated for some time, evidence suggests that renal medullary ET-A receptors may also mediate natriuresis.¹⁷⁸ This may partly explain the edema that can occur as a side effect of ET-A or dual ET receptor antagonism.

ROLE OF ENDOTHELIN IN ESSENTIAL HYPERTENSION

In view of its potent vasoconstrictive properties, it is not surprising that ET-1 has been implicated in the pathogenesis of hypertension. In preclinical models of hypertension, ET antagonism may ameliorate heart failure, vascular injury, and renal failure, as well as reduce the incidence of stroke.^{179,180} ET-A receptor antagonism has also been shown to normalize blood pressure in rats exposed to eucapnic intermittent hypoxia, which is analogous to sleep apnea in humans.¹⁸¹

PreproET-1 mRNA is increased in the endothelium of subcutaneous resistance arteries in patients with moderate to severe hypertension¹⁸² and according to a meta-analysis, plasma ET-1 concentrations are increased in individuals with hypertension.¹⁸³ However, plasma ET-1 levels are not universally elevated;¹⁷⁹ an increase is found more commonly in the presence of end-organ damage or in salt-depleted, salt-sensitive patients with a blunted renin response.¹⁸⁴ A major component of this increase in disease is often decreased clearance by the kidneys. These findings suggest that certain patient subgroups may be more responsive than others to ET receptor blockade. Females appear to be relatively protected from the pressor effects of ET-1 by virtue of both increased ET-B expression and a blunted hemodynamic response to ET-A receptor activation.¹⁸⁵

Clinical trials of the antihypertensive effects of ET receptor antagonism have been hampered by difficulties

with selectivity for the ET-A receptor, study design, dosing regimens, and adverse events.¹⁸⁶ Because the ET-B receptor exerts diuretic and natriuretic effects, induces vasodilation, and clears ET-1, selective ET-A receptor antagonists may be expected to demonstrate a more favorable antihypertensive profile.¹⁸⁶ Mixed ET-A/B and specific ET-A receptor antagonists are distinguished by their in vitro binding affinities with mixed ET-A/B receptor antagonists demonstrating selectivity for ET-A of <100-fold and ET-A selective antagonists having an affinity for the ET-A receptor of 100-fold or higher. However, it has been suggested that a 1000-fold or higher affinity may be required to induce ET-A receptor-specific effects in vivo.^{187,188} In an early study, treatment of patients with essential hypertension with the nonselective ET receptor antagonist bosentan decreased blood pressure as effectively as enalapril, without reflex neurohumoral activation, over a 4-week period.¹⁸⁹ Similarly, in 115 persons with resistant hypertension who were taking three or more agents, the selective ET-A receptor antagonist darusentan significantly reduced blood pressure at 10 weeks.¹⁹⁰ In a subsequent study of 379 individuals with resistant hypertension, darusentan treatment for 14 weeks reduced blood pressure by approximately 18/10 mm Hg with no evidence of dose dependence across a range of 50 to 100 mg/day.¹⁹¹ However, in a second study of similar design, a large placebo effect meant that darusentan treatment failed to achieve its primary endpoint of change in office blood pressure, and the development of the drug for this indication was halted.¹⁹² Interestingly, in both of these studies ambulatory blood pressure monitoring revealed a reduction in systolic blood pressure with active treatment.^{187,193} However, also in both studies, peripheral edema or fluid retention was more common in patients treated with darusentan than those receiving placebo.^{191,192}

ROLE OF ENDOTHELIN IN KIDNEY INJURY

Beyond its effects on the regulation of vascular tone, the ET system also likely plays a direct role in the pathogenesis of fibrotic injury in CKD.¹⁹⁴ In persons with CKD, plasma ET-1 concentrations are elevated, as a result of both increased production and decreased renal clearance.¹⁹⁵ Urinary levels of ET-1 are also increased, which is indicative of increased renal ET-1 expression.¹⁹⁶ One mechanism for increased renal ET-1 in CKD is a direct effect of urinary protein on ET-1 expression in tubule epithelial cells.^{197,198} Beyond the direct effects of urine protein, a number of proinflammatory factors induce ET-1 expression in the kidneys including hypoxia, angiotensin II, thrombin, thromboxane A₂, TGF- β , and shear stress (see Table 11.1).

Several distinct mechanisms may account for the injurious effects of ET-1 on the kidneys. Locally derived ET-1 has direct hemodynamic effects, increasing P_{GC} at high doses and causing vasoconstriction of the vasa recta and peritubular capillaries, with a resultant reduction in tissue oxygen tension. ET-1 acts as a chemoattractant for inflammatory cells, which may express the peptide themselves, stimulating interstitial fibroblast and mesangial cell proliferation and mediating the production of a number of factors associated with collagenous matrix deposition, including TGF- β , matrix metalloproteinase-1, and tissue inhibitors of metalloproteinases 1 and 2. In mesangial cells, ET-1 can

induce cytoskeletal remodeling and cell contraction¹⁹⁹ and, in these cells, ET-A receptor activation appears to be important in the development of Alport syndrome-associated glomerular disease.²⁰⁰

A particularly important role of ET-1 in mediating injury of glomerular podocytes is also beginning to emerge.²⁰¹ For instance, increased passage of protein across the filtration barrier causes podocyte cytoskeletal rearrangements and coincident upregulation of ET-1, which may act in an autocrine manner to further propagate ultrastructural injury in the same cells.²⁰²⁻²⁰⁴ ET-1 promotes podocyte dedifferentiation and migration by activating ET-A and increasing β -arrestin-1 expression, which ultimately results in EGFR transactivation, phosphorylation of β -catenin, and increased expression of the transcription factor Snail, an inducer of epithelial-mesenchymal transition.²⁰⁵ ET-1 induces calcium signaling in podocytes through both the ET-A and ET-B receptors and mice with deletion of both the ET-A receptor and ET-B receptor from podocytes are protected from the glomerular injury associated with streptozotocin-induced diabetes.²⁰⁶ Using these mice, investigators have subsequently gone on to demonstrate that ET-1 causes podocytes to release heparanase that damages the endothelial glycocalyx facilitating albumin passage across the filtration barrier.²⁰⁷

THE ENDOTHELIN SYSTEM IN CHRONIC KIDNEY DISEASE AND DIABETIC NEPHROPATHY

PRECLINICAL STUDIES OF ENDOTHELIN RECEPTOR ANTAGONISTS IN DIABETIC KIDNEY DISEASE

ET receptor antagonists have been employed to study the role of ETs in renal pathophysiology in a range of experimental models including the rat remnant kidney, lupus nephritis, and diabetes. In the remnant kidney model of progressive kidney disease, although beneficial effects have been reported with nonselective ET receptor antagonists,²⁰⁸ selective ET-A receptor inhibition appears to yield superior outcomes, with concomitant inhibition of ET-B receptors potentially abrogating any beneficial effects.²⁰⁹

Data with regard to an effect of high glucose concentrations on ET synthesis and secretion are conflicting. Mesangial cell p38 MAPK activation in response to ET-1, angiotensin II, and PDGF is enhanced in the presence of high glucose levels.²¹⁰ By contrast, mesangial contraction in response to ET-1 is diminished under high glucose conditions.^{211,212} Circulating ET-1 concentrations are elevated in animal models of both type 1 and type 2 diabetes. Increased expression of ET-1 and its receptors has been found in glomeruli and tubule epithelial cells,^{198,213} although increased expression of ET receptors has not been a universal finding.²¹⁴ Diabetes also causes an increase in renal ECE1 expression, the effect being synergistic with that of radiocontrast media.²¹⁵

A number of researchers have investigated the effect of both nonselective and selective ET-A receptor antagonists in experimental diabetic nephropathy. In streptozotocin-diabetic rats, the nonselective ET receptor antagonist bosentan has yielded conflicting results,^{216,217} whereas another nonselective ET receptor antagonist, PD142893, improved kidney function when administered to streptozotocin-diabetic rats that were already proteinuric.¹⁹⁸ In addition, acute ET-A receptor

antagonism has been shown to improve oxygen availability in the kidneys of streptozotocin-diabetic rats.²¹⁸ In Otsuka Long Evans Tokushima Fatty (OLETF) rats with type 2 diabetes, selective ET-A receptor blockade attenuated albuminuria, without affecting blood pressure, whereas ET-B receptor blockade had no effect.²¹⁹ In a study of streptozotocin-diabetic apolipoprotein E knockout mice, the kidney-protective effects of the predominant ET-A receptor antagonist avosentan were comparable with or superior to the ACE inhibitor quinapril.²²⁰ Supporting a protective role for the ET-B receptor, diabetic ET-B receptor-deficient rats developed severe hypertension and progressive kidney failure.²²¹

Accumulation of reactive oxygen species plays a major role in the pathogenesis of diabetic complications, particularly diabetic nephropathy,^{222,223} and several observations suggest that the ET system may contribute to oxidative stress. In low-renin hypertension, ET-1 increases superoxide in carotid arteries²²⁴ and ET-A receptor blockade decreases vascular superoxide generation.^{225,226} Similarly, ET-1 infusion increased urinary excretion of 8-isoprostane prostaglandin $F_{2\alpha}$ in rats, which is indicative of increased generation of reactive oxygen species.²²⁷ By contrast, however, other preclinical studies have suggested a predominantly proinflammatory role for ET-1 in diabetic nephropathy. For instance, the selective ET-A receptor antagonist ABT-627 prevented the development of albuminuria in streptozotocin-diabetic rats without an improvement in markers of oxidative stress but with a reduction in macrophage infiltration and urinary excretion of TGF- β and prostaglandin E_2 metabolites.²²⁸

CLINICAL STUDIES OF ENDOTHELIN RECEPTOR ANTAGONISTS IN CHRONIC KIDNEY DISEASE AND DIABETIC NEPHROPATHY

Both plasma and urinary ET-1 levels are increased in patients with CKD^{195,229,230} with plasma ET-1 levels inversely correlating with estimated GFR.²³¹ In a study of hypertensive patients with CKD, both selective ET-A receptor blockade and nonselective ET receptor blockade lowered blood pressure.²³² However, ET-A receptor blockade increased both RBF and effective filtration fraction and decreased RVR, whereas dual blockade had no effect.²³²

In a study of 22 nondiabetic individuals with CKD, intravenous infusion of the ET-A receptor antagonist BQ-123 reduced pulse-wave velocity and proteinuria to a greater extent than the calcium channel antagonist nifedipine, which comparably lowered blood pressure.²³³ These findings suggest a potential blood pressure-independent mechanism of action for the antiproteinuric effect observed. In a subsequent study by the same investigators, 27 subjects with proteinuric CKD were treated with the ET-A receptor antagonist sitaxsentan for 6 weeks in a three-way crossover study design. Sitaxsentan treatment was associated with a reduction in blood pressure, proteinuria, and pulse-wave velocity, whereas nifedipine reduced pulse-wave velocity and blood pressure but had no effect on urine protein excretion.²³⁴ Subsequently, sitaxsentan was also shown to restore the nocturnal dip in blood pressure in people with CKD.²³⁵ A fall in GFR with sitaxsentan therapy observed in this study is analogous to that seen with RAS blockade.²³⁴ Although no clinically significant adverse effects were seen,

sitaxsentan development has subsequently been halted due to hepatotoxicity.

The antiproteinuric effect of the ET-A receptor antagonist, zibotentan, has been evaluated in combination with the SGLT2 inhibitor, dapagliflozin, in persons with eGFR ≥ 20 mL/min/1.73 m² and urinary albumin-creatinine ratio (UACR) 150 to 1500 mg/g. The rationale for this trial design was to assess if the effects of zibotentan were additive to dapagliflozin and to explore whether the natriuretic effects of an SGLT2 inhibitor may ameliorate the fluid retention associated with ET-A receptor antagonists. The original study design included groups randomized to placebo, dapagliflozin, and dapagliflozin combined with three doses of zibotentan: 5 mg/day, 1.5 mg/day and 0.25 mg/day (in addition to RAAS inhibitor treatment if tolerated). Randomization to the placebo and high-dose (5 mg) zibotentan groups was discontinued on the advice of the data safety monitoring committee due to evidence of a high incidence of fluid retention and publication of trials establishing the kidney-protective efficacy of SGLT2 inhibitors. Among 447 participants randomized to the lower doses of zibotentan (mean GFR 46.7 mL/min/1.73 m²; median UACR 565.5 mg/g), zibotentan 1.5 mg/day and 0.25 mg/day plus dapagliflozin were associated with a 33.7% (90% CI, 42.5–23.5%; $P < 0.0001$) and 27.0% (90% CI, 38.4–13.6%; $P = 0.0022$) reductions in UACR versus dapagliflozin alone after 12 weeks. Fluid retention (increase in body weight $\geq 3\%$) was observed in 18%, 9%, and 8% in the three groups, respectively.²³⁶ A post hoc analysis of all the original groups that included 508 participants found that the risk of fluid retention increased with higher zibotentan dose and lower GFR but was decreased when zibotentan was combined with dapagliflozin.²³⁷

The effect of the ET-A receptor antagonist avosentan was examined, in addition to standard treatment with an ACE inhibitor or ARB, in a placebo-controlled trial of 286 patients with diabetic nephropathy and macroalbuminuria.²³⁸ At 12 weeks, avosentan was found to decrease urine albumin excretion rate without affecting blood pressure. These results led to the initiation of the ASCEND trial (a randomized, double-blind, placebo-controlled, parallel group study to assess the effect of the endothelin receptor antagonist avosentan on time to doubling of serum creatinine, end-stage renal disease, or death in patients with type 2 diabetes mellitus and diabetic nephropathy).²³⁹ ASCEND set out to examine the effects of avosentan, on top of RAS blockade, in 1392 individuals with type 2 diabetes and nephropathy but was terminated after a median duration of 4 months due to adverse events.²³⁹ In that study, despite a more than 40% reduction in urine albumin-to-creatinine ratio with avosentan, adverse events, predominantly fluid overload and congestive heart failure, occurred more frequently in those receiving active therapy than placebo.²³⁹

Since the publication of ASCEND,²⁴⁰ it has become apparent that the adverse outcomes were likely related to the relatively high dose of avosentan that was selected and that ET-A receptor antagonists likely have a relatively narrow therapeutic window. However, if this therapeutic window is appropriately targeted and patients carefully selected, then ET-A receptor antagonists may still find a clinical niche for the treatment of diabetic nephropathy. The Reducing Residual Albuminuria in Subjects with Diabetes and

Nephropathy with Atrasentan trial and an identical study conducted in Japan (RADAR/JAPAN) explored the effects of two different doses of the ET-A receptor antagonist atrasentan (0.75 and 1.25 mg/day, respectively) in 211 patients with type 2 diabetes and albuminuria (estimated GFR [eGFR] 30–75 mL/min/1.73 m²).²⁴¹ In comparison with placebo, 0.75 and 1.25 mg/day atrasentan, on top of maximum tolerated doses of ACE inhibitor or ARB, reduced urine albumin-to-creatinine ratio by an average of 35% and 38%, respectively, without major side effects.²⁴¹ Atrasentan treatment was also associated with decreases in 24-hour blood pressure, low-density lipoprotein cholesterol, and triglycerides.²⁴¹ Importantly, despite a comparable lowering of albuminuria to the 0.75 mg/day dose of atrasentan, the 1.25 mg/day atrasentan dose was accompanied by more fluid retention.²⁴¹ In a post hoc analysis, fluid retention was more likely in participants with lower eGFR and a higher dose of atrasentan, whereas the degree of urinary albumin lowering was not linked to the degree of fluid retention.²⁴² Thus it is likely that the antiproteinuric- and fluid-retaining effects of ET receptor antagonists are mediated by different mechanisms; plausibly, the former by vascular or glomerular actions, and the latter by direct effects of ET receptor blockade on sodium transport in the renal tubule.²⁴²

To reduce the risk of adverse effects, investigators included an “enrichment period” in the Study of Diabetic Nephropathy with Atrasentan (SONAR) trial, during which all participants received atrasentan. Only those who evidenced at least a 30% reduction in albuminuria and who did not develop edema progressed to the next phase of the trial to be randomized to continue atrasentan or placebo ($n = 2648$ of an initial 5117). The primary endpoint of doubling of serum creatinine, kidney failure, or death from kidney failure was reduced by 35% (HR 0.65; 95% CI, 0.49–0.88; $P = 0.0047$).²⁴³ Fluid retention and anemia occurred more frequently in the group who received atrasentan, but there was no statistically significant increase in hospital admissions with heart failure (HR 1.33; 95% CI, 0.85–2.07; $P = 0.208$). The relative and absolute effects of the 0.75 mg/day dose of atrasentan on kidney and heart failure events according to baseline eGFR and UACR were evaluated in a *post hoc* analysis of the SONAR trial. Atrasentan reduced the relative risk of the primary kidney outcome (composite of doubling of serum creatinine, kidney failure, or kidney death) consistently across baseline urine ACR and eGFR subgroups. The absolute risk reduction was greater among patients in the lowest eGFR and highest albuminuria category who were at highest baseline risk. Conversely, the relative and absolute risks of heart failure hospitalization were similar across baseline UACR and eGFR subgroups.²⁴⁴ (For further discussion of the treatment of diabetic kidney disease, see [Chapter 41](#))

ENDOTHELIN CONVERTING ENZYME (ECE)

COMBINED ECE AND NEPRILYSIN INHIBITION

Distinct from ET receptor antagonism, blockade of ET-1-induced signaling has been explored in the clinical setting with the use of the combined ECE and neprilysin inhibitor daglutril.²⁴⁵ In an 8-week, endothelin converting enzyme (ECE) crossover design study of participants with type 2 diabetes, blood pressure $< 140/90$ mm Hg, and urinary albumin excretion 20–999 $\mu\text{g}/\text{min}$, daglutril (300 mg/day)

did not significantly reduce albuminuria compared with placebo, although blood pressure was reduced.²⁴⁵ The failure to reach the primary endpoint of albuminuria reduction may relate to concurrent neprilysin inhibition, which may diminish ET-1 degradation.²⁴⁶ Alternatively, it may reflect an overall diminution of ET-1 with consequent decreased activation of ET-B, as well as ET-A.²⁴⁷

THE ENDOTHELIN SYSTEM AND OTHER KIDNEY DISEASES

In addition to diabetic and nondiabetic CKD, the role of the ET system has also been investigated in a number of other kidney diseases. Overall, these studies have suggested some degree of renoprotection with either selective ET-A or nonselective ET receptor inhibitors.

IgA NEPHROPATHY

Sparsentan is a dual endothelin and angiotensin receptor antagonist.²⁴⁸ In the phase 3 PROTECT trial, treatment with sparsentan was compared with irbesartan, an angiotensin II receptor blocker, in patients with IgA nephropathy.²⁴⁹ This double-blind, randomized trial was conducted across 134 clinical sites in 18 countries spanning the Americas, Asia, and Europe. Treatment with sparsentan led to greater proteinuria reduction and better preservation of kidney function than maximally titrated irbesartan. Patients receiving sparsentan experienced a slower decline in eGFR, with a *chronic* 2-year slope (weeks 6–110) of -2.7 mL/min per 1.73 m² per year compared with -3.8 mL/min per 1.73 m² per year in the irbesartan group (difference: 1.1 mL/min per 1.73 m² per year; 95% CI 0.1 – 2.1 ; $P=0.037$); the difference in *total* 2-year slope (day 1–week 110) was -2.9 mL/min per 1.73 m² per year versus -3.9 mL/min per 1.73 m² per year (difference: 1.0 mL/min per 1.73 m² per year; 95% CI -0.03 to 1.94 ; $P=0.058$).²⁵⁰

The ALIGN study evaluated the impact of atrasentan, a selective endothelin type A receptor antagonist, on albuminuria in IgA nephropathy using the 24-hour urinary protein-to-creatinine ratio as a surrogate endpoint. In this trial, patients with at least 1 g of daily urinary protein excretion and an eGFR of at least 30 mL/min per 1.73 m² were randomly assigned to receive either atrasentan (0.75 mg/day) or placebo for 132 weeks. While the long-term effect on kidney disease progression remains unclear, interim results showed a significant reduction in proteinuria with atrasentan compared with placebo at 36 weeks. Ongoing analysis is examining atrasentan's effect on eGFR.²⁵¹

SICKLE CELL DISEASE—ASSOCIATED NEPHROPATHY

Administration of the selective ET-A receptor antagonist ambrisentan preserved GFR and prevented the development of albuminuria in a humanized mouse model of sickle cell disease (SCD).²⁵² However, the kidney-protective effects of ambrisentan were only partially recapitulated by treatment with the combined ET-A/ET-B receptor antagonist A-182086, highlighting the importance of selectively targeting the ET-A receptor in SCD.²⁵²

RENOVASCULAR DISEASE

A series of studies in pigs provide support for ET-A receptor blockade in the treatment of renovascular disease. In an initial study, investigators treated pigs with unilateral renal

artery stenosis with an ET-A receptor antagonist beginning at the onset of renovascular disease and continuing for 6 weeks.²⁵³ In these experiments, researchers observed that ET-A receptor blockade preserved renal hemodynamics, renal function, and microvascular architecture in the stenotic kidney.²⁵³ Similarly, ET-A receptor blockade (but not ET-B receptor blockade) reversed microvascular rarefaction and diminished renal inflammation and fibrosis when it was initiated 6 weeks after the induction of renal artery stenosis.²⁵⁴ Finally, ET-A receptor blockade also led to an improvement in microvascular density and renal function recovery compared with placebo when it was administered following percutaneous transluminal renal angioplasty/stenting.²⁵⁵

ACUTE KIDNEY INJURY

ET-1 may play a role in sepsis-mediated acute kidney injury (AKI),²⁵⁶ although experimental findings have been conflicting, dependent to some extent on the ET receptor antagonist employed. For example, in a rat model of early normotensive endotoxemia, neither an ET-A receptor antagonist nor a combined ET-A/ET-B receptor blockade improved GFR,²⁵⁷ whereas ET-B receptor blockade alone resulted in a marked reduction in RBF.²⁵⁷ By contrast, in a porcine model of endotoxemic shock, the dual ET receptor antagonist tezosentan attenuated the decrease in RBF and increase in plasma creatinine.²⁵⁸ Pointing to a role of ET-1/ET-A signaling in the progression from AKI to CKD, transient unilateral renal ischemia-induced upregulation of ET-1 and ET-A receptor in mice and ET-A receptor antagonism (but not ET-B receptor antagonism) prevented progressive kidney injury.²⁵⁹

SYSTEMIC LUPUS ERYTHEMATOSUS

Urinary ET-1 excretion is correlated with disease activity in persons with systemic lupus erythematosus (SLE),²⁶⁰ and serum from such patients has been shown to stimulate ET-1 release from endothelial cells in culture.²⁶¹ In accordance with a pathogenetic role for the ET system in SLE, the ET-A receptor antagonist FR139317 attenuated kidney injury in a murine model of lupus nephritis. Moreover, endothelin receptor antagonism decreased renal immune cell infiltration in mice with SLE.²⁶²

PRIMARY FOCAL AND SEGMENTAL GLOMERULOSCLEROSIS

Sparsentan, a dual ET-A receptor/ARB antagonist, has also been evaluated for the treatment of primary focal and segmental glomerulosclerosis (FSGS). A phase 2 study examined the effect of three different doses of sparsentan (200 , 400 , and 800 mg/day) when compared with the ARB, irbesartan (300 mg/day), in 96 participants over an 8-week period and reported a mean reduction in proteinuria in sparsentan-treated patients of 45% in comparison with a 19% reduction in those receiving irbesartan.²⁶³ In a subsequent phase 3 trial, 371 participants with FSGS (without known secondary cause) were randomized to treatment with sparsentan or irbesartan for 108 weeks. At a prespecified interim analysis after 36 weeks, a greater proportion of those receiving sparsentan had achieved partial remission of proteinuria than those receiving

irbesartan (42% vs. 26%; $P = 0.009$), but there was no difference in the primary endpoint of GFR slope at 108 weeks.²⁶⁴

SCLERODERMA

The Zibotentan Better Renal Scleroderma Outcome Study (ZEBRA) was a three-part phase 2 study (ZEBRA 1, ZEBRA 2A, and ZEBRA 2B) exploring the safety and therapeutic potential of the ET-A receptor antagonist zibotentan in acute and chronic renal complications of scleroderma (NCT02047708). The primary outcome measure was the plasma level of soluble vascular cell adhesion molecule-1 as a biomarker of scleroderma renal involvement. Zibotentan was generally well tolerated. ZEBRA 1 did not show any effect of zibotentan on serum sVCAM-1 but was associated with numerical improvement in eGFR at 26 weeks that was more marked at 52 weeks.²⁶⁵

HEPATORENAL SYNDROME

Plasma ET-1 concentrations are increased in individuals with cirrhosis and ascites and in patients with type 2 hepatorenal syndrome (diuretic-resistant or refractory ascites with slowly progressive renal decline) in whom systemic vasodilation accompanies paradoxical renal vasoconstriction.²⁶⁶ To investigate the therapeutic potential of ET receptor antagonism in this setting, the combined ET-A/ET-B receptor blocker tezosentan was administered to six patients in an early-phase clinical trial.²⁶⁷ In this study, treatment was discontinued early in five patients, in one case because of systemic hypotension and in four because of concerns about worsening renal function.²⁶⁷ These adverse effects are consistent with a dose-dependent decline in kidney function in patients with acute heart failure treated with tezosentan, and they highlight the need for caution with the use of ET receptor antagonists in certain patient populations.

PREECLAMPSIA

A role for ET-1 in the development of preeclampsia is suggested by the observations that infusion of fms-like tyrosine kinase-1 and TNF- α into pregnant rats induced ET-A-dependent hypertension,²⁶⁸⁻²⁷⁰ whereas ET-A receptor antagonism attenuated placental ischemia-induced hypertension in a rat model.²⁷¹ Despite the mechanistic role of ET-1 in the pathogenesis of preeclampsia, however, ET receptor antagonists are unlikely to be used in this condition given their known teratogenicity.²⁶⁸

SAFETY PROFILE OF ENDOTHELIN RECEPTOR ANTAGONISTS

The therapeutic development of ET receptor antagonists has been slowed by the adverse side effect profile of available agents. Most notable has been the dose-dependent development of fluid retention, peripheral edema, and congestive heart failure despite the use of predominant ET-A receptor antagonists. The mechanisms that underlie the fluid retention associated with ET-A receptor antagonism have not been fully resolved. It has been suggested that the use of comparatively high doses of ET-A receptor antagonists may have resulted in concurrent ET-B receptor blockade. However, inhibition of nephron ET-A receptors may also be implicated.^{171,272} For instance,

mice with nephron or collecting duct ET-A receptor deletion were protected from the fluid retention associated with ET-A receptor blockade.²⁷³ Hepatotoxicity may be a class effect or may be restricted to particular subclasses of ET receptor antagonist. A rise in hepatic transaminases has been observed with both bosentan and sitaxsentan, which are both sulfonamide-based agents, but not with ambrisentan or darusentan, which are propionic acid based.^{187,190,191,274,275} As discussed earlier, teratogenicity would preclude the use of this class of agents in pregnancy, whereas the potential for testicular toxicity has also been described, although testicular damage has not been reported in patients taking ET receptor antagonists for the treatment of pulmonary hypertension.²⁴⁷

NATRIURETIC PEPTIDES

Clinical Relevance

Neprilysin inhibitors prevent the enzymatic degradation of the natriuretic peptides. When used alone they do not produce sustained antihypertensive effects, likely a consequence of compensatory upregulation of the renin-angiotensin system. The combination of neprilysin inhibition and angiotensin-converting enzyme inhibition is associated with an increased risk of angioedema. The combination of angiotensin receptor blockade and an inhibitor of neprilysin has a more favorable side effect profile and has demonstrated efficacy in the treatment of heart failure. The effect of combination angiotensin receptor blockade/neprilysin inhibition on hard renal outcomes is currently unknown.

The NPs are a family of vasoactive hormones that play a role in salt and water homeostasis. The family consists of at least five structurally related peptides: ANP, BNP, CNP, *Dendroaspis* natriuretic peptide (DNP), and urodilatin. ANP was originally isolated from human and rat atrial tissues in 1984.²⁷⁶ Since then, the NP family has been found to include several other members, all of which share a common 17-amino acid ring structure that is stabilized by a cysteine bridge and that contains several invariant amino acids.²⁷⁷ Both BNP²⁷⁸ and CNP²⁷⁹ were originally identified in porcine brain tissue, and DNP was first isolated from the venom of the green mamba snake *Dendroaspis angusticeps*.²⁸⁰ Urodilatin is an NH₂-terminally extended form of ANP that was initially described in human urine.²⁸¹ NP inactivation occurs through at least two distinct pathways: binding to a clearance receptor (natriuretic peptide receptor [NPR]-C) and enzymatic degradation. Other peptides that may be involved in salt and water balance include guanylin, uroguanylin, and adrenomedullin.

ANP and BNP act as endogenous antagonists of the RAS, mediating natriuresis, diuresis, vasodilation, and suppression of sympathetic activity, as well as inhibiting cell growth and decreasing secretion of aldosterone and renin.²⁸² The role of NPs in cardiovascular and kidney disease, particularly BNP, has led to their adoption into clinical practice as indicators of disease states and, to some extent, as therapeutic agents.

thus it may represent the evolutionary ancestor of ANP and BNP. BNP, like ANP, is synthesized as a prehormone, between 121 and 134 amino acids in length, according to species of origin. Human preproBNP (134 amino acids) is cleaved to produce the 108–amino acid precursor proBNP. Further processing leads to the production of the 32–amino acid, biologically active BNP (which corresponds to the C terminal of the precursor), as well as a 76–amino acid N-terminal fragment (NT-proBNP).²⁸⁴ Active BNP, NT-proBNP, and proBNP all circulate in the plasma. Circulating BNP contains the characteristic 17–amino acid ring structure closed by a disulfide bond between two cysteine residues, along with a nine–amino acid N-terminal tail and a six–amino acid C-terminal tail (see Fig. 11.7).²⁸⁵

The term *brain natriuretic peptide* is somewhat misleading, given that the primary sites of synthesis of BNP are the cardiac ventricles, and expression also occurs, to a lesser extent, in atrial cardiomyocytes. Like ANP, expression of BNP is regulated by changes in intracardiac pressure and stretch. However, unlike ANP, which is stored and released from secretory granules, BNP is regulated at the gene expression level and is synthesized and secreted in bursts. BNP expression is increased in heart failure, hypertension, and kidney failure. Its plasma half-life is approximately 22 minutes; by contrast, the half-life of circulating ANP is 3 to 5 minutes, and the half-life of the biologically inactive NT-proBNP is 120 minutes. This difference is relevant to the utility of NP measurement as a biologic marker of cardiorenal disease. Changes in pulmonary capillary wedge pressure may be reflected by plasma BNP concentrations every 2 hours and by NT-proBNP levels every 12 hours.^{286,287} The physiologic actions of BNP are similar to those of ANP including effects on the kidneys (natriuresis and diuresis), vasculature (hypotension), endocrine system (inhibition of plasma renin and aldosterone secretion), and the brain (central vasodepressor activity).

C-TYPE NATRIURETIC PEPTIDE

As is the case for ANP and BNP, CNP is derived from a prepropeptide that undergoes posttranslational proteolytic cleavage. The initial translation product preproCNP is 126 amino acids in length and is cleaved to produce the 103–amino acid prohormone. Cleavage of proCNP yields two mature peptides made up of 22 and 53 amino acids: CNP and NH₂-terminally extended forms of CNP, respectively. Of the 17 amino acids within the CNP ring structure, 11 are identical to those in the other NPs, although, uniquely, CNP lacks an amino tail at the carboxy terminus (see Fig. 11.7). Whereas ANP and BNP are ligands for a guanylyl cyclase-coupled receptor, the NPR-A receptor, CNP is a specific ligand for the NPR-B receptor. CNP primarily functions in an autocrine/paracrine manner with effects on vascular tone and muscle cell growth.²⁸⁸ Expression of the CNP gene by the endothelial cells, the presence of CNP receptors on vascular smooth muscle cells, and the antiproliferative effect of CNP on vascular smooth muscle cells suggest that CNP is produced by the endothelium and acts on adjacent cells, serving as an autocrine/paracrine endothelium-derived locally active vasoregulatory system. Accordingly, plasma concentrations of CNP are low, although they are increased in the conditions of heart failure and kidney failure. CNP is present in the heart,

kidneys, and endothelium, and its receptor is also expressed in abundance in the hypothalamus and pituitary gland, which suggests that the peptide may also play a role as a neuromodulator or neurotransmitter. Regulation of CNP expression is distinct from that of ANP and BNP and is controlled by a number of vasoactive mediators including insulin, vascular endothelial growth factor, TGF- β , TNF- α , and IL-1 β .²⁸⁸

The principal enzymes responsible for the conversion of proANP, proBNP, and proCNP to their active forms are the serine proteases, corin and furin.²⁸⁹ Corin converts proANP to ANP,²⁹⁰ furin converts proCNP to CNP,²⁹¹ and both corin and furin cleave proBNP.²⁹² Corin is highly expressed in the heart and to a lesser extent in the kidney²⁸⁹ and it is the rate-limiting enzyme in ANP activation.²⁹³ In response to pressure overload, corin-deficient mice develop hypertension together with cardiac hypertrophy and dysfunction.^{294,295} In the kidneys, corin colocalizes with ANP²⁹⁶ and decreased urinary corin excretion has been observed in patients with CKD.²⁹⁷ Interestingly, studies combining observations made in organ-specific corin-deficient mice together with human correlative experiments have identified a role for impaired uterine corin/ANP function in the pathogenesis of preeclampsia.²⁹⁸

DENDROASPIS NATRIURETIC PEPTIDE

The physiologic role of DNP has been controversial since its original identification in the venom of the Green Mamba snake, *D. angusticeps*, in 1992.^{280,299} DNP is a 38–amino acid peptide that shares the 17–amino acid ring structure common to all NPs, except that it has unique N- and C-terminal regions (see Fig. 11.7).²⁸⁷ Immunoreactivity for DNP has been reported in human plasma and atrial myocardium, and DNP has also been described in rat³⁰⁰ and rabbit³⁰¹ kidneys, rat colon,³⁰² rat aortic vascular smooth muscle cells,³⁰³ and pig ovarian granulosa cells.³⁰⁴ DNP binds to NPR-A³⁰⁵ and the clearance receptor NPR-C,³⁰⁶ which may be of particular relevance in view of the peptide's apparent resistance to enzymatic degradation.³⁰⁷ In dogs, either under normal conditions or in a pacing-induced heart failure model, administration of synthetic DNP decreased cardiac filling pressures; increased GFR, natriuresis, and diuresis; and lowered blood pressure, suppressing renin release and increasing plasma and urine cGMP levels.^{308,309} Despite these propitious findings, several aspects of the biologic role of DNP remain contentious. In particular, the gene for the peptide has not been identified in mammals and the fractionation of DNP from human samples has not been reported.²⁹⁹ These uncertainties have led some authors to question whether DNP is, in fact, expressed at all in humans.²⁹⁹

URODILATIN

Urodilatin is a structural homolog of ANP that shares the same 17–amino acid ring structure and COOH-terminal tail. It is synthesized in renal distal tubule cells and differentially processed to a 32–amino acid NH₂-terminally extended form of ANP.³¹⁰ Urodilatin is not found in plasma; instead, it acts in a paracrine manner within the kidneys on receptors in the glomeruli and IMCDs to promote natriuresis and diuresis. Urodilatin is upregulated in diabetic animals³¹¹ and in the remnant kidney³¹² and is relatively resistant to

enzymatic degradation, which may explain its more potent renal effects.

NATRIURETIC PEPTIDE RECEPTORS

NPs mediate their biologic effects by binding to three distinct guanylyl cyclase NPRs. The terminology can be somewhat confusing: NPR-A binds ANP and BNP, and NPR-B binds CNP, whereas NPR-C acts as a clearance receptor for all three peptides.

NPR-A and NPR-B are structurally similar but share only 44% homology in the extracellular ligand-binding segment; this difference is probably responsible for the differences in ligand specificity. Both NPR-A and NPR-B have a molecular weight of approximately 120 kDa and consist of a ligand-binding extracellular domain, a single transmembrane segment, an intracellular kinase domain, and an enzymatically active guanylyl cyclase domain.²⁷⁷ The kinase homology domain of NPR-A and NPR-B shares 30% homology with protein kinases but has no kinase activity. Ligand binding of NPR-A and NPR-B prevents the normal inhibitory action exerted by the kinase homology domain on the guanylyl cyclase domain, allowing the generation of cGMP, which acts as a second messenger responsible for most of the biologic effects of the NPs. NPR-C, in contrast to NPR-A and NPR-B, lacks both the kinase homology domain and the catalytic guanylyl cyclase domain and therefore does not signal through a second-messenger system. Instead, the receptor contains the extracellular ligand-binding segment, a transmembrane domain, and a 37-amino acid cytoplasmic domain containing a G protein-activating sequence.³¹³ In NPR-C-knockout mice, blood pressure is reduced and the plasma half-life of ANP is increased; this finding supports the role of NPR-C as a clearance receptor.³¹⁴

NPR-C binds all members of the NP family with high affinity. It is the most abundantly expressed of the NPRs—present in the kidneys, vascular endothelium, smooth muscle cells, and heart—and represents approximately 95% of the total receptor population. Preferential binding of NPR-C to ANP over BNP may explain the relatively increased plasma half-life of BNP.²⁸⁷ NPR-C clears NPs from circulation through a process of receptor-mediated endocytosis and lysosomal degradation before rapid

recycling of the internalized receptor to the cell surface. Although the primary function of NPR-C is as a clearance receptor, ligand binding may exert biologic effects on the cell through G protein-mediated inhibition of cAMP.³¹⁵ The biologic effects of NPs are largely dependent on the distribution of their receptors. NPR-A mRNA is present mainly in the kidneys, especially in the IMCD cells, although the receptor is also notably present within the glomeruli, renal vasculature, and proximal tubules. The distribution of NPR-B overlaps with that of NPR-A; the receptor is found in the kidneys, vasculature, and brain. However, in accordance with the paracrine effects of CNP on vascular tone, mitogenesis, and cell migration, NPR-B is expressed in greater abundance than NPR-A within the vascular endothelium and smooth muscle, whereas expression levels are relatively lower within the kidneys.

NEPRILYSIN

Receptor-mediated endocytosis probably accounts for about 50% of clearance of the NPs from the circulation; catalytic degradation by the enzyme neprilysin (NEP 24.11) is responsible for the majority of the rest, and direct renal excretion accounts for only a minor contribution.²⁸⁷ Receptor clearance probably plays an even smaller role in conditions associated with chronically elevated NP levels because of increased receptor occupancy and downregulation of NPR-C expression.

Neprilysin is a membrane-bound zinc metalloproteinase, originally termed *enkephalinase* because of its ability to degrade opioid receptors in the brain. The enzyme has structural and catalytic similarity to other metalloproteinases including aminopeptidase; ACE; ECE; and carboxypeptidases A, B, and E and, in addition to the NPs, numerous other substrates have been described for neprilysin (Table 11.2). The primary mechanism of action of neprilysin is to hydrolyze peptide bonds on the NH₂ side of hydrophobic amino acid residues. In the case of ANP, neprilysin cleaves the Cys¹⁰⁸-Phe¹⁰⁹ bond to disrupt the ring structure and inactivate the peptide. The Cys-Phe bond of BNP is relatively insensitive to enzymatic cleavage. Neprilysin has a nearly ubiquitous tissue distribution; expression has been demonstrated in the

Table 11.2 Peptides That Have Been Described as Substrates for Neprilysin

Atrial natriuretic peptide	Cholecystokinin	Interleukin-1 β
Brain natriuretic peptide	Corticotropin-releasing hormone	β -Lipotropin
C-type natriuretic peptide	Dynorphins	Luliberin
Endothelin-1	Endorphins	Luteinizing hormone-releasing hormone
Bradykinin and kallidin	Endothelin-2	α -Melanocyte-stimulating hormone
Substance P	Endothelin-3	Neurokinin A
Angiotensin I	Enkephalins	Neuropeptide Y
Angiotensin II	N-Formylmethionine-leucyl-phenylalanine	Neurotensin
Angiotensin 1-7	Fibroblast growth factor-2	Oxytocin
Adrenocorticotrophic hormone	Gastric-inhibitory peptide	Peptide YY
Adrenomedullin	Gastrin-releasing peptide	Secretin
Amyloid- β peptide	Glucagon	Somatostatin
Big endothelin-1	Gonadotropin-releasing hormone	Thymopentin
Bombesin-like peptides	Incretins	Vasoactive intestinal peptide
Calcitonin gene-related peptide	Insulin B chain	Vasopressin

kidneys, liver, heart, brain, lungs, gut, and adrenal glands. The metallopeptidase is present not only on the surface of endothelial cells but also on smooth muscle cells, fibroblasts, and cardiac myocytes;³¹⁶ it is most abundant in the brush border of the proximal tubules of the kidneys, where it rapidly degrades filtered ANP, preventing the peptide from reaching more distal luminal receptors.

ACTIONS OF THE NATRIURETIC PEPTIDES

RENAL EFFECTS OF THE NATRIURETIC PEPTIDES

The natriuretic and diuretic actions of the NPs are consequences of both vasomotor effects and direct effects on the renal tubule. Both ANP and BNP cause an increase in glomerular capillary hydrostatic pressure and a rise in GFR by inducing afferent arteriolar vasodilation and efferent arteriolar vasoconstriction. These contrasting effects of the NPs on the afferent and efferent arterioles differ from the actions of classical vasodilators such as bradykinin. In addition to direct effects on vascular tone, ANP can increase GFR through cGMP-mediated mesangial cell relaxation and consequent changes in the ultrafiltration coefficient. Plasma levels of ANP that do not increase GFR can induce natriuresis, indicating the potential for direct tubule effects, which may involve either locally produced NPs acting in a paracrine manner, such as urodilatin, or circulating NPs. A number of mechanisms may be responsible for the natriuresis including direct effects on sodium transport in tubule epithelial cells and indirect effects through inhibition of renin secretion after increased sodium delivery to the macula densa.

NPs also antagonize vasopressin in the cortical collecting ducts. Similar mechanisms probably underlie the response to ANP, BNP, and urodilatin. By contrast, CNP has little natriuretic or diuretic effect, which may indicate a requirement for the presence of the C-terminal extension of the peptide for renal effects. The NPs may have antifibrotic effects within the kidneys, as evidenced by an increase in renal fibrosis in NPR-A–knockout mice after unilateral ureteric obstruction.³¹⁷ In cultured proximal tubule cells, ANP attenuates high glucose–induced activation of TGF- β_1 , Smad, and collagen synthesis, which illustrates the potentially antifibrotic properties of the peptide in the context of diabetic nephropathy.³¹⁸

CARDIOVASCULAR EFFECTS OF THE NATRIURETIC PEPTIDES

All NPs have vasodilatory and hypotensive properties. Heterozygous mutant mice with a disrupted proANP gene display evidence of salt-sensitive hypertension,³¹⁹ whereas hypotension is a feature of transgenic mice overexpressing ANP.³²⁰ In a human patient population, a variant in the ANP promoter was associated with both lower levels of plasma ANP and increased susceptibility to early development of hypertension.³²¹ However, infusion of high concentrations of ANP can actually induce a rise in blood pressure, which suggests that counterregulatory baroreceptors may be activated.³²²

ANP lowers blood pressure through two major direct mechanisms. First, it increases vascular permeability with a shift of fluid from the intravascular to extravascular compartments by capillary hydraulic pressure. Second,

ANP increases venous capacitance and lowers preload.³²³ In addition, ANP and BNP antagonize the vasoconstrictive effects of the RAS, ET, and the sympathetic nervous system²⁸² by decreasing sympathetic peripheral vascular tone, thereby suppressing the release of catecholamines and reducing central sympathetic outflow.²⁸⁴ By lowering the activation threshold of vagal afferents, ANP prevents vasoconstriction and tachycardia that normally follow a reduction in preload and thereby produces a sustained drop in blood pressure. CNP is a more potent vasodilator than either ANP or BNP. In fact, CNP relaxes human subcutaneous resistance arteries, whereas ANP and BNP have no effect.³²⁴

NPs have a number of other effects on the cardiovascular system distinct from their action on vasomotor tone. For example, NPs play a major role in cardiac remodeling. Mice with genetic deficiencies of ANP exhibit an increase in cardiac mass,³¹⁹ whereas heart size is diminished in mice transgenically overexpressing ANP.³²⁰ The antimitogenic and antitrophic effects of NPs, which appear to be mediated by cGMP, have also been demonstrated in a range of cultured cell types, including cultured vascular cells, fibroblasts, and myocytes, and in vivo in response to balloon angioplasty. Further evidence for the role of ANP in mediating cardiac hypertrophy was obtained from population studies, in which variants in either the *NPPA* promoter (associated with reduced circulating ANP) or the NPR-A gene, *NPR1*, have been associated with left ventricular hypertrophy.^{325,326} BNP has been shown to have antifibrotic properties within the heart. In vitro, BNP antagonizes TGF- β –induced fibrosis in cardiac fibroblasts,³²⁷ and in vivo, targeted genetic disruption of BNP in mice is associated with an increase in cardiac fibrosis, in the absence of either hypertension or ventricular hypertrophy.³²⁸

Cardiac CNP is increased in heart failure, where it may play a role in ventricular remodeling.³²⁹ Comparison of plasma CNP levels in samples taken from the aorta and renal vein, at the time of diagnostic heart catheterization, has demonstrated that CNP is indeed synthesized and secreted by the kidney.³³⁰ Moreover, this effect was found to be blunted in patients with heart failure, potentially contributing to renal sodium retention.³³⁰ In rats subjected to unilateral ureteric obstruction, recombinant CNP decreased blood urea nitrogen and creatinine levels and attenuated renal fibrosis.³³¹

OTHER EFFECTS OF THE NATRIURETIC PEPTIDES

Even though they do not cross the blood-brain barrier, NPs exert important CNS effects that may augment their peripheral actions. ANP, BNP, and particularly CNP are all expressed within the brain. Circulating NPs may also exert central effects through actions at sites that are outside the blood-brain barrier. The NPR-B receptor is expressed throughout the CNS, which reflects the wide distribution of CNP, whereas the NPR-A receptor is expressed in areas adjacent to the third ventricle, which is indicative of a role of peripherally circulating ANP and BNP, as well as centrally expressed peptides. Complementing their natriuretic and diuretic effects, NPs inhibit both salt appetite and water drinking. ANP also prevents release of vasopressin and possibly adrenocorticotrophic hormone from the pituitary gland, whereas sympathetic tone is increased by the actions of the NPs on the brainstem.

Clinical and experimental evidence suggests that NPs play a role in mediating metabolism. Circulating levels of NPs are decreased in obese individuals³³² and among patients with the metabolic syndrome,^{333,334} correlating inversely with both plasma glucose and fasting insulin levels.³³⁵ In accordance with these epidemiologic observations, infusion of ANP activates hormone-sensitive lipase from fat cells, which is indicative of lipolysis.³³⁶ In vitro, ANP inhibits preadipocyte proliferation,³³⁷ the lipolytic properties of the peptide being mediated by cGMP phosphorylation.^{338,339}

Knockout mouse studies have revealed that CNP plays a predominant role in the regulation of skeletal growth, specifically cartilage homeostasis and endochondral bone formation.³⁴⁰ Mice with genetic deficiencies of either CNP or its receptor NPR-B lack growth of longitudinal bones and vertebrae and have a shortened life span as a consequence of respiratory insufficiency secondary to abnormal ossification of the skull and vertebrae.^{341,342} Transgenic mice that overexpress CNP are relatively protected from glucocorticoid-induced growth retardation.³⁴³ Mutations in the NPR-B gene have also been reported in patients with autosomal recessive skeletal dysplasia and acromesomelic dysplasia-type Maroteaux and obligate carriers of the mutations have heights that are below predicted levels.³⁴² Accordingly, CNP analog therapy is being investigated as a possible treatment for achondroplasia.³⁴⁴

NATRIURETIC PEPTIDES AS BIOMARKERS OF DISEASE

Both ANP and BNP have been studied as clinical biomarkers of heart failure and kidney failure. The short half-life of ANP (2–5 minutes) restricts its applicability.³⁴⁵ However, the biologically inactive NH₂-terminal 98-amino acid peptide ANP 1-98 does not bind to NPR-A or NPR-C and so remains in circulation longer than ANP does. In heart failure, ANP 1-98 levels closely reflect the degree of kidney function.³⁴⁶ Plasma concentrations of the midregional epitopes of the stable prohormones of both ANP and adrenomedullin are predictive of the progression of kidney function decline in patients with nondiabetic CKD.³⁴⁷ The prognostic performance of midregional proANP is not superior to that of NT-proBNP or BNP in hemodialysis patients,³⁴⁸ and measurement of ANP or one of its prohormone derivatives is currently not part of routine clinical care. Signal peptides from both ANP and BNP are present in venous blood and rise rapidly following myocardial infarction, suggesting that their detection may aid in the diagnosis of cardiac ischemia.^{349,350} Commercial assays are widely available for measurement of either BNP or the biologically inactive peptide fragment NT-proBNP. Correspondingly, since 2000, measurement of circulating BNP and NT-proBNP levels has been incorporated into several clinical practice guidelines for the management of heart failure. Important differences distinguish BNP and NT-proBNP from each other as clinical biomarkers. NT-proBNP is not removed from the circulation by binding to the clearance receptor NPR-C, and hence its circulating half-life of approximately 2 hours is significantly longer than that of BNP (approximately 20 minutes). In addition, both BNP and NT-proBNP are affected by kidney

impairment,³⁵¹ but the magnitude of the effect is greater for NT-proBNP.³⁵²

BRAIN NATRIURETIC PEPTIDE AND N-TERMINAL PROBRAIN NATRIURETIC PEPTIDE AS BIOMARKERS OF HEART FAILURE

Measurement of circulating levels of either BNP or NT-proBNP has effectively helped guide clinical practice in several aspects of the management of heart failure including diagnosis, screening, prognosis, and monitoring of therapy.²⁷⁷ The primary role of BNP measurement in the assessment of dyspnea is as a “ruling out” test: A plasma BNP level lower than 100 pg/mL has a negative predictive value for heart failure of 90%.³⁵³ In the ProBNP Investigation of Dyspnea in the Emergency Department (PRIDE) study, an NT-proBNP level lower than 300 pg/mL was optimal in ruling out heart failure, with a negative predictive value of 99%.³⁵⁴ Screening for BNP³⁵⁵ and NT-BNP³⁵⁶ levels has also proven useful in identifying individuals at risk for heart failure for whom aggressive medical therapy should be targeted. The utility of serial BNP or NT-BNP measurements in guiding the treatment of patients with heart failure was the subject of a 2016 Cochrane Review.³⁵⁷ This review concluded that low-quality evidence shows that NP-guided treatment is associated with a reduction in hospital admissions for heart failure and that low-quality evidence shows uncertainty with respect to the effect of NP-guided treatment on mortality or all-cause hospital admissions.³⁵⁷ In the interpretation of plasma levels of BNP and NT-proBNP, a number of other biologic variables should be taken into account. NP levels rise with age and are higher in women, the latter effect possibly secondary to estrogen regulation, inasmuch as hormone replacement therapy increases BNP levels.³⁵⁸ Conversely, NP levels fall with increasing obesity. Although BNP levels of heart failure patients are higher in Asian or African American patients than in Caucasian or Hispanic patients, they provide prognostic value regardless of race or ethnicity.³⁵⁹

ROLE OF BRAIN NATRIURETIC PEPTIDE AND N-TERMINAL PROBRAIN NATRIURETIC PEPTIDE AS BIOMARKERS IN KIDNEY DISEASE

The interpretation of NP concentrations in patients with kidney disease merits special consideration. NP levels are increased in individuals with impaired kidney function. This increase is probably multifactorial in origin and not solely the consequence of increased intravascular volume. Other factors that contribute to increased NP levels include decreased NP responsiveness, subclinical ventricular dysfunction, hypertension, left ventricular hypertrophy, subclinical ischemia, myocardial fibrosis, and RAS activation,³⁶⁰ as well as decreased filtration and reduced clearance by NPR-C and NEP.³⁶¹ Although, on the basis of observational studies, it has been widely considered that renal clearance plays a greater role in the removal of NT-proBNP from the circulation than removal of BNP, one study has challenged this view. By measuring both NT-proBNP and BNP in the renal arteries and veins of 165 subjects undergoing renal arteriography, investigators found that both NT-proBNP and BNP are equally dependent on renal clearance.³⁶² However, the NT-proBNP-to-BNP ratio did increase with declining GFR, which suggests that the two

peptides may be differentially cleared at GFRs lower than 30 mL/min/1.73 m².³⁶²

Even though both BNP and NT-proBNP are affected by kidney impairment, their clinical utility for the prediction of heart failure persists in CKD patients in the context of appropriately adjusted reference ranges. For example, in the Breathing Not Properly study, BNP cut point values were approximately threefold higher to diagnose heart failure in patients with an estimated GFR lower than 60 mL/min relative to the conventional cut point value of 100 pg/mL.³⁶³ In a cohort of 831 patients with dyspnea and a GFR <60 mL/min, both BNP and NT-proBNP were effective predictors of heart failure, although NT-proBNP was superior in predicting mortality.³⁶⁴ In asymptomatic patients with CKD, both BNP and NT-proBNP were equivalent and effective in indicating the presence of left ventricular hypertrophy or coronary artery disease.³⁶⁵ In patients with CKD, BNP and NT-proBNP may be predictive of the progression of kidney function decline^{366,367} and cardiovascular disease and mortality. In a nondialysis CKD population, NT-proBNP, but not BNP, was an independent predictor of death;³⁶⁸ in 994 black patients with hypertensive renal disease (GFR = 20 to 65 mL/min/1.73 m²), NT-proBNP was predictive of cardiovascular disease and mortality, particularly among individuals with proteinuria.³⁶⁹ In pediatric CKD patients, both BNP and proBNP (but not troponins I and T) were indicative of left ventricular hypertrophy or dysfunction.³⁷⁰

BNP and NT-proBNP have been studied extensively in dialysis recipients as both prognostic indicators and markers of volume status. The molecular weights of BNP (3.5 kDa) and NT-proBNP (8.35 kDa) are low enough that both peptides may be cleared by high-flux dialysis.^{371,372} Nevertheless, in contrast to ANP, which falls sharply after either hemodialysis or peritoneal dialysis, levels of BNP and NT-proBNP are less affected.^{371,373} The role of NP levels as indicators of volume status in either hemodialysis or peritoneal dialysis recipients is confounded by the common coexistence of left ventricular abnormalities.^{351,363,374-377} Both BNP and NT-proBNP levels are predictive of mortality, heart failure, and coronary artery disease in the population undergoing dialysis.³⁷⁸⁻³⁸³ However, no definite cut point values for diagnosing heart failure in dialysis patients have been defined.³⁶³

CIRCULATING C-TYPE NATRIURETIC PEPTIDE LEVELS AS A BIOMARKER FOR RISK OF MYOCARDIAL INFARCTION

Although CNP usually functions in a paracrine manner, its presence in the plasma may provide utility as a biomarker of cardiovascular risk. In a study of 1841 individuals from the general population, individuals with plasma CNP levels in the highest quartile were at increased risk of myocardial infarction and, unlike BNP levels, plasma CNP levels were unaffected by sex and only weakly associated with age.³⁸⁴

THERAPEUTIC USES OF NATRIURETIC PEPTIDES

Even though NP levels are increased in heart failure, their biologic effects are blunted. Intravenous administration of recombinant NPs increases their circulating levels several-fold, overcoming this resistance. As such, two recombinant

NPs are currently available as therapeutic agents for the treatment of heart failure: recombinant ANP (carperitide), which is available in Japan for the treatment of pulmonary edema, and recombinant BNP (nesiritide), which is licensed in several countries, including the United States, for the treatment of acute decompensated heart failure.

RECOMBINANT ATRIAL NATRIURETIC PEPTIDE

ANP has a short half-life and a high total body clearance. Its intravenous administration causes a reduction in blood pressure, diuresis, and natriuresis in healthy individuals; this response is reduced in the setting of acute heart failure. In a 6-year open-label study of 3777 patients with acute heart failure treated with carperitide, clinical improvement was reported in 82%.³⁸⁵ Whereas early experimental studies were suggestive of a potential benefit of exogenous ANP in AKI, results in patients have generally been disappointing. Nevertheless, the peptide may have a limited role in selected patient populations. For example, low-dose carperitide preserved kidney function in patients undergoing repair of abdominal aortic aneurysm³⁸⁶ and reduced the incidence of contrast-induced nephropathy in patients after coronary angiography.³⁸⁷ However, a meta-analysis suggested that recombinant ANP has no effect on mortality in patients with AKI, although a trend toward a reduction in the need for kidney replacement therapy was shown.³⁸⁸ In a separate meta-analysis of studies conducted in cardiovascular surgery patients, ANP infusion decreased peak serum creatinine concentration, incidence of arrhythmia, and need for kidney replacement therapy, whereas both ANP and BNP decreased the length of intensive care unit and hospital stay.³⁸⁹ Among 367 high-risk individuals undergoing coronary artery bypass grafting (CABG), recombinant ANP decreased the incidence of major adverse cardiovascular and cerebrovascular events and the need for dialysis, immediately and up to 2 years postoperatively, although survival was unaffected.³⁹⁰ Similarly, among CKD patients undergoing CABG, those receiving recombinant ANP experienced a smaller rise in serum creatinine, fewer cardiac events, and lower requirement for dialysis, although mortality did not differ from those that did not receive ANP.³⁹¹ However, when employed in an effort to treat rather than prevent AKI following cardiac surgery, recombinant ANP had no significant effect on kidney function, the need for kidney replacement, length of stay, or medical costs.³⁹²

RECOMBINANT BRAIN NATRIURETIC PEPTIDE

Nesiritide is recombinant human BNP, manufactured from *Escherichia coli* and identical in structure to native human BNP, with a mean terminal half-life of 18 minutes in patients with heart failure.³⁹³ Intravenous administration of nesiritide lowers pulmonary and systemic vascular resistance, decreases right atrial pressure, and increases cardiac output (presumably through effects on ventricular afterload) in a concentration-dependent fashion.³⁹⁴ In the kidneys, nesiritide increases RBF and GFR through both direct vasodilatory effects and indirect effects on cardiac output and norepinephrine inhibition.³⁹⁵ Diuresis and natriuresis may also occur, although these effects are modest and may not be seen at the approved doses. Additional effects of nesiritide may also include inhibition of renin secretion in

the kidneys and aldosterone production in the heart and adrenal glands.

In response to meta-analysis data suggesting that nesiritide treatment may be associated with a worsening of renal function and an increase in the rate of early death,^{396,397} the Acute Study of Clinical Effectiveness of Nesiritide in Decompensated Heart Failure (ASCEND-HF) trial was initiated.³⁹⁸ In this study of 7141 patients hospitalized with acute heart failure, nesiritide neither increased nor decreased the rate of death or rehospitalization, rates of worsening kidney function were unaffected, and there was a small but nonsignificant improvement in self-reported rates of dyspnea.³⁹⁸ On the basis of these results, the investigators concluded that nesiritide cannot be recommended for routine use in the broad population of patients with acute heart failure.³⁹⁸

THERAPEUTIC USES OF OTHER NATRIURETIC PEPTIDES

The effects of urodilatin (ularitide) have been assessed in both heart failure and AKI. However, the diuretic effect of urodilatin appears to be attenuated in heart failure patients, which reflects a blunted response, as observed for ANP and BNP.^{399,400} Similarly, as with ANP and BNP, hypotension appears to be a dose-limiting side effect of ularitide therapy.^{400,401} In the Safety and Efficacy of an Intravenous Placebo-Controlled Randomized Infusion of Ularitide in a Prospective Double-blind Study in Patients with Symptomatic, Decompensated Chronic Heart Failure (SIRIUS II) study, a phase II trial of 221 patients hospitalized for decompensated heart failure, a single 24-hour infusion of ularitide preserved short-term renal function.⁴⁰² The NP vessel dilator may offer theoretical advantages for the treatment of acute decompensated heart failure in comparison with current NP-based therapies.^{403,404} In particular, vessel dilator may produce a greater and more sustained natriuresis than does ANP or BNP, without a blunted response in patients with heart failure, and may also improve renal function in the setting of experimental acute renal injury.⁴⁰⁵ An alternative therapeutic approach is the development of novel chimeric peptides. For example, researchers have synthesized a peptide (cenderitide) that represents fusion of the 22-amino acid peptide CNP together with the 15-amino acid linear C terminus of DNP.⁴⁰⁶ In vitro, this peptide activates cGMP and attenuates cardiac fibroblast proliferation. In vivo, cenderitide is both natriuretic and diuretic and increases GFR with less hypotension than does BNP.^{406,407} Cenderitide is more resistant to degradation by neprilysin than the naturally occurring NPs and is eightfold more potent in inducing glomerular cGMP production than CNP.⁴⁰⁸

COMBINATION ANGIOTENSIN RECEPTOR BLOCKADE AND NEPRILYSIN INHIBITION

Notwithstanding concerns regarding the efficacy and cost-effectiveness of recombinant NP therapy, a major limitation is the requirement for systemic administration, which is unsuitable for chronic treatment. Alternative methods to increase the biologic activity of NPs may offer a more feasible approach for chronic therapy. In particular, inhibition of the enzymatic degradation of NPs by neprilysin has been the focus of drug discovery efforts for a number of years. Neprilysin is a zinc metallopeptidase

with catalytic similarity to ACE and with a wide tissue distribution, although abundant at the proximal tubule brush border. Several pharmacologic neprilysin inhibitors have been investigated (e.g., candoxatril, thiorphan, and phosphoramidon). Although these agents, in general, lead to an increase in plasma levels of the NPs and, under some experimental conditions, induce natriuresis and diuresis with peripheral vasodilation, results of clinical trials in hypertension and heart failure have generally been disappointing. Specifically, sustained antihypertensive effects have not been demonstrated, and some researchers have reported a paradoxical rise in blood pressure. This may be a consequence of the induction of both neprilysin and ACE expression with neprilysin inhibition⁴⁰⁹ and a consequent increase in angiotensin II levels.⁴¹⁰ The biologic actions of the NPs are, however, restored in the presence of an inhibited RAS and this has led to the development of two classes of agents: 1. vasopeptidase inhibitors that inhibit both neprilysin and ACE and 2. combined angiotensin receptor blockade/neprilysin inhibition, the latter having gained regulatory authority approval for the treatment of heart failure.

The rational design of vasopeptidase inhibitors—such as mixanpril (S21402), CGS30440, aladotril, MDL 100173, sampatrilat, and omapatrilat—was made possible because of the similar structural characteristics of the catalytic sites of both neprilysin and ACE.³¹⁶ Despite the theoretical advantages of vasopeptidase inhibitors, phase III clinical studies have not been able to demonstrate superiority of vasopeptidase inhibition over ACE inhibition, and an increase in the incidence of angioedema has raised safety concerns. For instance, in the Omapatrilat Cardiovascular Treatment vs. Enalapril (OCTAVE) trial of 25,302 hypertensive patients, angioedema occurred in 2.17% of omapatrilat-treated patients, in comparison with 0.68% of patients treated with the ACE inhibitor enalapril.⁴¹¹ In the Omapatrilat Versus Enalapril Randomized Trial of Utility in Reducing Events (OVERTURE) study, the incidence of angioedema was, again, increased among subjects receiving omapatrilat in comparison with those receiving enalapril (0.8% vs. 0.5%).⁴¹² The increased angioedema with vasopeptidase inhibition is likely a consequence of decreased degradation of bradykinin and substance P with combined inhibition of the two metallopeptidases.⁴¹³

The rationale that concurrent RAS blockade may potentiate the therapeutic effects of neprilysin inhibition, yet concurrent ACE inhibition increases the risk of angioedema, encouraged the development of a new class of drug that combines an ARB and neprilysin inhibitor. This class has been termed angiotensin receptor–neprilysin inhibitor (ARNi). In July 2015, the first in the class, valsartan/sacubitril gained approval from the U.S. Food and Drug Administration for the treatment of heart failure with reduced ejection fraction. Valsartan/sacubitril (LCZ696) is a single molecule composed of molecular moieties of the ARB, valsartan, and the neprilysin inhibitor prodrug sacubitril (formerly AHU-377) in a 1:1 ratio.⁴¹⁴ After ingestion, valsartan/sacubitril dissociates into valsartan and sacubitril and sacubitril is subsequently converted to its active form sacubitrilat (LBQ657) by esterases.⁴¹³ In a study of 1328 patients, valsartan/sacubitril conferred greater blood pressure lowering than valsartan alone with no cases

of angioedema reported.⁴¹⁵ In the Prospective comparison of ARNi with ARB on Management Of heart failure with preserved ejection fraction (PARAMOUNT) study of 301 individuals with heart failure with preserved ejection fraction, valsartan/sacubitril lowered NT-proBNP levels to a greater extent than valsartan after 12 weeks of treatment and was well tolerated.⁴¹⁶ Although the reduction in NT-proBNP was sustained at 36 weeks, the difference in NT-proBNP levels between participants randomized to valsartan/sacubitril and valsartan was no longer significant.⁴¹⁶ However, left atrial remodeling and heart failure symptoms were improved.⁴¹⁶

The case for regulatory authority approval for valsartan/sacubitril was based on the findings of the phase III prospective comparison of ARNi with ACEi (Determine Impact on Global Mortality and Morbidity in Heart Failure [PARADIGM-HF] trial).⁴¹⁷ PARADIGM-HF compared the effects of valsartan/sacubitril (200 mg twice daily) and enalapril (10 mg twice daily) in 8442 patients with heart failure (New York Heart Association class II–IV) and a reduced ejection fraction ($\leq 40\%$).⁴¹⁷ The primary outcome, a composite of cardiovascular death and hospitalization for heart failure, occurred in 21.8% of participants treated with valsartan/sacubitril and 26.5% of participants treated with enalapril (hazard ratio 0.80, confidence interval 0.73–0.87, $P < .001$). Hypotension was more common in participants receiving valsartan/sacubitril, whereas cough, hyperkalemia, and kidney impairment were more common in those receiving enalapril.⁴¹⁷ Importantly, there was no significant difference in the number of cases of angioedema between participants receiving valsartan/sacubitril than those receiving enalapril, although the number of cases of angioedema was numerically higher in the valsartan/sacubitril group (19 vs. 10, $P = 0.13$). The effect of valsartan/sacubitril in patients with heart failure and preserved ejection fraction was evaluated in the Prospective Comparison of ARNI with ARB Global Outcomes in Heart Failure with preserved Ejection Fraction (PARAGON-HF) trial (NCT01920711). There was no statistically significant reduction in the primary endpoint of hospital admission with heart failure or cardiovascular death (rate ratio 0.87; 95% CI, 0.75–1.01; $P = 0.06$).⁴¹⁸ Other molecules combining ARB and NEP inhibition are also currently under development.^{419,420}

Despite the promising findings of PARADIGM-HF, there are some difficulties with the design of the study and theoretical considerations around the use of neprilysin inhibitors in a broad population. In terms of the active comparator in PARADIGM-HF, it is noteworthy that valsartan/sacubitril was not compared with valsartan alone and that the dose of enalapril (10 mg twice daily) may have been insufficient. For instance, in the Cooperative North Scandinavian Enalapril Survival Study (CONSENSUS) of patients with heart failure, the target dose of enalapril was up to 20 mg twice daily.⁴²¹ Separately, whether the apparently low risk of angioedema observed in PARADIGM-HF translates to the real-world setting remains to be determined. Only 5% of participants in PARADIGM-HF were Black, a group at increased risk of angioedema associated with ACE inhibition or neprilysin inhibition.⁴¹³ Furthermore, 78% of participants in PARADIGM-HF had been previously treated with an ACE inhibitor and inclusion of a run-in period where all participants were exposed to

enalapril may have resulted in an underrepresentation in the number of cases of angioedema.⁴¹⁷ Other theoretical risk concerns surrounding chronic neprilysin inhibition largely relate to the other substrates that are normally degraded by neprilysin (see Table 11.2). On the basis of the breadth of activity of these substrates, the possibility has been raised that long-term neprilysin inhibition could have deleterious effects on bronchial reactivity, pain, inflammation, tumorigenesis, and neuronal function.⁴¹⁰ Of particular note has been the recognition that neprilysin is important in the metabolism of amyloid- β peptides and that its inhibition could predispose to the development of Alzheimer disease, age-related macular degeneration, and cerebral amyloid angiopathy, which may take many years to manifest.⁴¹⁰ Reassuringly, the PERSPECTIVE trial (NCT02884206) compared the efficacy and safety of valsartan/sacubitril with valsartan on cognitive function in patients with heart failure and preserved ejection fraction and showed that 3-year neprilysin inhibition treatment was not associated with increased A β accumulation, determined by PET, or with cognitive deterioration, which was reassuring.⁴²² Despite these theoretical concerns, valsartan/sacubitril treatment is now recommended in clinical guidelines for the treatment of heart failure with reduced ejection fraction.⁴²³

RENAL EFFECTS OF VALSARTAN/SACUBITRIL

The renal effects of valsartan/sacubitril were assessed in a post hoc analysis of participants in the PARAMOUNT trial of individuals with heart failure and preserved ejection fraction.⁴²⁴ Participants received treatment with valsartan/sacubitril titrated to 200 mg twice daily or valsartan titrated to 160 mg twice daily, which each gave similar levels of systemic exposure to valsartan.^{414,415} In the PARAMOUNT trial, eGFR declined less with valsartan/sacubitril treatment than it did with valsartan treatment over a 36-week period (-1.5 vs. -5.2 mL/min/1.73 m², $P = .002$), whereas the geometric mean of the urinary albumin-to-creatinine ratio increased from baseline in the valsartan/sacubitril group (2.4–2.9 mg/mmol) and was unchanged in the valsartan group (2.1–2.0 mg/mmol; P value for difference between groups = .016).⁴²⁴ The finding of relative preservation in eGFR with valsartan/sacubitril is consistent with the observation from the PARADIGM-HF trial that valsartan/sacubitril-treated patients experienced less kidney impairment that necessitated cessation of therapy than enalapril-treated patients.⁴¹⁷ Both the effects on eGFR and albuminuria are reminiscent of the effect of systemic ANP administration, raising the possibility that they are a consequence of increased levels of biologically active ANP with neprilysin inhibition.⁴²⁴ Participants in the PARAMOUNT trial were required to have an eGFR of at least 30 mL/min/1.73 m² at enrollment,⁴¹⁶ and thus the effects of combination ARB and neprilysin inhibition in more advanced renal disease are currently unknown, as are the effects on hard renal endpoints in at-risk populations.

The renal effects of valsartan/sacubitril have also been evaluated in persons with CKD in the absence of heart failure. In the U.K.'s Heart and Renal Protection (HARP) III trial, 414 participants with eGFR 60 to 45 mL/min/1.73 m² and UACR >177 mg/g (>20 mg/mmol) or eGFR 44 to 20 mL/min/1.73 m² regardless of albuminuria were randomized to treatment with valsartan/sacubitril (mean baseline measured 34 GFR

mL/min/1.73 m² or irbesartan (mean baseline measured GFR 34.7 mL/min/1.73 m²). After 12 months, there was no difference in the primary outcome of measured GFR or secondary outcome of UACR. There was also no difference in adverse events or incidence of severe hyperkalemia.⁴²⁵

While angiotensin receptor–neprilysin inhibitors (ARNIs) have become increasingly important in managing heart failure (HF) in patients with preserved kidney function, evidence regarding their benefits in HF patients with end-stage kidney disease (ESKD) on dialysis remains limited to two studies: a retrospective study showing improvements in left ventricular ejection fraction and a clinical trial demonstrating enhanced left ventricular echocardiographic parameters after 1 year of therapy.⁴²⁶ As of 2024, no randomized studies have compared ARNI with ACE inhibitors or ARBs in patients with CKD and an eGFR <30 mL/min per 1.73 m², regardless of heart failure status.

OTHER NATRIURETIC PEPTIDES

GUANYLIN AND UROGUANYLIN

The existence of intestinal NPs has been suggested by initial observations that sodium excretion is greater after an oral salt load than after an intravenous salt load.^{427,428} These intestinal peptides include guanylin and uroguanylin. However, a different study of 15 healthy volunteers found that sodium excretion was similar in response to either oral or intravenous sodium load during either a low- or high-sodium-containing diet.⁴²⁹ Moreover, serum concentrations of either prouroguanylin or proguanylin were unchanged following either oral or intravenous sodium load and showed no correlation with sodium excretion.⁴²⁹ Collectively, these observations challenge the notion of a gastrointestinal–renal natriuretic axis mediated by the guanylin peptide family.^{429,430} It thus appears likely that the natriuretic, kaliuretic, and diuretic effects of guanylin and uroguanylin, which occur without change in GFR or RBF, are mediated by local production of the peptides within the kidney.⁴³⁰

ADRENOMEDULLIN

Adrenomedullin is a 52-amino acid peptide originally isolated from human pheochromocytoma cells,⁴³¹ although it is synthesized mainly by vascular smooth muscle cells, endothelial cells, and macrophages⁴³² and is present in the plasma, vasculature, lungs, heart, and adipose tissue. The peptide is upregulated in patients with cardiovascular disease and has positive inotropic and vasodilatory properties. Systemic administration of adrenomedullin induces an NO-dependent natriuresis and an increase in GFR both under normal conditions and in patients with congestive heart failure; it also decreases plasma aldosterone levels without affecting renin activity. Individuals with type 2 diabetes and plasma levels of midregional proadrenomedullin (MR-proADM) peptide in the highest tertile are at an increased risk of severe nephropathy (doubling of plasma creatinine and/or end-stage renal disease), which may reflect a reactive rise in MR-proADM.⁴³³

KALLIKREIN–KININ SYSTEM

The kallikrein–kinin system (KKS) is a complex network of peptide hormones, receptors, and peptidases that is

evolutionarily conserved with homologs in nonmammalian species.⁴³⁴ Discovery of the KKS is attributed to Abelous and Bardier, who reported in 1909 that experimental injection of urine resulted in an acute fall in systemic blood pressure.⁴³⁵ Since that time, investigators have recognized that the physiologic actions of the KKS also include regulation of tissue blood flow, transepithelial water and electrolyte transport, cellular growth, capillary permeability, and inflammatory responses. The main components of the KKS are the enzyme kallikrein, its substrate kininogen, effector hormones known as *kinins* (especially bradykinin and kallidin [also termed *lys-bradykinin*]), and their inactivating enzymes, which include kininases I and II (ACE) and neprilysin.

Kinins exert their biologic effects through binding to two receptors: the bradykinin B1 receptor (B1R) and bradykinin B2 receptor (B2R). The B2R is widely expressed and mediates all the physiologic actions of kinins under normal conditions. The B1R is activated predominantly by des-Arg-bradykinin, a natural degradation product of bradykinin, generated by cleavage of the peptide by kininase I. The KKS may be subdivided into a circulatory (plasma) KKS and a tissue (including renal) KKS, which may be distinguished by their principal effector molecules, bradykinin and kallidin, respectively. In the kidneys, the kinins play a significant role in the modulation of renal hemodynamics and salt and water homeostasis.

COMPONENTS OF THE KALLIKREIN–KININ SYSTEM

KININOGEN

Humans possess a single kininogen gene, *KN1*, which is localized to chromosome 3q26 and encodes both high-molecular-weight (HMW) kininogens (626 amino acids, 88–120 kDa) and low-molecular-weight (LMW) kininogens (409 amino acids, 50–68 kDa) through alternate splicing from 11 exons spread over a 27-kb genomic region. A second kininogen gene has been identified in mice.⁴³⁶ In humans, kininogen deficiency may be relatively asymptomatic;⁴³⁷ the kininogen-deficient Brown Norway Katholiek rat strain, however, shows increased sensitivity to the pressor effects of salt, angiotensin II, and mineralocorticoid.^{438,439}

KALLIKREIN

HMW and LMW kininogen are cleaved by the serine protease kallikrein. The name “kallikrein” is derived from the Greek term *kallikreas*, meaning “pancreas,” after the work of Frey and colleagues,⁴⁴⁰ in the 1930s, who extracted a kinin-producing enzyme from the pancreas of dogs. Since then, 15 tissue kallikreins have been identified, although, in humans, only one (KLK1) is involved in local kinin production. The human kallikrein genes are clustered on chromosome 19 at loci q13.3–13.4. Plasma kallikrein is found in the circulation and is involved largely with the coagulation cascade and activation of neutrophils. The tissue kallikreins are acid glycoproteins that are variably and extensively glycosylated. Human renal kallikrein is synthesized as a zymogen (prekallikrein) with a 17-amino acid signal peptide and a 7-amino acid activation sequence, which must be cleaved to activate the enzyme. In most mammals including humans, tissue kallikrein cleaves

kallidin (lys-bradykinin) from kininogens, whereas plasma kallikrein releases bradykinin.

Although the physiologic effects of kallikrein have been attributed to increased kinin generation, the enzyme may also have direct effects on the B2R, as well as actions independent of the kinin receptors.^{441,442} For example, in kininogen-deficient Brown Norway Katholiek rats, local injection of kallikrein into the myocardium after coronary artery ligation had a cardioprotective effect that was abolished by the NO synthase inhibitor N ω -nitro-L-arginine methyl ester and the selective B2R inhibitor icatibant (Hoe 140).⁴⁴¹ As a serine protease, kallikrein may also elicit kinin receptor-independent effects on endothelial cell migration and survival through cleavage of growth factors and matrix metalloproteinases.⁴⁴³ Transgenic mice overexpressing human kallikrein exhibit a sustained reduction in systemic blood pressure throughout their life span, which is indicative of the lack of sufficient compensatory mechanisms to reverse the hypotensive effect of kallikrein.⁴⁴⁴ In humans, polymorphisms of the kallikrein gene *KLK1* or its promoter can impair enzymatic activity, potentially influencing both kinin-dependent and kinin-independent effects. Among normotensive men with a common loss-of-function *KLK1* polymorphism (R53H), an increase in wall shear stress and a paradoxical reduction in artery diameter and lumen were noted, although flow-mediated and endothelium-independent vasodilation were unaffected.⁴⁴⁵

KININS

The kinins are bradykinin and kallidin in humans and bradykinin and kallidin-like peptide in rodents.⁴⁴⁶ Plasma aminopeptidase can convert kallidin (10 amino acids: Lys-Arg-Pro-Pro-Gly-Phe-Ser-Pro-Phe-Arg) to bradykinin (9 amino acids: Arg-Pro-Pro-Gly-Phe-Ser-Pro-Phe-Arg) by cleavage of the first N-terminal lysine residue. Cleavage of the carboxy-terminal arginine residue by kininase I (carboxypeptidase-N) and carboxypeptidase-M generates their des-Arg derivatives, which are agonists of the B1R.⁴⁴⁶ Removal of two C-terminal amino acids (Phe and Arg) by ACE (kininase II), neprilysin, or ECE is responsible for inactivation of the peptides.⁴⁴⁶

BRADYKININ RECEPTORS

B1R and B2R share 36% homology, and both are GPCRs with seven-transmembrane domains. The genes for the two receptors are in tandem on a compact locus (14q23) separated by only 12 kb.⁴⁴⁷ The B2R is the principal receptor mediating the actions of both kinins, is expressed in abundance by vascular endothelial cells and is present in most tissues including those of the kidneys, heart, skeletal muscle, CNS, vas deferens, trachea, intestines, uterus, and bladder. In general, the distribution and action of B1Rs are similar to those of the B2Rs. The B1R, by contrast, is expressed at low levels under normal conditions but is upregulated in response to inflammatory stimuli (e.g., lipopolysaccharide, endotoxins, cytokines such as IL-1 β and TNF- α)⁴⁴⁸ and in the setting of diabetes⁴⁴⁹ and ischemia-reperfusion injury.⁴⁵⁰ B2R binds both bradykinin and kallidin, whereas bradykinin has almost no effect at the B1R. The carboxypeptidase required to generate the des-Arg B1R-active kinin fragments is closely associated with the B1R on the cell surface.⁴⁵¹ This

association would enable B2R agonists to rapidly activate B1Rs, particularly in response to inflammation.⁴⁵¹

Ligand binding of both receptor subtypes induces activation of phospholipase C, which results in intracellular calcium mobilization through production of inositol 1,4,5-triphosphate and DAG via activation of G proteins including G α_q and G α_i . The physiologic effects of bradykinin receptor activation are mediated through generation of both endothelial NO synthase-derived NO and prostaglandins. B2R activation leads to a rise in intracellular calcium concentrations in vascular endothelial cells.⁴⁴⁶ However, bradykinin-induced vasodilation is not abolished by coadministration of NO synthase and COX inhibitors, which indicates that additional effectors are also likely to be involved, possibly an endothelium-derived hyperpolarizing factor. In addition, through binding to both B1R⁴⁵² and B2R,⁴⁵³ bradykinin also increases the expression of inducible NO synthase (iNOS), at least in rodents. It is difficult to induce the iNOS gene in human tissues, especially the vascular endothelium. Mice that have genetic deficiencies of B2R,⁴⁵⁴ B1R,⁴⁵⁵ or both receptors⁴⁵⁶ have been generated; the reported phenotypes of the different knockout strains have been varied, which may be a result of different genetic backgrounds, or, in the case of the single knockouts, differing compensatory effects of the remaining receptor. For example, some studies of B2R-deficient mice revealed an increase in resting systemic blood pressure, an exaggerated pressor response to angiotensin II⁴⁵⁷ and salt sensitivity,⁴⁵⁸ whereas others revealed no difference in resting blood pressure between B2R- or B1R-deficient mice and wild-type animals.^{455,459} Double B2R-/B1R-knockout mice were also reported to have resting blood pressure identical to that in wild-type mice and were resistant to lipopolysaccharide-induced hypotension.^{456,460} By contrast, transgenic mice expressing the human B2R had a lower resting blood pressure than did wild-type controls.⁴⁶¹ Transgenic mice expressing the rat B1R (as well as their native murine B2R) were normotensive but showed an exaggerated hypotensive response to lipopolysaccharide and, unexpectedly, a hypertensive response to des-Arg bradykinin.⁴⁶²

KALLISTATIN

Kallistatin is an endogenous serpin inhibitor of kallikrein that acts by forming a heat-stable complex with the enzyme. Surprisingly, administration of human kallistatin to rodents induced vasodilation and a decline in systemic blood pressure, which was unaltered by either an NO synthase inhibitor or the B2R antagonist icatibant; this suggests that the vasodilatory properties of kallistatin may be mediated through a smooth muscle mechanism independent of bradykinin receptor activation.⁴⁶³

KININASES

With the exception of the metabolites des-Arg-bradykinin and des-Arg-kallidin, kinin-cleavage products are biologically inactive. Kinins are cleaved by a number of enzymes including carboxypeptidases, ACE, and neprilysin. ACE also truncates its own reaction product, bradykinin-(1–7), further to form bradykinin-(1–5). Neprilysin, like ACE, cleaves bradykinin at the 7 to 8 position and has a broad substrate specificity (see Table 11.2). The amino terminal of bradykinin possesses two proline residues and is susceptible to cleavage by the

proline-specific exopeptidase aminopeptidase P. The resultant peptide, bradykinin-(2–9), may be further cleaved by proteases that include the endothelial enzyme dipeptidyl peptidase-4, which reduces this metabolite to bradykinin-(4–9).

PLASMA AND TISSUE KALLIKREIN–KININ SYSTEM

The two independent KKSs in humans (plasma and tissue) can be distinguished by the specific subtypes of kallikreins, kininogens, and kinins involved. The circulating plasma KKS includes HMW kininogen and plasma prekallikrein, both of which are synthesized in the liver and secreted in the plasma, in which kallikrein is generated by the cell matrix-associated prekallikrein activator prolylcarboxypeptidase.⁴⁶⁴ Of importance is that bradykinin is the main effector molecule of the plasma KKS. The tissue-specific KKS consists of locally synthesized or liver-derived kininogen (HMW and LMW), tissue kallikrein, and the effector molecules kallidin in humans and kallidin-like peptide in rodents. The half-life of kinins is 10 to 30 seconds, but in tissues with high kallikrein content including the kidneys, local and plasma-derived LMW kininogen can be continuously cleaved to produce kallidin. Fig. 11.8 illustrates the enzymatic cascades of the plasma and tissue KKSs.

RENAL KALLIKREIN–KININ SYSTEM

The tissue KKS contributes to the physiologic functions of the kidneys with effects on RVR, natriuresis, diuresis, and

other vasoactive mediators, such as renin and angiotensin, eicosanoids, catecholamines, NO, vasopressin, and ET. In the kidneys, large quantities of kininogen and kallikrein are synthesized by the tubule epithelium and excreted in the urine. Locally formed kinin is also detectable in the urine, renal interstitial fluid, and renal venous blood. In human kidneys, kallikrein is localized to the connecting tubules with close anatomic association between the kallikrein-expressing tubules and afferent arterioles of the JGA. Results of some studies suggest that renal kallikrein mRNA is also detectable by in situ hybridization at the glomerular vascular pole. This anatomic association highlights the physiologic relationship between the KKS and the RAS and is consistent with a paracrine function for the KKS in the regulation of RBF, GFR, and renin release. In this regard, it has been suggested that, through effects on prostaglandin production, kinins may lower tubuloglomerular feedback sensitivity.

Expression of kallikrein within the kidneys is altered during development and is regulated by estrogen and progesterone, salt intake, thyroid hormone, and glucocorticoid.^{465–468} The enzyme is not normally filtered at the glomerulus in the absence of glomerular injury. Kininogens are localized mostly to connecting tubule principal cells near kallikrein, which can be found in the connecting tubules of the same nephron. Once activated, renal kallikrein cleaves both HMW and LMW kininogens to release kallidin. The majority of the physiologic effects of kinins are mediated through activation of constitutively expressed B2Rs, with little or no B1R mRNA detectable in normal kidneys. In

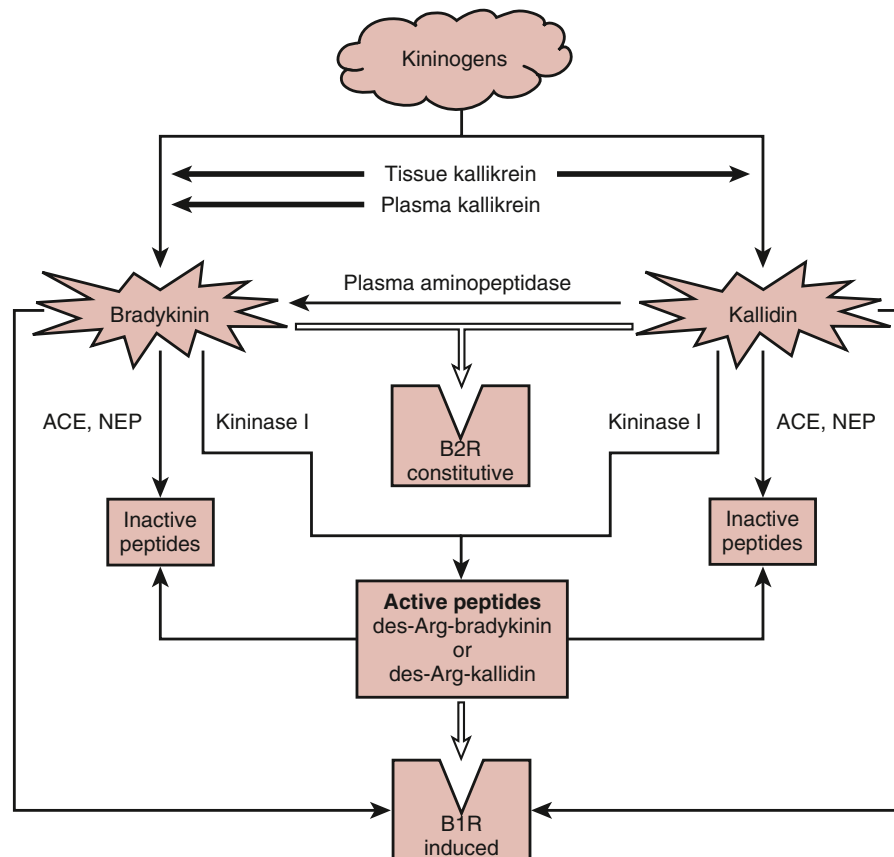


Fig. 11.8 Enzymatic cascade of the kallikrein–kinin system. ACE, Angiotensin-converting enzyme; B1R, bradykinin B1 receptor; B2R, bradykinin B2 receptor; NEP, neutral endopeptidase.

rats, administration of lipopolysaccharide, however, induces expression of B1R throughout the nephron (except the outer medullary collecting ducts), with strong expression in the efferent arteriole, medullary limb, and distal tubule.⁴⁶⁹

The KKS is involved in the regulation of both renal hemodynamics and tubule function. Diuretic and natriuretic effects play a pivotal role in the contribution of renal KKS to fluid and electrolyte balance. Kinins have been reported to increase RBF and papillary blood flow and to mediate the hyperfiltration induced by a high-protein diet. Kinins also inhibit conductive sodium entry in the IMCDs,⁴⁷⁰ and B2R-deficient mice demonstrate increased urinary concentration in response to vasopressin, which indicates that, through the B2R, endogenous kinins oppose the antidiuretic effect of vasopressin.⁴⁷¹ Kinins may therefore affect sodium reabsorption through direct effects on sodium transport along the nephron, through vasodilatory effects, and through changes in the osmotic gradient of the renal medulla. In addition to the effects on renal vascular tone, salt homeostasis, and water homeostasis, experiments with the B2R antagonist icatibant have yielded evidence that kinins may also have antihypertrophic and antiproliferative properties in mesangial cells, fibroblasts, and renomedullary interstitial cells. The antiproliferative effect of bradykinin in mesangial cells may be mediated through interaction of the B2R with the protein-tyrosine phosphatase SH2 domain-containing phosphatase-2.⁴⁷²

REGULATION OF TUBULE TRANSPORT BY TISSUE KALLIKREIN

Independent of its ability to generate kinin, tissue kallikrein also exerts separate effects on tubule solute transport by regulating the activity of the epithelial Na⁺ channel (ENaC), the colonic H⁺, K⁺ ATPase, and the epithelial calcium channel TRPV5 (transient receptor potential channel vanilloid subtype 5).⁴⁷³ The connecting tubules secrete a large amount of tissue kallikrein, which, through its enzymatic activity, can alter the function of ion transporters expressed on the luminal surface of cells downstream of its site of secretion.⁴⁷³ For instance, tissue kallikrein may participate in the proteolytic processing of ENaC increasing its activity, whereas tissue kallikrein-deficient mice have decreased ENaC activity.⁴⁷⁴ Despite decreased ENaC activity, however, coincident upregulation of ENaC-independent electroneutral NaCl absorption ensures that tissue kallikrein is not essential for sodium homeostasis.^{473,475} The cortical collecting ducts from tissue kallikrein-deficient mice also demonstrate enhanced activity of the colonic H⁺, K⁺ ATPase in intercalated cells, resulting in net K⁺ absorption.^{473,476} Finally, tissue kallikrein functions to stabilize the TRPV5 channel at the plasma membrane, promoting Ca²⁺ reabsorption, whereas tissue kallikrein knockout mice exhibit robust hypercalciuria.⁴⁷⁷ Distal tubule defects in potassium⁴⁷⁸ and calcium⁴⁷⁹ handling have also been reported in humans with the loss-of-function R53H polymorphism in the tissue kallikrein gene.

THE KALLIKREIN-KININ SYSTEM IN KIDNEY DISEASE

HYPERTENSION

Although it has been known for many years that kinin infusion results in an acute drop in systemic blood pressure

by reducing peripheral resistance, the role of the KKS in mediating primary or secondary hypertension has yet to be fully established. Decreased activity of kallikrein has been reported in the urine of hypertensive patients and hypertensive rats. An inverse relationship between urinary kallikrein excretion and blood pressure in humans may be suggestive of a role for the renal KKS in protecting against hypertension.⁴⁸⁰ However, an alternative interpretation may be that preexisting or hypertension-induced renal disease may itself lead to a reduction in renal kallikrein excretion. In the Dahl salt-sensitive rat model of hypertension, ACE inhibitors attenuate the progression of proteinuria and hypertensive nephrosclerosis better than do ARBs.⁴⁸¹ That this difference may be mediated by enhanced kinin activity with ACE inhibition is supported by the observations that infusion of either kallikrein⁴⁸² or bradykinin⁴⁸³ in this model attenuated glomerulosclerosis without affecting blood pressure. Studies in two-kidney, one-clip hypertension have yielded conflicting results. The incidence of two-kidney, one-clip hypertension was increased in B2R-deficient mice, in comparison with wild-type animals.⁴⁸⁴ By contrast, with regard to the response between tissue kallikrein-deficient mice and wild-type animals, there was no difference with respect to kidney size, renin release, systemic blood pressure increase, and cardiac remodeling.⁴⁸⁵

Despite the uncertainty about the role of the KKS in mediating the pathogenesis of hypertension, a variety of genetic mutations of the KKS have been associated with hypertension in animal models and humans.⁴⁸⁶ Inactivating mutations in the kallikrein gene have been identified in spontaneously hypertensive rats,⁴⁸⁷ and an association between mutations in the regulatory region of the kallikrein gene *KLK1* and hypertension has also been described in African Americans⁴⁸⁸ and Chinese Han people.⁴⁸⁹ The loss-of-function *KLK1* R53H mutation is found in 5% to 7% of the Caucasian population,⁴⁹⁰ but this one single-nucleotide polymorphism (SNP) has not in itself been found to markedly alter blood pressure.⁴⁹¹ ACE polymorphisms, responsible for different plasma levels of the enzyme and, accordingly, altered kinin levels, have been identified as independent risk factors for progression of various diseases including diabetic nephropathy, but they do not affect blood pressure. Finally, a number of SNPs in both the *B2R* and *B1R* genes have been associated with hypertension^{492,493} and coronary risk in hypertensive individuals.⁴⁹⁴

DIABETIC NEPHROPATHY

Observations in both experimental animal models and in humans indicate a role for altered KKS activity in the pathogenesis of diabetic nephropathy, although results in experimental studies have been conflicting. The KKS is markedly altered in rats with streptozotocin-induced diabetes, and changes are correlated with those in renal plasma flow and GFR.⁴⁹⁵ Renal and urinary levels of active kallikrein are increased in rats with moderate hyperglycemia in association with reduced RVR, increased GFR, and increased renal plasma flow and treatment of diabetic rats with the kallikrein inhibitor aprotinin or with a B2R antagonist-reduced RBF and GFR.⁴⁹⁵ By contrast, in non-insulin-treated streptozotocin-treated rats with severe hyperglycemia and hypofiltration, kallikrein excretion and expression were reduced.^{495,496} In addition to its hemodynamic effects, the KKS may also play a

kidney-protective role in diabetic nephropathy through its antiinflammatory and antiproliferative properties.⁴⁴⁸

Results of receptor antagonist studies initially suggested that the KKS had a limited role in preserving kidney structure and function in diabetic nephropathy: Treatment of diabetic rats with icatibant had no effect on glomerular structure or on albuminuria, nor did it alter the attenuating effect of ACE inhibition on either of these parameters.⁴⁹⁷ In contrast to this finding, however, the results of more contemporary work suggest that the beneficial effects of ACE inhibitors in experimental diabetic nephropathy may be attenuated by coadministration of a B2R antagonist.⁴⁹⁸⁻⁵⁰⁰ In Akita-diabetic mice lacking the B2R, there was a marked increase in mesangial sclerosis and a worsening of albuminuria,⁵⁰¹ in association with an increase in oxidative stress and mitochondrial damage;⁵⁰² however, another study reported contrary results in that B2R-knockout mice were relatively protected from the renal injury caused by streptozotocin-induced diabetes.⁵⁰³ Upregulation of B1R occurs in response to B2R knockout and could plausibly contribute to renal pathology or alternatively confer a kidney-protective benefit. In support of the latter thesis, Akita-diabetic mice deficient in both B2R and B1R exhibited augmented kidney injury in comparison with those lacking B2R alone.⁵⁰⁴

In further support of a kidney-protective effect of the KKS in diabetic nephropathy, one study showed that induction of diabetes by streptozotocin in mice caused a twofold increase in mRNA for kininogen, tissue kallikrein, kinins, and kinin receptors, with a doubling in albumin excretion in kallikrein-knockout mice in comparison with wild-type animals.⁵⁰⁵ In another study, gene delivery of human tissue kallikrein with an adeno-associated virus vector attenuated kidney injury in diabetes and decreased urinary albumin excretion.⁵⁰⁶ When exogenous pancreatic kallikrein was administered to diabetic mice, it caused a reduction in albuminuria, renal fibrosis, inflammation, and oxidative stress.⁵⁰⁷ Conversely, however, kallistatin, which decreases kallikrein activity, also attenuated kidney injury in diabetic mice when it was overexpressed using ultrasound microbubble-mediated gene transfer.⁵⁰⁸

Urinary kallikrein excretion in patients with type 1 diabetes demonstrates a similar association with GFR as observed in rats with streptozotocin-induced diabetes.⁵⁰⁹ Active kallikrein excretion is increased in hyperfiltering individuals in comparison with both patients with type 1 diabetes who have a normal GFR and normal controls, and it is correlated with both GFR and distal tubule sodium reabsorption.⁵⁰⁹ Results of genetic association studies in patients with diabetes have, however, been conflicting: One study demonstrated an association between B2R polymorphisms and albuminuria in 49 patients with type 1 diabetes and 112 patients with type 2 diabetes,⁵¹⁰ whereas another revealed no association between either B1R or B2R polymorphisms and incipient or overt nephropathy in 285 patients with type 2 diabetes.⁵¹¹ Plasma levels of HMW kininogen fragments were observed to be elevated among individuals with type 1 diabetes and progressive renal decline.⁵¹²

ISCHEMIC RENAL INJURY

In models of ischemia-reperfusion injury, ACE inhibitors appear to be superior to ARBs in protecting against tubule

necrosis, loss of endothelial function, and excretory dysfunction.⁵¹³ This superiority may be attributed to enhanced kinin activity with ACE inhibition, inasmuch as the effect is negated by B2R antagonists and inhibitors of NO synthase.^{514,515} Bradykinin suppresses the opening of mitochondrial pores,⁵¹⁶ and NO suppresses oxidative metabolism; both observations indicate that the KKS may exert its protective effects in ischemia-reperfusion injury through attenuation of oxidative damage. In mice with genetic deficiencies in either the B2R alone or both B1R and B2R, ischemic damage was enhanced in comparison with wild-type mice; injury was most severe in mice that lacked both receptors.⁴⁶⁰ By contrast, tissue kallikrein infusion aggravated renal ischemia-reperfusion injury in rats,⁵¹⁷ whereas expression of the human kallistatin gene with an adenoviral vector protected mice from renal ischemia-reperfusion injury.⁵¹⁸ Thus although physiologic kinin levels may be protective in this setting, higher levels may be detrimental, possibly through pathologic reperfusion.⁴⁴⁶

CHRONIC KIDNEY DISEASE

In the remnant kidney model of progressive renal disease, adenovirus-mediated or adeno-associated virus-mediated gene delivery of kallikrein attenuated the decline in kidney function.⁵¹⁹ In the model of unilateral ureteric obstruction, both genetic ablation of the B2R and pharmacologic blockade of the B2R increased tubulointerstitial fibrosis.⁵²⁰ By contrast, expression of the B1R is increased after unilateral ureteric obstruction,⁵²¹ and treatment with a nonpeptide B1R antagonist-reduced macrophage infiltration and fibrosis.⁵²¹ In the same model, B1R-deficient mice similarly showed less upregulation of inflammatory cytokines, reduced albumin excretion, and diminished fibrosis in comparison with wild-type mice.⁵²² In an Adriamycin-induced mouse model of FSGS, B1R antagonist therapy attenuated and B1R agonist therapy aggravated renal dysfunction.⁵²³ Together, these observations suggest that although the B2R is kidney protective, under some circumstances (and in contrast to the observations made in Akita-diabetic B1R/B2R knockout mice) compensatory B1R upregulation may contribute to the pathogenesis of renal fibrosis. In humans, polymorphisms in both the *B1R* gene^{524,525} and *B2R* gene^{524,526} have been associated with the development of ESRD.

LUPUS NEPHRITIS AND ANTIGLOMERULAR BASEMENT MEMBRANE DISEASE

Evidence has linked the KKS to the pathogenesis of immune-mediated nephritides, SLE, Goodpasture syndrome (antiglomerular basement membrane [GBM] disease), and spontaneous lupus nephritis. Mice strains differ in their susceptibility to anti-GBM antibody-induced nephritis. Comparison of disease-sensitive and control strains, by microarray analysis of renal cortical tissue, revealed that 360 gene transcripts were differentially expressed.⁵²⁷ Of the underexpressed genes, one-fifth belonged to the kallikrein gene family.⁵²⁷ Furthermore, in disease-sensitive mice, B2R antagonism augmented proteinuria after anti-GBM challenge, whereas bradykinin administration attenuated disease.⁵²⁷ In the same study, SNPs in the *KLK1* and *KLK3* promoters were also described in patients with SLE and lupus nephritis.⁵²⁷ Extending their work further, the same

investigators showed that adenoviral delivery of the *KLK1* gene attenuated kidney injury in congenic mice possessing a lupus-susceptibility interval on chromosome 7.⁵²⁸

ANTINEUTROPHIL CYTOPLASMIC ANTIBODY-ASSOCIATED VASCULITIS

Granulomatosis with polyangiitis (GPA) may be associated with necrotizing glomerulonephritis. The major antigenic target in GPA is neutrophil-derived proteinase 3 (PR3). Incubation of PR3 with HMW kininogen resulted in the generation of a novel tridecapeptide kinin, termed “PR3-kinin.”⁵²⁹ PR3-kinin binds to B1R directly and can also activate B2R after further processing to form bradykinin.⁵²⁹ These observations suggest that, in GPA, PR3 may activate the kinin pathway in a kallikrein-independent manner. B1R upregulation has been observed in biopsies from patients with Henoch-Schönlein purpura nephropathy or with antineutrophil cytoplasmic antibody-associated vasculitis.⁵³⁰ Similarly, B1R upregulation was also observed in a murine serum-induced glomerulonephritis model, whereas treatment with a B1R antagonist attenuated kidney function decline.⁵³⁰

 Visit Elsevier eBooks+ ([eBooks.Health.Elsevier.com](https://www.elsevier.com/ebooksplus)) for a complete set of references and Board Review Questions.

KEY REFERENCES

2. Anderson S, Meyer TW, Rennke HG, Brenner BM. Control of glomerular hypertension limits glomerular injury in rats with reduced renal mass. *J Clin Invest*. 1985;76(2):612–619.
5. Griendling KK, Murphy TJ, Alexander RW. Molecular biology of the renin-angiotensin system. *Circulation*. 1993;87(6):1816–1828. <https://doi.org/10.1161/01.cir.87.6.1816>.
20. Nguyen G, Delarue F, Burckle C, Bouzahir L, Giller T, Sraer JD. Pivotal role of the renin/prorenin receptor in angiotensin II production and cellular responses to renin. *J Clin Invest*. 2002;109(11):1417–1427.
24. Schnermann J. Juxtaglomerular cell complex in the regulation of renal salt excretion. *Am J Physiol*. 1998;274(2):R263–R279. <https://doi.org/10.1152/ajpregu.1998.274.2.R263>.
28. Vargas SL, Toma I, Kang JJ, Meer EJ, Peti-Peterdi J. Activation of the succinate receptor GPR91 in macula densa cells causes renin release. *J Am Soc Nephrol*. 2009;20(5):1002–1011.
33. Campbell DJ, Nussberger J, Stowasser M, et al. Activity assays and immunoassays for plasma Renin and prorenin: information provided and precautions necessary for accurate measurement. *Clin Chem*. 2009;55(5):867–877. <https://doi.org/10.1373/clinchem.2008.118000>.
40. Gloy J, Henger A, Fischer KG, et al. Angiotensin II modulates cellular functions of podocytes. *Kidney Int Suppl*. 1998;67:S168–S170. <https://doi.org/10.1046/j.1523-1755.1998.06736.x>.
41. Velez JC. The importance of the intrarenal renin-angiotensin system. *Nat Clin Pract Nephrol*. 2009;5(2):89–100.
44. Murasawa S, Mori Y, Nozawa Y, et al. Angiotensin II type 1 receptor-induced extracellular signal-regulated protein kinase activation is mediated by Ca²⁺/calmodulin-dependent transactivation of epidermal growth factor receptor. *Circ Res*. 1998;82(12):1338–1348.
48. Lautrette A, Li S, Alili R, et al. Angiotensin II and EGF receptor cross-talk in chronic kidney diseases: a new therapeutic approach. *Nat Med*. 2005;11(8):867–874.
51. Kelly DJ, Cox AJ, Gow RM, Zhang Y, Kemp BE, Gilbert RE. Platelet-derived growth factor receptor transactivation mediates the trophic effects of angiotensin II in vivo. *Hypertension*. 2004;44(2):195–202.
68. Kennedy CR, Burns KD. Angiotensin II as a mediator of renal tubular transport. *Contrib Nephrol*. 2001;135:47–62.
80. Advani A, Kelly DJ, Cox AJ, et al. The (Pro)renin receptor: site-specific and functional linkage to the vacuolar H⁺-ATPase in the kidney. *Hypertension*. 2009;54(2):261–269.
89. Donoghue M, Hsieh F, Baronas E, et al. A novel angiotensin-converting enzyme-related carboxypeptidase (ACE2) converts angiotensin I to angiotensin 1–9. *Circ Res*. 2000;87(5):E1–E9.
98. Li W, Moore MJ, Vasilieva N, et al. Angiotensin-converting enzyme 2 is a functional receptor for the SARS coronavirus. *Nature*. 2003;426(6965):450–454.
106. Ferrario CM. Angiotensin-converting enzyme 2 and angiotensin-(1–7): an evolving story in cardiovascular regulation. *Hypertension*. 2006;47(3):515–521.
111. Dell'Italia LJ, Ferrario CM. The never-ending story of angiotensin peptides: beyond angiotensin I and II. *Circ Res*. 2013;112(8):1086–1087. <https://doi.org/10.1161/CIRCRESAHA.113.301246>.
112. Paul M, Poyan Mehr A, Kreutz R. Physiology of local renin-angiotensin systems. *Physiol Rev*. 2006;86(3):747–803. <https://doi.org/10.1152/physrev.00036.2005>.
114. Braam B, Mitchell KD, Fox J, Navar LG. Proximal tubular secretion of angiotensin II in rats. *Am J Physiol*. 1993;264(5 Pt 2):F891–F898.
126. Blantz RC, Konnen KS, Tucker BJ. Angiotensin II effects upon the glomerular microcirculation and ultrafiltration coefficient of the rat. *J Clin Invest*. 1976;57(2):419–434.
127. Anderson S, Rennke HG, Brenner BM. Therapeutic advantage of converting enzyme inhibitors in arresting progressive renal disease associated with systemic hypertension in the rat. *J Clin Invest*. 1986;77(6):1993–2000.
130. Kagami S, Border WA, Miller DE, Noble NA. Angiotensin II stimulates extracellular matrix protein synthesis through induction of transforming growth factor-beta expression in rat glomerular mesangial cells. *J Clin Invest*. 1994;93(6):2431–2437.
134. Langham RG, Kelly DJ, Gow RM, et al. Transforming growth factor-beta in human diabetic nephropathy: effects of ACE inhibition. *Diabetes Care*. 2006;29(12):2670–2675.
135. Houlihan CA, Akdeniz A, Tsalamandris C, Cooper ME, Jerums G, Gilbert RE. Urinary transforming growth factor-beta excretion in patients with hypertension, type 2 diabetes, and elevated albumin excretion rate: effects of angiotensin receptor blockade and sodium restriction. *Diabetes Care*. 2002;25(6):1072–1077.
136. Mifsud SA, Allen TJ, Bertram JF, et al. Podocyte foot process broadening in experimental diabetic nephropathy: amelioration with renin-angiotensin blockade. *Diabetologia*. 2001;44(7):878–882.
138. Langham RG, Kelly DJ, Cox AJ, et al. Proteinuria and the expression of the podocyte slit diaphragm protein, nephrin, in diabetic nephropathy: effects of angiotensin converting enzyme inhibition. *Diabetologia*. 2002;45(11):1572–1576.
189. Krum H, Viskoper RJ, Lacourciere Y, Budde M, Charlton V. The effect of an endothelin-receptor antagonist, bosentan, on blood pressure in patients with essential hypertension. Bosentan Hypertension Investigators. *N Engl J Med*. 1998;338(12):784–790.
191. Weber MA, Black H, Bakris G, et al. A selective endothelin-receptor antagonist to reduce blood pressure in patients with treatment-resistant hypertension: a randomised, double-blind, placebo-controlled trial. *Lancet*. 2009;374(9699):1423–1431.
234. Dhaun N, MacIntyre IM, Kerr D, et al. Selective endothelin-A receptor antagonism reduces proteinuria, blood pressure, and arterial stiffness in chronic proteinuric kidney disease. *Hypertension*. 2011;57(4):772–779.
238. Wenzel RR, Littke T, Kuranoff S, et al. Avasentan reduces albumin excretion in diabetics with macroalbuminuria. *J Am Soc Nephrol*. 2009;20(3):655–664.
239. Mann JF, Green D, Jamerson K, et al. Avasentan for overt diabetic nephropathy. *J Am Soc Nephrol*. 2010;21(3):527–535. <https://doi.org/10.1681/ASN.2009060593>.
241. de Zeeuw D, Coll B, Andress D, et al. The endothelin antagonist atrasentan lowers residual albuminuria in patients with type 2 diabetic nephropathy. *J Am Soc Nephrol*. 2014;25(5):1083–1093.
244. Waijjer SW, Gansevoort RT, Bakris GL, et al. The effect of atrasentan on kidney and heart failure outcomes by baseline albuminuria and kidney function: a post hoc analysis of the SONAR randomized trial. *Clin J Am Soc Nephrol*. 2021;16(12):1824–1832.
242. Kohan DE, Lambers Heerspink HJ, Coll B, et al. Predictors of atrasentan-associated fluid retention and change in albuminuria in patients with diabetic nephropathy. *Clin J Am Soc Nephrol*. 2015;10(9):1568–1574.
249. Rovin BH, Barratt J, Heerspink HJL, et al. Efficacy and safety of sparsentan versus irbesartan in patients with IgA nephropathy (PROTECT): 2-year results from a randomised, active-controlled, phase 3 trial. *Lancet*. 2023;402(10417):2077–2090.

354. Januzzi Jr JL, Camargo CA, Anwaruddin S, et al. The N-terminal Pro-BNP investigation of dyspnea in the emergency department (PRIDE) study. *Am J Cardiol*. 2005;95(8):948–954.
355. Ledwidge M, Gallagher J, Conlon C, et al. Natriuretic peptide-based screening and collaborative care for heart failure: the STOP-HF randomized trial. *JAMA*. 2013;310(1):66–74.
356. Huelsmann M, Neuhold S, Resl M, et al. PONTIAC (NT-proBNP selected prevention of cardiac events in a population of diabetic patients without a history of cardiac disease): a prospective randomized controlled trial. *J Am Coll Cardiol*. 2013;62(15):1365–1372.
397. Sackner-Bernstein JD, Skopicki HA, Aaronson KD. Risk of worsening renal function with nesiritide in patients with acutely decompensated heart failure. *Circulation*. 2005;111(12):1487–1491.
398. O'Connor CM, Starling RC, Hernandez AF, et al. Effect of nesiritide in patients with acute decompensated heart failure. *N Engl J Med*. 2011;365(1):32–43.
410. Campbell DJ. Long-term neprilysin inhibition—implications for ARNIs. *Nat Rev Cardiol*. 2017;14(3):171–186.
411. Kostis JB, Packer M, Black HR, Schmieder R, Henry D, Levy E. Omapatrilat and enalapril in patients with hypertension: the Omapatrilat Cardiovascular Treatment vs. Enalapril (OCTAVE) trial. *Am J Hypertens*. 2004;17(2):103–111.
412. Packer M, Califf RM, Konstam MA, et al. Comparison of omapatrilat and enalapril in patients with chronic heart failure: the omapatrilat versus enalapril randomized trial of utility in reducing events (OVERTURE). *Circulation*. 2002;106(8):920–926.
413. Hubers SA, Brown NJ. Combined angiotensin receptor antagonism and neprilysin inhibition. *Circulation*. 2016;133(11):1115–1124.
415. Ruilope LM, Dukat A, Bohm M, Lacourciere Y, Gong J, Lefkowitz MP. Blood-pressure reduction with LCZ696, a novel dual-acting inhibitor of the angiotensin II receptor and neprilysin: a randomised, double-blind, placebo-controlled, active comparator study. *Lancet*. 2010;375(9722):1255–1266.
416. Solomon SD, Zile M, Pieske B, et al. The angiotensin receptor neprilysin inhibitor LCZ696 in heart failure with preserved ejection fraction: a phase 2 double-blind randomised controlled trial. *Lancet*. 2012;380(9851):1387–1395.
417. McMurray JJ, Packer M, Desai AS, et al. Angiotensin-neprilysin inhibition versus enalapril in heart failure. *N Engl J Med*. 2014;371(11):993–1004.
424. Voors AA, Gori M, Liu LC, et al. Renal effects of the angiotensin receptor neprilysin inhibitor LCZ696 in patients with heart failure and preserved ejection fraction. *Eur J Heart Fail*. 2015;17(5):510–517.
460. Kakoki M, McGarrah RW, Kim HS, Smithies O. Bradykinin B1 and B2 receptors both have protective roles in renal ischemia/reperfusion injury. *Proc Natl Acad Sci USA*. 2007;104(18):7576–7581.
512. Merchant ML, Niewczas MA, Ficociello LH, et al. Plasma kininogen and kininogen fragments are biomarkers of progressive renal decline in type 1 diabetes. *Kidney Int*. 2013;83(6):1177–1184.

REFERENCES

- Yanai K, Ishibashi K, Morishita Y. Systematic review and meta-analysis of renin-angiotensin-aldosterone system blocker effects on the development of cardiovascular disease in patients with chronic kidney disease. *Front Pharmacol*. 2021;12:662544.
- Anderson S, Meyer TW, Rennke HG, Brenner BM. Control of glomerular hypertension limits glomerular injury in rats with reduced renal mass. *J Clin Invest*. 1985;76(2):612–619.
- Zicha J, Vaněčková I. Altered balance between vasoconstrictor and vasodilator systems in experimental hypertension. *Physiol Res*. 2024;73(6):901–928.
- Gerstein HC, Santaguida P, Raina P, et al. Annual incidence and relative risk of diabetes in people with various categories of dysglycemia: a systematic overview and meta-analysis of prospective studies. *Diabetes Res Clin Pract*. 2007;78(3):305–312.
- Griendling KK, Murphy TJ, Alexander RW. Molecular biology of the renin-angiotensin system. *Circulation*. 1993;87(6):1816–1828. <https://doi.org/10.1161/01.cir.87.6.1816>.
- Clauser E, Gaillard I, Wei L, Corvol P. Regulation of angiotensinogen gene. *Am J Hypertens*. 1989;2(5 Pt 1):403–410. <https://doi.org/10.1093/ajh/2.5.403>.
- Cruz-López EO, Ye D, Wu C, et al. Angiotensinogen suppression: a new tool to treat cardiovascular and renal disease. *Hypertension*. 2022;79(10):2115–2126. <https://doi.org/10.1161/hypertension-aha.122.18731>.
- Høj Nielsen A, Knudsen F. Angiotensinogen is an acute-phase protein in man. *Scand J Clin Lab Invest*. 1987;47(2):175–178.
- Ron D, Brasier AR, Habener JF. Angiotensinogen gene-inducible enhancer-binding protein 1, a member of a new family of large nuclear proteins that recognize nuclear factor kappa B-binding sites through a zinc finger motif. *Mol Cell Biol*. 1991;11(5):2887–2895.
- Dzau VJ, Burt DW, Pratt RE. Molecular biology of the renin-angiotensin system. *Am J Physiol*. 1988;255(4 Pt 2):F563–F573. <https://doi.org/10.1152/ajprenal.1988.255.4.F563>.
- Schweda F, Friis U, Wagner C, Skott O, Kurtz A. Renin release. *Physiology*. 2007;22:310–319.
- Neves FA, Duncan KG, Baxter JD. Cathepsin B is a prorenin processing enzyme. *Hypertension*. 1996;27(3 Pt 2):514–517.
- Reudelhuber TL, Ramla D, Chiu L, Mercure C, Seidah NG. Proteolytic processing of human prorenin in renal and non-renal tissues. *Kidney Int*. 1994;46(6):1522–1524.
- Saris JJ, Derkx FH, De Bruin RJ, et al. High-affinity prorenin binding to cardiac man-6-P/IGF-II receptors precedes proteolytic activation to renin. *Am J Physiol Heart Circ Physiol*. 2001;280(4):H1706–H1715.
- van den Eijnden MM, Saris JJ, de Bruin RJ, et al. Prorenin accumulation and activation in human endothelial cells: importance of mannose 6-phosphate receptors. *Arterioscler Thromb Vasc Biol*. 2001;21(6):911–916.
- Danser AH, Deinum J. Renin, prorenin and the putative (pro) renin receptor. *Hypertension*. 2005;46(5):1069–1076. <https://doi.org/10.1161/01.HYP.0000186329.92187.2e>.
- Lumbers ER. Activation of renin in human amniotic fluid by low pH. *Enzymologia*. 1971;40(6):329–336.
- Skinner SL, Cran EJ, Gibson R, Taylor R, Walters WA, Catt KJ. Angiotensins I and II, active and inactive renin, renin substrate, renin activity, and angiotensinase in human liquor amnii and plasma. *Am J Obstet Gynecol*. 1975;121(5):626–630.
- Pitarresi TM, Rubattu S, Henrikson R, Sealey JE. Reversible cryoactivation of recombinant human prorenin. *J Biol Chem*. 1992;267(17):11753–11759.
- Nguyen G, Delarue F, Burckle C, Bouzahir L, Giller T, Sraer JD. Pivotal role of the renin/prorenin receptor in angiotensin II production and cellular responses to renin. *J Clin Invest*. 2002;109(11):1417–1427.
- Skinner SL, McCubbin JW, Page IH. Control of renin secretion. *Circ Res*. 1964;15:64–76.
- Hackenthal E, Paul M, Ganten D, Taugner R. Morphology, physiology, and molecular biology of renin secretion. *Physiol Rev*. 1990;70(4):1067–1116.
- Yang X, Zeng H, Wang L, Luo S, Zhou Y. Activation of Piezo1 downregulates renin in juxtaglomerular cells and contributes to blood pressure homeostasis. *Cell Biosci*. 2022;12(1):197.
- Schnermann J. Juxtaglomerular cell complex in the regulation of renal salt excretion. *Am J Physiol*. 1998;274(2):R263–R279. <https://doi.org/10.1152/ajpregu.1998.274.2.R263>.
- Oliver WJ, Cohen EL, Neel JV. Blood pressure, sodium intake, and sodium related hormones in the Yanomamo Indians, a “no-salt” culture. *Circulation*. 1975;52(1):146–151.
- Hebert SC. Physiology: orphan detectors of metabolism. *Nature*. 2004;429(6988):143–145.
- He W, Miao FJ, Lin DC, et al. Citric acid cycle intermediates as ligands for orphan G-protein-coupled receptors. *Nature*. 2004;429(6988):188–193.
- Vargas SL, Toma I, Kang JJ, Meer EJ, Peti-Peterdi J. Activation of the succinate receptor GPR91 in macula densa cells causes renin release. *J Am Soc Nephrol*. 2009;20(5):1002–1011.
- Li YC, Kong J, Wei M, Chen ZF, Liu SQ, Cao LP. 1,25-Dihydroxyvitamin D(3) is a negative endocrine regulator of the renin-angiotensin system. *J Clin Invest*. 2002;110(2):229–238.
- Yuan W, Pan W, Kong J, et al. 1,25-dihydroxyvitamin D3 suppresses renin gene transcription by blocking the activity of the cyclic AMP response element in the renin gene promoter. *J Biol Chem*. 2007;282(41):29821–29830.
- Lutscher JA, Kraemer FB, Wilson DM, Schwartz HC, Bryer-Ash M. Increased plasma inactive renin in diabetes mellitus. A marker of microvascular complications. *N Engl J Med*. 1985;312(22):1412–1417.
- Allen TJ, Cooper ME, Gilbert RE, Winikoff J, Skinn SL, Jerums G. Serum total renin is increased before microalbuminuria in diabetes. *Kidney Int*. 1996;50(3):902–907.
- Campbell DJ, Nussberger J, Stowasser M, et al. Activity assays and immunoassays for plasma Renin and prorenin: information provided and precautions necessary for accurate measurement. *Clin Chem*. 2009;55(5):867–877.
- Menard J, Guyene TT, Peyrard S, Azizi M. Conformational changes in prorenin during renin inhibition in vitro and in vivo. *J Hypertens*. 2006;24(3):529–534.
- Kumar RS, Thekkumkara TJ, Sen GC. The mRNAs encoding the two angiotensin-converting isozymes are transcribed from the same gene by a tissue-specific choice of alternative transcription initiation sites. *J Biol Chem*. 1991;266(6):3854–3862.
- Atanassova N, Lakova E, Bratchkova Y, Krasteva G, Donchev M. Expression of testicular angiotensin-converting enzyme in adult spontaneously hypertensive rats. *Folia Histochem Cytobiol*. 2009;47(1):117–122.
- Perich RB, Jackson B, Rogerson F, Mendelsohn FA, Paxton D, Johnston CI. Two binding sites on angiotensin-converting enzyme: evidence from radioligand binding studies. *Mol Pharmacol*. 1992;42(2):286–293.
- Lai KN, Leung JC, Lai KB, To WY, Yeung VT, Lai FM. Gene expression of the renin-angiotensin system in human kidney. *J Hypertens*. 1998;16(1):91–102.
- Mehta PK, Griendling KK. Angiotensin II cell signaling: physiological and pathological effects in the cardiovascular system. *Am J Physiol Cell Physiol*. 2007;292(1):C82–C97.
- Gloy J, Henger A, Fischer KG, et al. Angiotensin II modulates cellular functions of podocytes. *Kidney Int Suppl*. 1998;67:S168–S170. <https://doi.org/10.1046/j.1523-1755.1998.06736.x>.
- Velez JC. The importance of the intrarenal renin-angiotensin system. *Nat Clin Pract Nephrol*. 2009;5(2):89–100.
- Zhuo J, Alcorn D, Allen AM, Mendelsohn FA. High resolution localization of angiotensin II receptors in rat renal medulla. *Kidney Int*. 1992;42(6):1372–1380.
- Touyz RM, Schiffrin EL. Signal transduction mechanisms mediating the physiological and pathophysiological actions of angiotensin II in vascular smooth muscle cells. *Pharmacol Rev*. 2000;52(4):639–672.
- Murasawa S, Mori Y, Nozawa Y, et al. Angiotensin II type 1 receptor-induced extracellular signal-regulated protein kinase activation is mediated by Ca²⁺/calmodulin-dependent transactivation of epidermal growth factor receptor. *Circ Res*. 1998;82(12):1338–1348.
- Eguchi S, Numaguchi K, Iwasaki H, et al. Calcium-dependent epidermal growth factor receptor transactivation mediates the angiotensin II-induced mitogen-activated protein kinase activation in vascular smooth muscle cells. *J Biol Chem*. 1998;273(15):8890–8896.

46. Heeneman S, Haendeler J, Saito Y, Ishida M, Berk BC. Angiotensin II induces transactivation of two different populations of the platelet-derived growth factor beta receptor. Key role for the p66 adaptor protein Shc. *J Biol Chem*. 2000;275(21):15926–15932.
47. Linseman DA, Benjamin CW, Jones DA. Convergence of angiotensin II and platelet-derived growth factor receptor signaling cascades in vascular smooth muscle cells. *J Biol Chem*. 1995;270(21):12563–12568.
48. Lautrette A, Li S, Alili R, et al. Angiotensin II and EGF receptor cross-talk in chronic kidney diseases: a new therapeutic approach. *Nat Med*. 2005;11(8):867–874. <https://doi.org/10.1038/nm1275>.
49. Wassef L, Kelly DJ, Gilbert RE. Epidermal growth factor receptor inhibition attenuates early kidney enlargement in experimental diabetes. *Kidney Int*. 2004;66(5):1805–1814.
50. Eskildsen-Helmond YE, Mulvany MJ. Pressure-induced activation of extracellular signal-regulated kinase 1/2 in small arteries. *Hypertension*. 2003;41(4):891–897.
51. Kelly DJ, Cox AJ, Gow RM, Zhang Y, Kemp BE, Gilbert RE. Platelet-derived growth factor receptor transactivation mediates the trophic effects of angiotensin II in vivo. *Hypertension*. 2004;44(2):195–202.
52. Mogi M, Iwai M, Horiuchi M. New insights into the regulation of angiotensin receptors. *Curr Opin Nephrol Hypertens*. 2009;18(2):138–143.
53. Daviet L, Lehtonen JY, Tamura K, Griesse DP, Horiuchi M, Dzau VJ. Cloning and characterization of ATRAP, a novel protein that interacts with the angiotensin II type 1 receptor. *J Biol Chem*. 1999;274(24):17058–17062.
54. Guo DF, Chenier I, Lavoie JL, et al. Development of hypertension and kidney hypertrophy in transgenic mice overexpressing ARAP1 gene in the kidney. *Hypertension*. 2006;48(3):453–459.
55. Prinster SC, Hague C, Hall RA. Heterodimerization of G protein-coupled receptors: specificity and functional significance. *Pharmacol Rev*. 2005;57(3):289–298.
56. Hansen JL, Theilade J, Haunso S, Sheikh SP. Oligomerization of wild type and nonfunctional mutant angiotensin II type I receptors inhibits galphaq protein signaling but not ERK activation. *J Biol Chem*. 2004;279(23):24108–24115.
57. AbdAlla S, Lother H, Quitterer U. AT1-receptor heterodimers show enhanced G-protein activation and altered receptor sequestration. *Nature*. 2000;407(6800):94–98.
58. Zeng C, Liu Y, Wang Z, et al. Activation of D3 dopamine receptor decreases angiotensin II type 1 receptor expression in rat renal proximal tubule cells. *Circ Res*. 2006;99(5):494–500.
59. Zeng C, Wang Z, Asico LD, et al. Aberrant ETB receptor regulation of AT receptors in immortalized renal proximal tubule cells of spontaneously hypertensive rats. *Kidney Int*. 2005;68(2):623–631.
60. Mogi M, Iwai M, Horiuchi M. Emerging concepts of regulation of angiotensin II receptors: new players and targets for traditional receptors. *Arterioscler Thromb Vasc Biol*. 2007;27(12):2532–2539.
61. Yasuda N, Miura S, Akazawa H, et al. Conformational switch of angiotensin II type 1 receptor underlying mechanical stress-induced activation. *EMBO Rep*. 2008;9(2):179–186.
62. Zou Y, Akazawa H, Qin Y, et al. Mechanical stress activates angiotensin II type 1 receptor without the involvement of angiotensin II. *Nat Cell Biol*. 2004;6(6):499–506. <https://doi.org/10.1038/ncb1137>.
63. Yatabe J, Sanada H, Yatabe MS, et al. Angiotensin II type 1 receptor blocker attenuates the activation of ERK and NADPH oxidase by mechanical strain in mesangial cells in the absence of angiotensin II. *Am J Physiol Ren Physiol*. 2009;296(5):F1052–F1060.
64. Wallukat G, Homuth V, Fischer T, et al. Patients with preeclampsia develop agonistic autoantibodies against the angiotensin AT1 receptor. *J Clin Invest*. 1999;103(7):945–952.
65. Dragun D, Muller DN, Brasen JH, et al. Angiotensin II type 1-receptor activating antibodies in renal-allograft rejection. *N Engl J Med*. 2005;352(6):558–569.
66. Toke A, Meyer TW. Hemodynamic effects of angiotensin II in the kidney. *Contrib Nephrol*. 2001;135:34–46.
67. Navar LG, Inscho EW, Majid SA, Imig JD, Harrison-Bernard LM, Mitchell KD. Paracrine regulation of the renal microcirculation. *Physiol Rev*. 1996;76(2):425–536.
68. Kennedy CR, Burns KD. Angiotensin II as a mediator of renal tubular transport. *Contrib Nephrol*. 2001;135:47–62.
69. Rothenberger F, Velic A, Stehberger PA, Kovacicova J, Wagner CA. Angiotensin II stimulates vacuolar H⁺-ATPase activity in renal acid-secretory intercalated cells from the outer medullary collecting duct. *J Am Soc Nephrol*. 2007;18(7):2085–2093.
70. Pech V, Zheng W, Pham TD, Verlander JW, Wall SM. Angiotensin II activates H⁺-ATPase in type A intercalated cells. *J Am Soc Nephrol*. 2008;19(1):84–91.
71. Koike G, Horiuchi M, Yamada T, Szpirer C, Jacob HJ, Dzau VJ. Human type 2 angiotensin II receptor gene: cloned, mapped to the X chromosome, and its mRNA is expressed in the human lung. *Biochem Biophys Res Commun*. 1994;203(3):1842–1850. <https://doi.org/10.1006/bbrc.1994.2402>.
72. Reudelhuber TL. The continuing saga of the AT2 receptor: a case of the good, the bad, and the innocuous. *Hypertension*. 2005;46(6):1261–1262.
73. Savoia C, Touyz RM, Volpe M, Schiffrin EL. Angiotensin type 2 receptor in resistance arteries of type 2 diabetic hypertensive patients. *Hypertension*. 2007;49(2):341–346.
74. Padia SH, Howell NL, Siragy HM, Carey RM. Renal angiotensin type 2 receptors mediate natriuresis via angiotensin III in the angiotensin II type 1 receptor-blocked rat. *Hypertension*. 2006;47(3):537–544.
75. Miyata N, Park F, Li XF, Cowley Jr AW. Distribution of angiotensin AT1 and AT2 receptor subtypes in the rat kidney. *Am J Physiol*. 1999;277(3):F437–F446.
76. Carey RM, Padia SH. Angiotensin AT2 receptors: control of renal sodium excretion and blood pressure. *Trends Endocrinol Metabol*. 2008;19(3):84–87.
77. Padia SH, Kemp BA, Howell NL, et al. Intrarenal aminopeptidase N inhibition augments natriuretic responses to angiotensin III in angiotensin type 1 receptor-blocked rats. *Hypertension*. 2007;49(3):625–630.
78. Huang C, Kim Y, Caramori ML, et al. Diabetic nephropathy is associated with gene expression levels of oxidative phosphorylation and related pathways. *Diabetes*. 2006;55(6):1826–1831.
79. van den Heuvel M, Batenburg WW, Danser AH. Diabetic complications: a role for the prorenin-(pro)renin receptor-TGF-beta1 axis? *Mol Cell Endocrinol*. 2009;302(2):213–218.
80. Advani A, Kelly DJ, Cox AJ, et al. The (Pro)renin receptor: site-specific and functional linkage to the vacuolar H⁺-ATPase in the kidney. *Hypertension*. 2009;54(2):261–269.
81. Burckle C, Bader M. Prorenin and its ancient receptor. *Hypertension*. 2006;48(4):549–551.
82. Ichihara A, Hayashi M, Kaneshiro Y, et al. Inhibition of diabetic nephropathy by a decoy peptide corresponding to the “handle” region for nonproteolytic activation of prorenin. *J Clin Invest*. 2004;114(8):1128–1135. <https://doi.org/10.1172/JCI21398>.
83. Strausberg RL, Feingold EA, Grouse LH, et al. Generation and initial analysis of more than 15,000 full-length human and mouse cDNA sequences. *Proc Natl Acad Sci USA*. 2002;99(26):16899–16903.
84. Campbell DJ. Critical review of prorenin and (pro)renin receptor research. *Hypertension*. 2008;51(5):1259–1264.
85. Bader M. The second life of the (pro)renin receptor. *J Renin Angiotensin Aldosterone Syst*. 2007;8(4):205–208.
86. Ludwig J, Kerscher S, Brandt U, et al. Identification and characterization of a novel 9.2-kDa membrane sector-associated protein of vacuolar proton-ATPase from chromaffin granules. *J Biol Chem*. 1998;273(18):10939–10947.
87. Ichihara A. (Pro)renin receptor and vacuolar H⁽⁺⁾-ATPase. *Keio J Med*. 2012;61(3):73–78.
88. Tipnis SR, Hooper NM, Hyde R, Karran E, Christie G, Turner AJ. A human homolog of angiotensin-converting enzyme. Cloning and functional expression as a captopril-insensitive carboxypeptidase. *J Biol Chem*. 2000;275(43):33238–33243.
89. Donoghue M, Hsieh F, Baronas E, et al. A novel angiotensin-converting enzyme-related carboxypeptidase (ACE2) converts angiotensin I to angiotensin 1-9. *Circ Res*. 2000;87(5):E1–E9.
90. Turner AJ, Tipnis SR, Guy JL, Rice G, Hooper NM. ACEH/ACE2 is a novel mammalian metallopeptidase and a homologue of angiotensin-converting enzyme insensitive to ACE inhibitors. *Can J Physiol Pharmacol*. 2002;80(4):346–353.
91. Ingelfinger JR. Angiotensin-converting enzyme 2: implications for blood pressure and kidney disease. *Curr Opin Nephrol Hypertens*. 2009;18(1):79–84.

92. Der Sarkissian S, Grobe JL, Yuan L, et al. Cardiac overexpression of angiotensin converting enzyme 2 protects the heart from ischemia-induced pathophysiology. *Hypertension*. 2008;51(3):712–718.
93. Masson R, Nicklin SA, Craig MA, et al. Onset of experimental severe cardiac fibrosis is mediated by overexpression of Angiotensin-converting enzyme 2. *Hypertension*. 2009;53(4):694–700.
94. Soler MJ, Wysocki J, Battle D. Angiotensin-converting enzyme 2 and the kidney. *Exp Physiol*. 2008;93(5):549–556. <https://doi.org/10.1113/expphysiol.2007.041350>.
95. Soler MJ, Wysocki J, Battle D. ACE2 alterations in kidney disease. *Nephrol Dial Transplant*. 2013;28(11):2687–2697.
96. Wong DW, Oudit GY, Reich H, et al. Loss of angiotensin-converting enzyme-2 (Ace2) accelerates diabetic kidney injury. *Am J Pathol*. 2007;171(2):438–451.
97. Oudit GY, Liu GC, Zhong J, et al. Human recombinant ACE2 reduces the progression of diabetic nephropathy. *Diabetes*. 2010;59(2):529–538. <https://doi.org/10.2337/db09-1218>.
98. Li W, Moore MJ, Vasilieva N, et al. Angiotensin-converting enzyme 2 is a functional receptor for the SARS coronavirus. *Nature*. 2003;426(6965):450–454.
99. Fyhrquist F, Sajjonmaa O. Renin-angiotensin system revisited. *J Intern Med*. 2008;264(3):224–236.
100. Zini S, Fournie-Zaluski MC, Chauvel E, Roques BP, Corvol P, Llorens-Cortes C. Identification of metabolic pathways of brain angiotensin II and III using specific aminopeptidase inhibitors: predominant role of angiotensin III in the control of vasopressin release. *Proc Natl Acad Sci USA*. 1996;93(21):11968–11973.
101. Gammelgaard I, Wamberg S, Bie P. Systemic effects of angiotensin III in conscious dogs during acute double blockade of the renin-angiotensin-aldosterone-system. *Acta Physiol*. 2006;188(2):129–138.
102. Stragier B, De Bundel D, Sarre S, et al. Involvement of insulin-regulated aminopeptidase in the effects of the renin-angiotensin fragment angiotensin IV: a review. *Heart Fail Rev*. 2008;13(3):321–337.
103. Esteban V, Ruperez M, Sanchez-Lopez E, et al. Angiotensin IV activates the nuclear transcription factor-kappaB and related proinflammatory genes in vascular smooth muscle cells. *Circ Res*. 2005;96(9):965–973.
104. Kerins DM, Hao Q, Vaughan DE. Angiotensin induction of PAI-1 expression in endothelial cells is mediated by the hexapeptide angiotensin IV. *J Clin Invest*. 1995;96(5):2515–2520.
105. Tallant EA, Ferrario CM, Gallagher PE. Angiotensin-(1-7) inhibits growth of cardiac myocytes through activation of the mas receptor. *Am J Physiol Heart Circ Physiol*. 2005;289(4):H1560–H1566.
106. Ferrario CM. Angiotensin-converting enzyme 2 and angiotensin-(1-7): an evolving story in cardiovascular regulation. *Hypertension*. 2006;47(3):515–521.
107. Strawn WB, Ferrario CM, Tallant EA. Angiotensin-(1-7) reduces smooth muscle growth after vascular injury. *Hypertension*. 1999;33(1 Pt 2):207–211.
108. Dharmani M, Mustafa MR, Achike FI, Sim MK. Effect of des-aspargate-angiotensin I on the actions of angiotensin II in the isolated renal and mesenteric vasculature of hypertensive and STZ-induced diabetic rats. *Regul Pept*. 2005;129(1–3):213–219.
109. Nagata S, Kato J, Sasaki K, Minamino N, Eto T, Kitamura K. Isolation and identification of proangiotensin-12, a possible component of the renin-angiotensin system. *Biochem Biophys Res Commun*. 2006;350(4):1026–1031.
110. Westwood BM, Chappell MC. Divergent pathways for the angiotensin-(1-12) metabolism in the rat circulation and kidney. *Peptides*. 2012;35(2):190–195.
111. Dell'Italia LJ, Ferrario CM. The never-ending story of angiotensin peptides: beyond angiotensin I and II. *Circ Res*. 2013;112(8):1086–1087.
112. Paul M, Poyan Mehr A, Kreutz R. Physiology of local renin-angiotensin systems. *Physiol Rev*. 2006;86(3):747–803.
113. Kobori H, Nangaku M, Navar LG, Nishiyama A. The intrarenal renin-angiotensin system: from physiology to the pathobiology of hypertension and kidney disease. *Pharmacol Rev*. 2007;59(3):251–287.
114. Braam B, Mitchell KD, Fox J, Navar LG. Proximal tubular secretion of angiotensin II in rats. *Am J Physiol*. 1993;264(5 Pt 2):F891–F898. <https://doi.org/10.1152/ajprenal.1993.264.5.F891>.
115. Ingelfinger JR, Zuo WM, Fon EA, Ellison KE, Dzau VJ. In situ hybridization evidence for angiotensinogen messenger RNA in the rat proximal tubule. An hypothesis for the intrarenal renin angiotensin system. *J Clin Invest*. 1990;85(2):417–423.
116. Nishiyama A, Seth DM, Navar LG. Renal interstitial fluid concentrations of angiotensins I and II in anesthetized rats. *Hypertension*. 2002;39(1):129–134.
117. Sequeira Lopez ML, Pentz ES, Nomasa T, Smithies O, Gomez RA. Renin cells are precursors for multiple cell types that switch to the renin phenotype when homeostasis is threatened. *Dev Cell*. 2004;6(5):719–728.
118. Gilbert RE, Wu LL, Kelly DJ, et al. Pathological expression of renin and angiotensin II in the renal tubule after subtotal nephrectomy. Implications for the pathogenesis of tubulointerstitial fibrosis. *Am J Pathol*. 1999;155(2):429–440.
119. Holdsworth SR, Summers SA. Role of mast cells in progressive renal diseases. *J Am Soc Nephrol*. 2008;19(12):2254–2261.
120. Silver RB, Reid AC, Mackins CJ, et al. Mast cells: a unique source of renin. *Proc Natl Acad Sci USA*. 2004;101(37):13607–13612.
121. Haller H, Lindschau C, Erdmann B, Quass P, Luft FC. Effects of intracellular angiotensin II in vascular smooth muscle cells. *Circ Res*. 1996;79(4):765–772.
122. Redding KM, Chen BL, Singh A, et al. Transgenic mice expressing an intracellular fluorescent fusion of angiotensin II demonstrate renal thrombotic microangiopathy and elevated blood pressure. *Am J Physiol Heart Circ Physiol*. 2010;298(6):H1807–H1818.
123. Cook JL, Re RN. Lessons from in vitro studies and a related intracellular angiotensin II transgenic mouse model. *Am J Physiol Regul Integr Comp Physiol*. 2012;302(5):R482–R493.
124. Abadir PM, Foster DB, Crow M, et al. Identification and characterization of a functional mitochondrial angiotensin system. *Proc Natl Acad Sci USA*. 2011;108(36):14849–14854.
125. Alshahrani S. Renin-angiotensin-aldosterone pathway modulators in chronic kidney disease: a comparative review. *Front Pharmacol*. 2023;14:1101068.
126. Blantz RC, Konnen KS, Tucker BJ. Angiotensin II effects upon the glomerular microcirculation and ultrafiltration coefficient of the rat. *J Clin Invest*. 1976;57(2):419–434.
127. Anderson S, Rennke HG, Brenner BM. Therapeutic advantage of converting enzyme inhibitors in arresting progressive renal disease associated with systemic hypertension in the rat. *J Clin Invest*. 1986;77(6):1993–2000.
128. Zatz R, Dunn BR, Meyer TW, Anderson S, Rennke HG, Brenner BM. Prevention of diabetic glomerulopathy by pharmacological amelioration of glomerular capillary hypertension. *J Clin Invest*. 1986;77(6):1925–1930.
129. Naaman SC, Bakris GL. Diabetic nephropathy: update on pillars of therapy slowing progression. *Diabetes Care*. 2023;46(9):1574–1586.
130. Kagami S, Border WA, Miller DE, Noble NA. Angiotensin II stimulates extracellular matrix protein synthesis through induction of transforming growth factor-beta expression in rat glomerular mesangial cells. *J Clin Invest*. 1994;93(6):2431–2437.
131. Wolf G, Mueller E, Stahl RA, Ziyadeh FN. Angiotensin II-induced hypertrophy of cultured murine proximal tubular cells is mediated by endogenous transforming growth factor-beta. *J Clin Invest*. 1993;92(3):1366–1372.
132. Wu LL, Cox A, Roe CJ, Dziadek M, Cooper ME, Gilbert RE. Transforming growth factor beta 1 and renal injury following subtotal nephrectomy in the rat: role of the renin-angiotensin system. *Kidney Int*. 1997;51(5):1553–1567.
133. Gilbert RE, Cox A, Wu LL, et al. Expression of transforming growth factor-beta1 and type IV collagen in the renal tubulointerstitium in experimental diabetes: effects of ACE inhibition. *Diabetes*. 1998;47(3):414–422.
134. Langham RG, Kelly DJ, Gow RM, et al. Transforming growth factor-beta in human diabetic nephropathy: effects of ACE inhibition. *Diabetes Care*. 2006;29(12):2670–2675.
135. Houlihan CA, Akdeniz A, Tsalamandris C, Cooper ME, Jerums G, Gilbert RE. Urinary transforming growth factor-beta excretion in patients with hypertension, type 2 diabetes, and elevated albumin excretion rate: effects of angiotensin receptor blockade and sodium restriction. *Diabetes Care*. 2002;25(6):1072–1077.
136. Mifsud SA, Allen TJ, Bertram JF, et al. Podocyte foot process broadening in experimental diabetic nephropathy: amelioration

- with renin-angiotensin blockade. *Diabetologia*. 2001;44(7):878–882.
137. Suzuki K, Han GD, Miyauchi N, et al. Angiotensin II type 1 and type 2 receptors play opposite roles in regulating the barrier function of kidney glomerular capillary wall. *Am J Pathol*. 2007;170(6):1841–1853.
 138. Langham RG, Kelly DJ, Cox AJ, et al. Proteinuria and the expression of the podocyte slit diaphragm protein, nephrin, in diabetic nephropathy: effects of angiotensin converting enzyme inhibition. *Diabetologia*. 2002;45(11):1572–1576.
 139. Wu LL, Yang N, Roe CJ, et al. Macrophage and myofibroblast proliferation in remnant kidney: role of angiotensin II. *Kidney Int Suppl*. 1997;63:S221–S225.
 140. Ruiz-Ortega M, Lorenzo O, Rupérez M, Esteban V, Mezzano S, Egido J. Renin-angiotensin system and renal damage: emerging data on angiotensin II as a proinflammatory mediator. *Contrib Nephrol*. 2001;135:123–137.
 141. Esteban V, Heringer-Walther S, Sterner-Kock A, et al. Angiotensin-(1-7) and the G protein-coupled receptor MAS are key players in renal inflammation. *PLoS One*. 2009;4(4):e5406.
 142. Guzik TJ, Hoch NE, Brown KA, et al. Role of the T cell in the genesis of angiotensin II induced hypertension and vascular dysfunction. *J Exp Med*. 2007;204(10):2449–2460.
 143. Kvakani H, Kleinewietfeld M, Qadri F, et al. Regulatory T cells ameliorate angiotensin II-induced cardiac damage. *Circulation*. 2009;119(22):2904–2912. <https://doi.org/10.1161/CIRCULATIONAHA.108.832782>.
 144. Lush DJ, King JA, Fray JC. Pathophysiology of low renin syndromes: sites of renal renin secretory impairment and prorenin overexpression. *Kidney Int*. 1993;43(5):983–999.
 145. Paulsen EP, Seip RL, Ayers CR, Croft BY, Kaiser DL. Plasma renin activity and albumin excretion in teenage type I diabetic subjects. A prospective study. *Hypertension*. 1989;13(6 Pt 2):781–788.
 146. Levy SB, Lilley JJ, Frigon RP, Stone RA. Urinary kallikrein and plasma renin activity as determinants of renal blood flow. The influence of race and dietary sodium intake. *J Clin Invest*. 1977;60(1):129–138.
 147. Sato A, Nishimoto M. Clinical effect of nonsteroidal mineralocorticoid receptor (MR) antagonists in the treatment of diabetic kidney disease: expectations as a new therapeutic strategy. *Hypertens Res*. 2022;45(8):1310–1321. <https://doi.org/10.1038/s41440-022-00940-1>.
 148. Peti-Peterdi J, Kang JJ, Toma I. Activation of the renal renin-angiotensin system in diabetes—new concepts. *Nephrol Dial Transplant*. 2008;23(10):3047–3049.
 149. Kang JJ, Toma I, Sipos A, Meer EJ, Vargas SL, Peti-Peterdi J. The collecting duct is the major source of prorenin in diabetes. *Hypertension*. 2008;51(6):1597–1604.
 150. Jais X, D'Armini AM, Jansa P, et al. Bosentan for treatment of inoperable chronic thromboembolic pulmonary hypertension: BENEFIT (Bosentan Effects in iNoperable Forms of chronic Thromboembolic pulmonary hypertension), a randomized, placebo-controlled trial. *J Am Coll Cardiol*. 2008;52(25):2127–2134.
 151. Galie N, Manes A, Negro L, Palazzini M, Bacchi-Reggiani ML, Branzi A. A meta-analysis of randomized controlled trials in pulmonary arterial hypertension. *Eur Heart J*. 2009;30(4):394–403.
 152. Barst RJ, Langleben D, Badesch D, et al. Treatment of pulmonary arterial hypertension with the selective endothelin-A receptor antagonist sitaxsentan. *J Am Coll Cardiol*. 2006;47(10):2049–2056.
 153. Galie N, Rubin L, Hoeper M, et al. Treatment of patients with mildly symptomatic pulmonary arterial hypertension with bosentan (EARLY study): a double-blind, randomised controlled trial. *Lancet*. 2008;371(9630):2093–2100.
 154. Olsson KM, Hoeper MM. Novel approaches to the pharmacotherapy of pulmonary arterial hypertension. *Drug Discov Today*. 2009;14(5–6):284–290. <https://doi.org/10.1016/j.drudis.2008.12.003>.
 155. Dhaun N, Webb DJ, Kluth DC. Endothelin-1 and the kidney—beyond BP. *Br J Pharmacol*. 2012;167(4):720–731.
 156. Morita S, Kitamura K, Yamamoto Y, et al. Immunoreactive endothelin in human kidney. *Ann Clin Biochem*. 1991;28(Pt 3):267–271.
 157. Marsden PA, Dorfman DM, Collins TM, Brenner BM, Orkin SH, Ballermann BJ. Regulated expression of endothelin 1 in glomerular capillary endothelial cells. *Am J Physiol*. 1991;261(1 Pt 2):F117–F125.
 158. Ohta K, Hirata Y, Imai T, et al. Cytokine-induced release of endothelin-1 from porcine renal epithelial cell line. *Biochem Biophys Res Commun*. 1990;169(2):578–584.
 159. Sakamoto H, Sasaki S, Hirata Y, et al. Production of endothelin-1 by rat cultured mesangial cells. *Biochem Biophys Res Commun*. 1990;169(2):462–468.
 160. Kohan DE. The renal medullary endothelin system in control of sodium and water excretion and systemic blood pressure. *Curr Opin Nephrol Hypertens*. 2006;15(1):34–40.
 161. Kohan DE. Endothelin synthesis by rabbit renal tubule cells. *Am J Physiol*. 1991;261(2 Pt 2):F221–F226.
 162. Kohan DE. Endothelins in the normal and diseased kidney. *Am J Kidney Dis*. 1997;29(1):2–26.
 163. Abbasi Z, Winaver J, Rubinstein I, et al. Renal endothelin-converting enzyme in rats with congestive heart failure. *J Cardiovasc Pharmacol*. 1998;31(suppl 1):S31–S34.
 164. Pupilli C, Romagnani P, Lasagni L, et al. Localization of endothelin-converting enzyme-1 in human kidney. *Am J Physiol*. 1997;273(5):F749–F756.
 165. de Nucci G, Thomas R, D'Orleans-Juste P, et al. Pressor effects of circulating endothelin are limited by its removal in the pulmonary circulation and by the release of prostacyclin and endothelium-derived relaxing factor. *Proc Natl Acad Sci USA*. 1988;85(24):9797–9800.
 166. Dhaun N, Goddard J, Webb DJ. The endothelin system and its antagonism in chronic kidney disease. *J Am Soc Nephrol*. 2006;17(4):943–955.
 167. Luscher TF, Barton M. Endothelins and endothelin receptor antagonists: therapeutic considerations for a novel class of cardiovascular drugs. *Circulation*. 2000;102(19):2434–2440.
 168. Yamamoto T, Hirohama T, Uemura H. Endothelin B receptor-like immunoreactivity in podocytes of the rat kidney. *Arch Histol Cytol*. 2002;65(3):245–250. <https://doi.org/10.1679/aohc.65.245>.
 169. Katoh T, Chang H, Uchida S, Okuda T, Kurokawa K. Direct effects of endothelin in the rat kidney. *Am J Physiol*. 1990;258(2 Pt 2):F397–F402. <https://doi.org/10.1152/ajprenal.1990.258.2.F397>.
 170. Tomobe Y, Miyauchi T, Saito A, et al. Effects of endothelin on the renal artery from spontaneously hypertensive and Wistar Kyoto rats. *Eur J Pharmacol*. 1988;152(3):373–374.
 171. Kohan DE, Pollock DM. Endothelin antagonists for diabetic and non-diabetic chronic kidney disease. *Br J Clin Pharmacol*. 2013;76(4):573–579. <https://doi.org/10.1111/bcp.12064>.
 172. Konishi F, Okada Y, Takaoka M, Garipey CE, Yanagisawa M, Matsumura Y. Role of endothelin ET(B) receptors in the renal hemodynamic and excretory responses to big endothelin-1. *Eur J Pharmacol*. 2002;451(2):177–184.
 173. Hunter RW, Moorhouse R, Farrah TE, et al. First-in-Man demonstration of direct endothelin-mediated natriuresis and diuresis. *Hypertension*. 2017;70(1):192–200.
 174. Vanni S, Polidori G, Cecioni I, Serni S, Carini M, Modesti PA. ET(B) receptor in renal medulla is enhanced by local sodium during low salt intake. *Hypertension*. 2002;40(2):179–185.
 175. Ge Y, Bagnall A, Stricklett PK, et al. Collecting duct-specific knock-out of the endothelin B receptor causes hypertension and sodium retention. *Am J Physiol Ren Physiol*. 2006;291(6):F1274–F1280.
 176. Ahn D, Ge Y, Stricklett PK, et al. Collecting duct-specific knock-out of endothelin-1 causes hypertension and sodium retention. *J Clin Invest*. 2004;114(4):504–511. <https://doi.org/10.1172/JCI21064>.
 177. Chu TS, Peng Y, Cano A, Yanagisawa M, Alpern RJ. Endothelin(B) receptor activates NHE-3 by a Ca²⁺-dependent pathway in OKP cells. *J Clin Invest*. 1996;97(6):1454–1462.
 178. Nakano D, Pollock DM. Contribution of endothelin A receptors in endothelin 1-dependent natriuresis in female rats. *Hypertension*. 2009;53(2):324–330.
 179. Schiffrin EL. Role of endothelin-1 in hypertension and vascular disease. *Am J Hypertens*. 2001;14(6 Pt 2):83S–89S. [https://doi.org/10.1016/s0895-7061\(01\)02074-x](https://doi.org/10.1016/s0895-7061(01)02074-x).
 180. Touyz RM, Turgeon A, Schiffrin EL. Endothelin-A-receptor blockade improves renal function and doubles the lifespan of stroke-prone spontaneously hypertensive rats. *J Cardiovasc Pharmacol*. 2000;36(5 suppl 1):S300–S304.
 181. Allahdadi KJ, Cherng TW, Pai H, et al. Endothelin type A receptor antagonist normalizes blood pressure in rats exposed to eucapnic intermittent hypoxia. *Am J Physiol Heart Circ*

- Physiol.* 2008;295(1):H434–H440. <https://doi.org/10.1152/ajpheart.91477.2007>.
182. Schiffrin EL, Deng LY, Sventek P, Day R. Enhanced expression of endothelin-1 gene in resistance arteries in severe human essential hypertension. *J Hypertens.* 1997;15(1):57–63.
 183. Xu M, Lu YP, Hasan AA, Hochoer B. Plasma ET-1 concentrations are elevated in patients with hypertension—meta-analysis of clinical studies. *Kidney Blood Press Res.* 2017;42(2):304–313. <https://doi.org/10.1159/000477572>.
 184. Elijovich F, Laffer CL, Amador E, Gavras H, Bresnahan MR, Schiffrin EL. Regulation of plasma endothelin by salt in salt-sensitive hypertension. *Circulation.* 2001;103(2):263–268.
 185. Kittikulsuth W, Sullivan JC, Pollock DM. ET-1 actions in the kidney: evidence for sex differences. *Br J Pharmacol.* 2013;168(2):318–326.
 186. Meyers KE, Sethna C. Endothelin antagonists in hypertension and kidney disease. *Pediatr Nephrol.* 2013;28(5):711–720.
 187. Moorhouse RC, Webb DJ, Kluth DC, Dhaun N. Endothelin antagonism and its role in the treatment of hypertension. *Curr Hypertens Rep.* 2013;15(5):489–496.
 188. Maguire JJ, Kuc RE, Davenport AP. Defining the affinity and receptor sub-type selectivity of four classes of endothelin antagonists in clinically relevant human cardiovascular tissues. *Life Sci.* 2012;91(13–14):681–686.
 189. Krum H, Viskoper RJ, Lacourciere Y, Budde M, Charlon V. The effect of an endothelin-receptor antagonist, bosentan, on blood pressure in patients with essential hypertension. Bosentan Hypertension Investigators. *N Engl J Med.* 1998;338(12):784–790.
 190. Black HR, Bakris GL, Weber MA, et al. Efficacy and safety of darusentan in patients with resistant hypertension: results from a randomized, double-blind, placebo-controlled dose-ranging study. *J Clin Hypertens.* 2007;9(10):760–769.
 191. Weber MA, Black H, Bakris G, et al. A selective endothelin-receptor antagonist to reduce blood pressure in patients with treatment-resistant hypertension: a randomised, double-blind, placebo-controlled trial. *Lancet.* 2009;374(9699):1423–1431.
 192. Bakris GL, Lindholm LH, Black HR, et al. Divergent results using clinic and ambulatory blood pressures: report of a darusentan-resistant hypertension trial. *Hypertension.* 2010;56(5):824–830.
 193. Webb DJ. DORADO: opportunity postponed: lessons from studies of endothelin receptor antagonists in treatment-resistant hypertension. *Hypertension.* 2010;56(5):806–807.
 194. Gerstung M, Roth T, Dienes HP, Licht C, Fries JW. Endothelin-1 induces NF-kappaB via two independent pathways in human renal tubular epithelial cells. *Am J Nephrol.* 2007;27(3):294–300.
 195. Koyama H, Tabata T, Nishizawa Y, Inoue T, Morii H, Yamaji T. Plasma endothelin levels in patients with uraemia. *Lancet.* 1989;1(8645):991–992. [https://doi.org/10.1016/s0140-6736\(89\)92631-7](https://doi.org/10.1016/s0140-6736(89)92631-7).
 196. Zoccali C, Leonardis D, Parlongo S, Mallamaci F, Postorino M. Urinary and plasma endothelin 1 in essential hypertension and in hypertension secondary to renoparenchymal disease. *Nephrol Dial Transplant.* 1995;10(8):1320–1323.
 197. Benigni A, Zoja C, Corna D, et al. A specific endothelin subtype A receptor antagonist protects against injury in renal disease progression. *Kidney Int.* 1993;44(2):440–444.
 198. Benigni A, Colosio V, Brena C, Bruzzi I, Bertani T, Remuzzi G. Unselective inhibition of endothelin receptors reduces renal dysfunction in experimental diabetes. *Diabetes.* 1998;47(3):450–456.
 199. Sorokin A, Kohan DE. Physiology and pathology of endothelin-1 in renal mesangium. *Am J Physiol Ren Physiol.* 2003;285(4):F579–F589. <https://doi.org/10.1152/ajprenal.00019.2003>.
 200. Dufek B, Meehan DT, Delimont D, et al. Endothelin A receptor activation on mesangial cells initiates Alport glomerular disease. *Kidney Int.* 2016;90(2):300–310.
 201. Barbosa GSB, Câmara NOS, Ledesma FL, Duarte Neto AN, Dias CB. Vascular injury in glomerulopathies: the role of the endothelium. *Front Nephrol.* 2024;4:1396588.
 202. Morigi M, Buelli S, Angioletti S, et al. In response to protein load podocytes reorganize cytoskeleton and modulate endothelin-1 gene: implication for permesective dysfunction of chronic nephropathies. *Am J Pathol.* 2005;166(5):1309–1320.
 203. Morigi M, Buelli S, Zanchi C, et al. Shigatoxin-induced endothelin-1 expression in cultured podocytes autocrinally mediates actin remodeling. *Am J Pathol.* 2006;169(6):1965–1975.
 204. Fligny C, Barton M, Tharaux PL. Endothelin and podocyte injury in chronic kidney disease. *Contrib Nephrol.* 2011;172:120–138. <https://doi.org/10.1159/000328692>.
 205. Buelli S, Rosano L, Gagliardini E, et al. beta-arrestin-1 drives endothelin-1-mediated podocyte activation and sustains renal injury. *J Am Soc Nephrol.* 2014;25(3):523–533. <https://doi.org/10.1681/ASN.2013040362>.
 206. Lenoir O, Milon M, Virsolvy A, et al. Direct action of endothelin-1 on podocytes promotes diabetic glomerulosclerosis. *J Am Soc Nephrol.* 2014;25(5):1050–1062. <https://doi.org/10.1681/ASN.2013020195>.
 207. Garsen M, Lenoir O, Rops AL, et al. Endothelin-1 induces proteinuria by heparanase-mediated disruption of the glomerular glycocalyx. *J Am Soc Nephrol.* 2016;27(12):3545–3551. <https://doi.org/10.1681/ASN.2015091070>.
 208. Nabokov A, Amann K, Wagner J, Gehlen F, Munter K, Ritz E. Influence of specific and non-specific endothelin receptor antagonists on renal morphology in rats with surgical renal ablation. *Nephrol Dial Transplant.* 1996;11(3):514–520.
 209. Shimizu T, Hata S, Kuroda T, Mihara S, Fujimoto M. Different roles of two types of endothelin receptors in partial ablation-induced chronic renal failure in rats. *Eur J Pharmacol.* 1999;381(1):39–49.
 210. Tsiani E, Lekas P, Fantus IG, Dlugosz J, Whiteside C. High glucose-enhanced activation of mesangial cell p38 MAPK by ET-1, ANG II, and platelet-derived growth factor. *Am J Physiol Endocrinol Metab.* 2002;282(1):E161–E169.
 211. Dlugosz JA, Munk S, Ispanovic E, Goldberg HJ, Whiteside CI. Mesangial cell filamentous actin disassembly and hypocontractility in high glucose are mediated by PKC-zeta. *Am J Physiol Ren Physiol.* 2002;282(1):F151–F163.
 212. Whiteside CI, Dlugosz JA. Mesangial cell protein kinase C isozyme activation in the diabetic milieu. *Am J Physiol Ren Physiol.* 2002;282(6):F975–F980.
 213. Chen S, Evans T, Deng D, Cukiernik M, Chakrabarti S. Hyperhexosemia induced functional and structural changes in the kidneys: role of endothelins. *Nephron.* 2002;90(1):86–94.
 214. Jandeleit-Dahm K, Allen TJ, Youssef S, Gilbert RE, Cooper ME. Is there a role for endothelin antagonists in diabetic renal disease? *Diabetes Obes Metab.* 2000;2(1):15–24.
 215. Khamaisi M, Raz I, Shilo V, et al. Diabetes and radiocontrast media increase endothelin converting enzyme-1 in the kidney. *Kidney Int.* 2008;74(1):91–100. <https://doi.org/10.1038/ki.2008.112>.
 216. Kelly DJ, Skinner SL, Gilbert RE, Cox AJ, Cooper ME, Wilkinson-Berka JL. Effects of endothelin or angiotensin II receptor blockade on diabetes in the transgenic (mRen-2)27 rat. *Kidney Int.* 2000;57(5):1882–1894.
 217. Cosenzi A, Bernobich E, Trevisan R, Milutinovic N, Borri A, Bellini G. Nephroprotective effect of bosentan in diabetic rats. *J Cardiovasc Pharmacol.* 2003;42(6):752–756.
 218. Franzen S, Palm F. Endothelin type A receptor inhibition normalises intrarenal hypoxia in rats used as a model of type 1 diabetes by improving oxygen delivery. *Diabetologia.* 2015;58(10):2435–2442.
 219. Sugimoto K, Fujimori A, Yuyama H, Tahara A, Fujimura A. Renal protective effect of YM598, a selective endothelin type A receptor antagonist. *J Cardiovasc Pharmacol.* 2004;44(suppl 1):S451–S454.
 220. Watson AM, Li J, Schumacher C, et al. The endothelin receptor antagonist avosentan ameliorates nephropathy and atherosclerosis in diabetic apolipoprotein E knockout mice. *Diabetologia.* 2010;53(1):192–203. <https://doi.org/10.1007/s00125-009-1540-3>.
 221. Pfaff T, Thone-Reineke C, Theilig F, et al. Diabetic endothelin B receptor-deficient rats develop severe hypertension and progressive renal failure. *J Am Soc Nephrol.* 2006;17(4):1082–1089.
 222. Brownlee M. Biochemistry and molecular cell biology of diabetic complications. *Nature.* 2001;414(6865):813–820.
 223. Advani A, Gilbert RE, Thai K, et al. Expression, localization, and function of the thioredoxin system in diabetic nephropathy. *J Am Soc Nephrol.* 2009;20(4):730–741.
 224. Li L, Fink GD, Watts SW, et al. Endothelin-1 increases vascular superoxide via endothelin(A)-NADPH oxidase pathway in low-renin hypertension. *Circulation.* 2003;107(7):1053–1058.
 225. Elmarakby AA, Dabbs Loomis E, Pollock JS, Pollock DM. ETA receptor blockade attenuates hypertension and decreases reactive oxygen species in ETB receptor-deficient rats. *J Cardiovasc*

- Pharmacol.* 2004;44(suppl 1):S7–S10. <https://doi.org/10.1097/01.fjc.0000166205.66555.40>.
226. Callera GE, Touyz RM, Teixeira SA, et al. ETA receptor blockade decreases vascular superoxide generation in DOCA-salt hypertension. *Hypertension*. 2003;42(4):811–817.
 227. Sedek MH, Llinas MT, Drummond H, et al. Role of reactive oxygen species in endothelin-induced hypertension. *Hypertension*. 2003;42(4):806–810.
 228. Sasser JM, Sullivan JC, Hobbs JL, et al. Endothelin A receptor blockade reduces diabetic renal injury via an anti-inflammatory mechanism. *J Am Soc Nephrol*. 2007;18(1):143–154.
 229. Grenda R, Wuhl E, Litwin M, et al. Urinary excretion of endothelin-1 (ET-1), transforming growth factor- β (TGF- β) and vascular endothelial growth factor (VEGF165) in paediatric chronic kidney diseases: results of the ESCAPE trial. *Nephrol Dial Transplant*. 2007;22(12):3487–3494.
 230. Goddard J, Johnston NR, Cumming AD, Webb DJ. Fractional urinary excretion of endothelin-1 is reduced by acute ETB receptor blockade. *Am J Physiol Ren Physiol*. 2007;293(5):F1433–F1438.
 231. Smeijer JD, Kohan DE, Dhaun N, Noronha IL, Liew A, Heerspink HJL. Endothelin receptor antagonists in chronic kidney disease. *Nat Rev Nephrol*. 2024; z.
 232. Goddard J, Johnston NR, Hand MF, et al. Endothelin-A receptor antagonism reduces blood pressure and increases renal blood flow in hypertensive patients with chronic renal failure: a comparison of selective and combined endothelin receptor blockade. *Circulation*. 2004;109(9):1186–1193.
 233. Dhaun N, Macintyre IM, Melville V, et al. Blood pressure-independent reduction in proteinuria and arterial stiffness after acute endothelin-A receptor antagonism in chronic kidney disease. *Hypertension*. 2009;54(1):113–119.
 234. Dhaun N, MacIntyre IM, Kerr D, et al. Selective endothelin-A receptor antagonism reduces proteinuria, blood pressure, and arterial stiffness in chronic proteinuric kidney disease. *Hypertension*. 2011;57(4):772–779.
 235. Dhaun N, Moorhouse R, MacIntyre IM, et al. Diurnal variation in blood pressure and arterial stiffness in chronic kidney disease: the role of endothelin-1. *Hypertension*. 2014;64(2):296–304.
 236. Heerspink HJL, Kiyosue A, Wheeler DC, et al. Zibotentan in combination with dapagliflozin compared with dapagliflozin in patients with chronic kidney disease (ZENITH-CKD): a multicentre, randomised, active-controlled, phase 2b, clinical trial. *Lancet*. 2023;402(10416):2004–2017.
 237. Smeijer JD, Wasehuus VS, Dhaun N, et al. Effects of zibotentan alone and in combination with dapagliflozin on fluid retention in patients with CKD. *J Am Soc Nephrol*. 2024;35(10):1381–1390.
 238. Wenzel RR, Litke T, Kuranoff S, et al. Avasentan reduces albumin excretion in diabetics with macroalbuminuria. *J Am Soc Nephrol*. 2009;20(3):655–664.
 239. Mann JF, Green D, Jamerson K, et al. Avasentan for overt diabetic nephropathy. *J Am Soc Nephrol*. 2010;21(3):527–535.
 240. Empitu MA, Rinastiti P, Kadariswantiningsih IN. Targeting endothelin signaling in podocyte injury and diabetic nephropathy-diabetic kidney disease. *J Nephrol*. 2025;38(1):49–60.
 241. de Zeeuw D, Coll B, Andress D, et al. The endothelin antagonist atrasentan lowers residual albuminuria in patients with type 2 diabetic nephropathy. *J Am Soc Nephrol*. 2014;25(5):1083–1093. <https://doi.org/10.1681/ASN.2013080830>.
 242. Kohan DE, Lambers Heerspink HJ, Coll B, et al. Predictors of atrasentan-associated fluid retention and change in albuminuria in patients with diabetic nephropathy. *Clin J Am Soc Nephrol*. 2015;10(9):1568–1574.
 243. Heerspink HJL, Parving HH, Andress DL, et al. Atrasentan and renal events in patients with type 2 diabetes and chronic kidney disease (SONAR): a double-blind, randomised, placebo-controlled trial. *Lancet*. 2019;393(10184):1937–1947.
 244. Waijjer SW, Gansevoort RT, Bakris GL, et al. The effect of atrasentan on kidney and heart failure outcomes by baseline albuminuria and kidney function: a post hoc analysis of the SONAR randomised trial. *Clin J Am Soc Nephrol*. 2021;16(12):1824–1832.
 245. Parvanova A, van der Meer IM, Iliev I, et al. Effect on blood pressure of combined inhibition of endothelin-converting enzyme and neutral endopeptidase with daglutril in patients with type 2 diabetes who have albuminuria: a randomised, crossover, double-blind, placebo-controlled trial. *Lancet Diabetes Endocrinol*. 2013;1(1):19–27. [https://doi.org/10.1016/S2213-8587\(13\)70029-9](https://doi.org/10.1016/S2213-8587(13)70029-9).
 246. Czopek A, Moorhouse R, Webb DJ, Dhaun N. Therapeutic potential of endothelin receptor antagonism in kidney disease. *Am J Physiol Regul Integr Comp Physiol*. 2016;310(5):R388–R397.
 247. Kohan DE, Barton M. Endothelin and endothelin antagonists in chronic kidney disease. *Kidney Int*. 2014;86(5):896–904.
 248. Floege J, Bernier-Jean A, Barratt J, Rovin B. Treatment of patients with IgA nephropathy: a call for a new paradigm. *Kidney Int*. 2025;107(4):640–651.
 249. Rovin BH, Barratt J, Heerspink HJL, et al. Efficacy and safety of sparsentan versus irbesartan in patients with IgA nephropathy (PROTECT): 2-year results from a randomised, active-controlled, phase 3 trial. *Lancet*. 2023;402(10417):2077–2090.
 250. Al-Horani RA. Potential new treatments for chronic kidney diseases: a concise review. *Curr Pharm Des*. 2025.
 251. Heerspink HJL, Jardine M, Kohan DE, et al. Atrasentan in patients with IgA nephropathy. *N Engl J Med*. 2025;392(6):544–554.
 252. Kasztan M, Fox BM, Speed JS, et al. Long-term endothelin-A receptor antagonism provides robust renal protection in humanized sickle cell disease mice. *J Am Soc Nephrol*. 2017;28(8):2443–2458.
 253. Kelsen S, Hall JE, Chade AR. Endothelin-A receptor blockade slows the progression of renal injury in experimental renovascular disease. *Am J Physiol Ren Physiol*. 2011;301(1):F218–F225.
 254. Chade AR, Stewart NJ, Peavy PR. Disparate effects of single endothelin-A and -B receptor blocker therapy on the progression of renal injury in advanced renovascular disease. *Kidney Int*. 2014;85(4):833–844.
 255. Chade AR, Tullos N, Stewart NJ, Surles B. Endothelin-a receptor antagonism after renal angioplasty enhances renal recovery in renovascular disease. *J Am Soc Nephrol*. 2015;26(5):1071–1080.
 256. Tschakowsky K, Sagner S, Lehnert N, Kaul M, Ritter J. Endothelin in septic patients: effects on cardiovascular and renal function and its relationship to proinflammatory cytokines. *Crit Care Med*. 2000;28(6):1854–1860.
 257. Nitsescu N, Grimberg E, Ricksten SE, Herlitz H, Guron G. Endothelin B receptors preserve renal blood flow in a normotensive model of endotoxin-induced acute kidney dysfunction. *Shock*. 2008;29(3):402–409.
 258. Fenhammar J, Andersson A, Frithiof R, et al. The endothelin receptor antagonist tezosentan improves renal microcirculation in a porcine model of endotoxemic shock. *Acta Anaesthesiol Scand*. 2008;52(10):1385–1393.
 259. Zager RA, Johnson AC, Andress D, Becker K. Progressive endothelin-1 gene activation initiates chronic/end-stage renal disease following experimental ischemic/reperfusion injury. *Kidney Int*. 2013;84(4):703–712.
 260. Dhaun N, Lilikarntakul P, Macintyre IM, et al. Urinary endothelin-1 in chronic kidney disease and as a marker of disease activity in lupus nephritis. *Am J Physiol Ren Physiol*. 2009;296(6):F1477–F1483.
 261. Yoshio T, Masuyama J, Mimori A, Takeda A, Minota S, Kano S. Endothelin-1 release from cultured endothelial cells induced by sera from patients with systemic lupus erythematosus. *Ann Rheum Dis*. 1995;54(5):361–365.
 262. Nakamura T, Ebihara I, Tomino Y, Koide H. Effect of a specific endothelin A receptor antagonist on murine lupus nephritis. *Kidney Int*. 1995;47(2):481–489.
 263. Trachtman H, Nelson P, Adler S, et al. DUET: a phase 2 study evaluating the efficacy and safety of sparsentan in patients with FSGS. *J Am Soc Nephrol*. 2018;29(11):2745–2754.
 264. Rheault MN, Alpers CE, Barratt J, et al. Sparsentan versus irbesartan in focal segmental glomerulosclerosis. *N Engl J Med*. 2023;389(26):2436–2445.
 265. Stern EP, Host LV, Wanjiku I, et al. Zibotentan in systemic sclerosis-associated chronic kidney disease: a phase II randomised placebo-controlled trial. *Arthritis Res Ther*. 2022;24(1):130.
 266. Cardenas A, Gines P. Hepatorenal syndrome. *Clin Liver Dis*. 2006;10(2):371–385, ix–x.
 267. Wong F, Moore K, Dingemans J, Jalan R. Lack of renal improvement with nonselective endothelin antagonism with tezosentan in type 2 hepatorenal syndrome. *Hepatology*. 2008;47(1):160–168.
 268. Speed JS, Pollock DM. Endothelin, kidney disease, and hypertension. *Hypertension*. 2013;61(6):1142–1145.
 269. Murphy SR, LaMarca BB, Cockrell K, Granger JP. Role of endothelin in mediating soluble fms-like tyrosine kinase 1-induced hypertension in pregnant rats. *Hypertension*. 2010;55(2):394–398.

270. LaMarca B, Speed J, Fournier L, et al. Hypertension in response to chronic reductions in uterine perfusion in pregnant rats: effect of tumor necrosis factor- α blockade. *Hypertension*. 2008;52(6):1161–1167.
271. Tam TKB, George E, Cockrell K, et al. Endothelin type A receptor antagonist attenuates placental ischemia-induced hypertension and uterine vascular resistance. *Am J Obstet Gynecol*. 2011;204(4):330–330.
272. Kohan DE, Rossi NF, Inscho EW, Pollock DM. Regulation of blood pressure and salt homeostasis by endothelin. *Physiol Rev*. 2011;91(1):1–77.
273. Stuart D, Chapman M, Rees S, Woodward S, Kohan DE. Myocardial, smooth muscle, nephron, and collecting duct gene targeting reveals the organ sites of endothelin A receptor antagonist fluid retention. *J Pharmacol Exp Ther*. 2013;346(2):182–189.
274. Humbert M, Segal ES, Kiely DG, Carlsen J, Schwierin B, Hoepfer MM. Results of European post-marketing surveillance of bosentan in pulmonary hypertension. *Eur Respir J*. 2007;30(2):338–344.
275. Nakov R, Pfarr E, Eberle S, Investigators H. Darusentan: an effective endothelinA receptor antagonist for treatment of hypertension. *Am J Hypertens*. 2002;15(7 Pt 1):583–589.
276. Kangawa K, Tawaragi Y, Oikawa S, et al. Identification of rat gamma atrial natriuretic polypeptide and characterization of the cDNA encoding its precursor. *Nature*. 1984;312(5990):152–155.
277. Woodard GE, Rosado JA. Recent advances in natriuretic peptide research. *J Cell Mol Med*. 2007;11(6):1263–1271.
278. Sudoh T, Kangawa K, Minamino N, Matsuo H. A new natriuretic peptide in porcine brain. *Nature*. 1988;332(6159):78–81.
279. Sudoh T, Minamino N, Kangawa K, Matsuo H. C-type natriuretic peptide (CNP): a new member of natriuretic peptide family identified in porcine brain. *Biochem Biophys Res Commun*. 1990;168(2):863–870.
280. Schweitz H, Vigne P, Moinier D, Frelin C, Lazdunski M. A new member of the natriuretic peptide family is present in the venom of the green mamba (*Dendroaspis angusticeps*). *J Biol Chem*. 1992;267(20):13928–13932.
281. Schulz-Knappe P, Forssmann K, Herbst F, Hock D, Pipkorn R, Forssmann WG. Isolation and structural analysis of “urodilatin”, a new peptide of the cardiodilatin-(ANP)-family, extracted from human urine. *Klin Wochenschr*. 1988;66(17):752–759.
282. Silver MA. The natriuretic peptide system: kidney and cardiovascular effects. *Curr Opin Nephrol Hypertens*. 2006;15(1):14–21.
283. Vesely DL, Perez-Lamboy GI, Schocken DD. Long-acting natriuretic peptide, vessel dilator, and kaliuretic peptide enhance urinary excretion rate of albumin, total protein, and beta(2)-microglobulin in patients with congestive heart failure. *J Card Fail*. 2001;7(1):55–63.
284. de Lemos JA, McGuire DK, Drazner MH. B-type natriuretic peptide in cardiovascular disease. *Lancet*. 2003;362(9380):316–322.
285. Valli N, Gobinet A, Bordenave L. Review of 10 years of the clinical use of brain natriuretic peptide in cardiology. *J Lab Clin Med*. 1999;134(5):437–444.
286. McCullough PA, Omland T, Maisel AS. B-type natriuretic peptides: a diagnostic breakthrough for clinicians. *Rev Cardiovasc Med*. 2003;4(2):72–80.
287. Vanderheyden M, Bartunek J, Goethals M. Brain and other natriuretic peptides: molecular aspects. *Eur J Heart Fail*. 2004;6(3):261–268.
288. Woodard GE, Rosado JA, Brown J. Expression and control of C-type natriuretic peptide in rat vascular smooth muscle cells. *Am J Physiol Regul Integr Comp Physiol*. 2002;282(1):R156–R165.
289. Armaly Z, Assady S, Abassi Z. Corin: a new player in the regulation of salt-water balance and blood pressure. *Curr Opin Nephrol Hypertens*. 2013;22(6):713–722.
290. Yan W, Wu F, Morser J, Wu Q. Corin, a transmembrane cardiac serine protease, acts as a pro-atrial natriuretic peptide-converting enzyme. *Proc Natl Acad Sci USA*. 2000;97(15):8525–8529. <https://doi.org/10.1073/pnas.150149097>.
291. Wu C, Wu F, Pan J, Morser J, Wu Q. Furin-mediated processing of Pro-C-type natriuretic peptide. *J Biol Chem*. 2003;278(28):25847–25852. <https://doi.org/10.1074/jbc.M301223200>.
292. Semenov AG, Tamm NN, Seferian KR, et al. Processing of pro-B-type natriuretic peptide: furin and corin as candidate convertases. *Clin Chem*. 2010;56(7):1166–1176. <https://doi.org/10.1373/clinchem.2010.143883>.
293. Theilig F, Wu Q. ANP-induced signaling cascade and its implications in renal pathophysiology. *Am J Physiol Ren Physiol*. 2015;308(10):F1047–F1055.
294. Chan JC, Knudson O, Wu F, Morser J, Dole WP, Wu Q. Hypertension in mice lacking the proatrial natriuretic peptide convertase corin. *Proc Natl Acad Sci USA*. 2005;102(3):785–790.
295. Buckley CL, Stokes AJ. Corin-deficient W-sh mice poorly tolerate increased cardiac afterload. *Regul Pept*. 2011;172(1–3):44–50. <https://doi.org/10.1016/j.regpep.2011.08.006>.
296. Polzin D, Kaminski HJ, Kastner C, et al. Decreased renal corin expression contributes to sodium retention in proteinuric kidney diseases. *Kidney Int*. 2010;78(7):650–659.
297. Fang C, Shen L, Dong L, et al. Reduced urinary corin levels in patients with chronic kidney disease. *Clin Sci*. 2013;124(12):709–717. <https://doi.org/10.1042/cs20120517>.
298. Cui Y, Wang W, Dong N, et al. Role of corin in trophoblast invasion and uterine spiral artery remodelling in pregnancy. *Nature*. 2012;484(7393):246–250.
299. Richards AM, Lainchbury JG, Nicholls MG, Cameron AV, Yandle TG. Dendroaspis natriuretic peptide: endogenous or dubious?. *Lancet*. 2002;359(9300):5–6.
300. Kim SW, Lee JU, Kim SZ, Choi KC, Kim NH, Cho KW. Enhanced Dendroaspis natriuretic peptide immunoreactivity in experimental ureteral obstruction. *Nephron*. 2002;92(2):369–372.
301. Kim SM, Kim SY, Kim SH, Cho KW, Kim SZ. Renal actions of dendroaspis natriuretic peptide in rabbits. *Peptides*. 2012;33(1):59–66.
302. Kim JH, Yang SH, Yu MY, Lee HK, Kim SY, Kim SH. Dendroaspis natriuretic peptide system and its paracrine function in rat colon. *Regul Pept*. 2004;120(1–3):93–98.
303. Woodard GE, Rosado JA, Brown J. Dendroaspis natriuretic peptide-like immunoreactivity and its regulation in rat aortic vascular smooth muscle. *Peptides*. 2002;23(1):23–29.
304. Piao FL, Park SH, Han JH, Cao C, Kim SZ, Kim SH. Dendroaspis natriuretic peptide and its functions in pig ovarian granulosa cells. *Regul Pept*. 2004;118(3):193–198.
305. Singh G, Maguire JJ, Kuc RE, Skepper JN, Fidock M, Davenport AP. Characterization of the snake venom ligand [125I]-DNP binding to natriuretic peptide receptor-A in human artery and potent DNP mediated vasodilatation. *Br J Pharmacol*. 2006;149(7):838–844.
306. Johns DG, Ao Z, Heidrich BJ, et al. Dendroaspis natriuretic peptide binds to the natriuretic peptide clearance receptor. *Biochem Biophys Res Commun*. 2007;358(1):145–149.
307. Chen HH, Lainchbury JG, Burnett Jr JC. Natriuretic peptide receptors and neutral endopeptidase in mediating the renal actions of a new therapeutic synthetic natriuretic peptide dendroaspis natriuretic peptide. *J Am Coll Cardiol*. 2002;40(6):1186–1191.
308. Lisy O, Jougasaki M, Heublein DM, et al. Renal actions of synthetic dendroaspis natriuretic peptide. *Kidney Int*. 1999;56(2):502–508.
309. Lisy O, Lainchbury JG, Leskinen H, Burnett Jr JC. Therapeutic actions of a new synthetic vasoactive and natriuretic peptide, dendroaspis natriuretic peptide, in experimental severe congestive heart failure. *Hypertension*. 2001;37(4):1089–1094.
310. Forssmann W, Meyer M, Forssmann K. The renal urodilatin system: clinical implications. *Cardiovasc Res*. 2001;51(3):450–462.
311. Shin SJ, Lee YJ, Tan MS, Hsieh TJ, Tsai JH. Increased atrial natriuretic peptide mRNA expression in the kidney of diabetic rats. *Kidney Int*. 1997;51(4):1100–1105.
312. Totsune K, Mackenzie HS, Totsune H, Troy JL, Lytton J, Brenner BM. Upregulation of atrial natriuretic peptide gene expression in remnant kidney of rats with reduced renal mass. *J Am Soc Nephrol*. 1998;9(9):1613–1619.
313. Kone BC. Molecular biology of natriuretic peptides and nitric oxide synthases. *Cardiovasc Res*. 2001;51(3):429–441.
314. Matsukawa N, Grzesik WJ, Takahashi N, et al. The natriuretic peptide clearance receptor locally modulates the physiological effects of the natriuretic peptide system. *Proc Natl Acad Sci USA*. 1999;96(13):7403–7408.
315. Murthy KS, Teng BQ, Zhou H, Jin JG, Grider JR, Makhlof GM. G(i-1)/G(i-2)-dependent signaling by single-transmembrane natriuretic peptide clearance receptor. *Am J Physiol Gastrointest Liver Physiol*. 2000;278(6):G974–G980.
316. Corti R, Burnett Jr JC, Rouleau JL, Ruschitzka F, Luscher TF. Vaso-peptidase inhibitors: a new therapeutic concept in cardiovascular disease? *Circulation*. 2001;104(15):1856–1862.

317. Nishikimi T, Inaba-Iemura C, Ishimura K, et al. Natriuretic peptide/natriuretic peptide receptor-A (NPR-A) system has inhibitory effects in renal fibrosis in mice. *Regul Pept.* 2009;154(1-3):44-53.
318. Lo CS, Chen ZH, Hsieh TJ, Shin SJ. Atrial natriuretic peptide attenuates high glucose-activated transforming growth factor-beta, Smad and collagen synthesis in renal proximal tubular cells. *J Cell Biochem.* 2008;103(6):1999-2009.
319. John SW, Kregel JH, Oliver PM, et al. Genetic decreases in atrial natriuretic peptide and salt-sensitive hypertension. *Science.* 1995;267(5198):679-681.
320. Steinhilber ME, Cochran KL, Field LJ. Hypotension in transgenic mice expressing atrial natriuretic factor fusion genes. *Hypertension.* 1990;16(3):301-307.
321. Rubattu S, Evangelista A, Barbato D, et al. Atrial natriuretic peptide (ANP) gene promoter variant and increased susceptibility to early development of hypertension in humans. *J Hum Hypertens.* 2007;21(10):822-824.
322. Levin ER, Gardner DG, Samson WK. Natriuretic peptides. *N Engl J Med.* 1998;339(5):321-328. <https://doi.org/10.1056/NEJM199807303390507>.
323. Vesely DL. Natriuretic peptides and acute renal failure. *Am J Physiol Ren Physiol.* 2003;285(2):F167-F177.
324. Garcha RS, Hughes AD. CNP, but not ANP or BNP, relax human isolated subcutaneous resistance arteries by an action involving cyclic GMP and BKCa channels. *J Renin Angiotensin Aldosterone Syst.* 2006;7(2):87-91.
325. Nakayama T, Soma M, Takahashi Y, Rehemudula D, Kanmatsuse K, Furuya K. Functional deletion mutation of the 5'-flanking region of type A human natriuretic peptide receptor gene and its association with essential hypertension and left ventricular hypertrophy in the Japanese. *Circ Res.* 2000;86(8):841-845.
326. Rubattu S, Bigatti G, Evangelista A, et al. Association of atrial natriuretic peptide and type A natriuretic peptide receptor gene polymorphisms with left ventricular mass in human essential hypertension. *J Am Coll Cardiol.* 2006;48(3):499-505.
327. Kapoun AM, Liang F, O'Young G, et al. B-type natriuretic peptide exerts broad functional opposition to transforming growth factor-beta in primary human cardiac fibroblasts: fibrosis, myofibroblast conversion, proliferation, and inflammation. *Circ Res.* 2004;94(4):453-461.
328. Tamura N, Ogawa Y, Chusho H, et al. Cardiac fibrosis in mice lacking brain natriuretic peptide. *Proc Natl Acad Sci USA.* 2000;97(8):4239-4244. <https://doi.org/10.1073/pnas.070371497>.
329. Del Ry S. C-type natriuretic peptide: a new cardiac mediator. *Peptides.* 2013;40:93-98.
330. Kalra PR, Clague JR, Coats AJ, Anker SD, Poole-Wilson PA, Struthers AD. C-type natriuretic peptide production by the human kidney is blunted in chronic heart failure. *Clin Sci.* 2009;118(1):71-77.
331. Hu P, Zhang XC, Kong HB, Xia X, Hu B, Qin YH. Exogenous C-type natriuretic peptide infusion ameliorates unilateral ureteral obstruction-induced tubulointerstitial fibrosis in rats. *Lab Invest.* 2015;95(3):263-272.
332. Wang TJ, Larson MG, Levy D, et al. Impact of obesity on plasma natriuretic peptide levels. *Circulation.* 2004;109(5):594-600.
333. Olsen MH, Hansen TW, Christensen MK, et al. N-terminal pro brain natriuretic peptide is inversely related to metabolic cardiovascular risk factors and the metabolic syndrome. *Hypertension.* 2005;46(4):660-666.
334. Rubattu S, Sciarretta S, Ciavarella GM, et al. Reduced levels of N-terminal-proatrial natriuretic peptide in hypertensive patients with metabolic syndrome and their relationship with left ventricular mass. *J Hypertens.* 2007;25(4):833-839.
335. Wang TJ, Larson MG, Keyes MJ, Levy D, Benjamin EJ, Vasan RS. Association of plasma natriuretic peptide levels with metabolic risk factors in ambulatory individuals. *Circulation.* 2007;115(11):1345-1353.
336. Birkenfeld AL, Boschmann M, Moro C, et al. Lipid mobilization with physiological atrial natriuretic peptide concentrations in humans. *J Clin Endocrinol Metab.* 2005;90(6):3622-3628.
337. Sarzani R, Marcucci P, Salvi F, et al. Angiotensin II stimulates and atrial natriuretic peptide inhibits human visceral adipocyte growth. *Int J Obes.* 2008;32(2):259-267.
338. Galitzky J, Sengenès C, Thalamos C, et al. The lipid-mobilizing effect of atrial natriuretic peptide is unrelated to sympathetic nervous system activation or obesity in young men. *J Lipid Res.* 2001;42(4):536-544.
339. Sengenès C, Bouloumié A, Hauner H, et al. Involvement of a cGMP-dependent pathway in the natriuretic peptide-mediated hormone-sensitive lipase phosphorylation in human adipocytes. *J Biol Chem.* 2003;278(49):48617-48626.
340. Pejchalova K, Krejci P, Wilcox WR. C-natriuretic peptide: an important regulator of cartilage. *Mol Genet Metab.* 2007;92(3):210-215.
341. Tamura N, Doolittle LK, Hammer RE, Shelton JM, Richardson JA, Garbers DL. Critical roles of the guanylyl cyclase B receptor in endochondral ossification and development of female reproductive organs. *Proc Natl Acad Sci USA.* 2004;101(49):17300-17305.
342. Bartels CF, Bukulmez H, Padayatti P, et al. Mutations in the transmembrane natriuretic peptide receptor NPR-B impair skeletal growth and cause acromesomelic dysplasia, type Maroteaux. *Am J Hum Genet.* 2004;75(1):27-34.
343. Ueda Y, Yasoda A, Yamashita Y, et al. C-type natriuretic peptide restores impaired skeletal growth in a murine model of glucocorticoid-induced growth retardation. *Bone.* 2016;92:157-167.
344. Legeai-Mallet L. C-type natriuretic peptide analog as therapy for achondroplasia. *Endocr Dev.* 2016;30:98-105.
345. Doust JA, Glasziou PP, Pietrzak E, Dobson AJ. A systematic review of the diagnostic accuracy of natriuretic peptides for heart failure. *Arch Intern Med.* 2004;164(18):1978-1984.
346. Trof RJ, Di Maggio F, Leemreis J, Groeneveld AB. Biomarkers of acute renal injury and renal failure. *Shock.* 2006;26(3):245-253. <https://doi.org/10.1097/01.shk.0000225415.5969694.ce>.
347. Dieplinger B, Mueller T, Kollerits B, et al. Pro-A-type natriuretic peptide and pro-adrenomedullin predict progression of chronic kidney disease: the MMKD Study. *Kidney Int.* 2009;75(4):408-414.
348. Artunc F, Mueller C, Breidhardt T, et al. Comparison of the diagnostic performance of three natriuretic peptides in hemodialysis patients: which is the appropriate biomarker? *Kidney Blood Press Res.* 2012;36(1):172-181.
349. Siriwardena M, Kleffmann T, Ruygrok P, et al. B-type natriuretic peptide signal peptide circulates in human blood: evaluation as a potential biomarker of cardiac ischemia. *Circulation.* 2010;122(3):255-264.
350. Pemberton CJ, Siriwardena M, Kleffmann T, et al. First identification of circulating prepro-A-type natriuretic peptide (preproANP) signal peptide fragments in humans: initial assessment as cardiovascular biomarkers. *Clin Chem.* 2012;58(4):757-767.
351. Khalifeh N, Haider D, Horl WH. Natriuretic peptides in chronic kidney disease and during renal replacement therapy: an update. *J Invest Med.* 2009;57(1):33-39.
352. Lee DS, Vasan RS. Novel markers for heart failure diagnosis and prognosis. *Curr Opin Cardiol.* 2005;20(3):201-210.
353. Silver MA, Maisel A, Yancy CW, et al. BNP Consensus Panel 2004: a clinical approach for the diagnostic, prognostic, screening, treatment monitoring, and therapeutic roles of natriuretic peptides in cardiovascular diseases. *Congest Heart Fail.* 2004;10(5 suppl 3):1-30.
354. Januzzi Jr JL, Camargo CA, Anwaruddin S, et al. The N-terminal Pro-BNP investigation of dyspnea in the emergency department (PRIDE) study. *Am J Cardiol.* 2005;95(8):948-954.
355. Ledwidge M, Gallagher J, Conlon C, et al. Natriuretic peptide-based screening and collaborative care for heart failure: the STOP-HF randomized trial. *JAMA.* 2013;310(1):66-74.
356. Huelsmann M, Neuhold S, Resl M, et al. PONTIAC (NT-proBNP selected prevention of cardiac events in a population of diabetic patients without a history of cardiac disease): a prospective randomized controlled trial. *J Am Coll Cardiol.* 2013;62(15):1365-1372.
357. McLellan J, Heneghan CJ, Perera R, et al. B-type natriuretic peptide-guided treatment for heart failure. *Cochrane Database Syst Rev.* 2016;12(12):CD008966. <https://doi.org/10.1002/14651858.CD008966.pub2>.
358. McKie PM, Burnett Jr JC. B-type natriuretic peptide as a biomarker beyond heart failure: speculations and opportunities. *Mayo Clin Proc.* 2005;80(8):1029-1036.
359. Krim SR, Vivo RP, Krim NR, et al. Racial/Ethnic differences in B-type natriuretic peptide levels and their association with care and outcomes among patients hospitalized with heart failure: findings from Get with the Guidelines-Heart Failure. *JACC Heart Fail.* 2013;1(4):345-352.

360. Munagala VK, Burnett Jr JC, Redfield MM. The natriuretic peptides in cardiovascular medicine. *Curr Probl Cardiol*. 2004;29(12):707–769. <https://doi.org/10.1016/j.cpcardiol.2004.07.002>.
361. McDonald K, Dahlström U, Aspromonte N, et al. B-type natriuretic peptide: application in the community. *Congest Heart Fail*. 2008;14(4 suppl 1):12–16.
362. van Kimmenade RR, Januzzi Jr JL, Bakker JA, et al. Renal clearance of B-type natriuretic peptide and amino terminal pro-B-type natriuretic peptide: a mechanistic study in hypertensive subjects. *J Am Coll Cardiol*. 2009;53(10):884–890.
363. Vanderheyden M, Bartunek J, Filippatos G, Goethals M, Vlem BV, Maisel A. Cardiovascular disease in patients with chronic renal impairment: role of natriuretic peptides. *Congest Heart Fail*. 2008;14(4 suppl 1):38–42.
364. deFilippi CR, Seliger SL, Maynard S, Christenson RH. Impact of renal disease on natriuretic peptide testing for diagnosing decompensated heart failure and predicting mortality. *Clin Chem*. 2007;53(8):1511–1519.
365. Khan IA, Fink J, Nass C, Chen H, Christenson R, deFilippi CR. N-terminal pro-B-type natriuretic peptide and B-type natriuretic peptide for identifying coronary artery disease and left ventricular hypertrophy in ambulatory chronic kidney disease patients. *Am J Cardiol*. 2006;97(10):1530–1534.
366. Spanaus KS, Kronenberg F, Ritz E, et al. B-type natriuretic peptide concentrations predict the progression of nondiabetic chronic kidney disease: the Mild-to-Moderate Kidney Disease Study. *Clin Chem*. 2007;53(7):1264–1272.
367. Yasuda K, Kimura T, Sasaki K, et al. Plasma B-type natriuretic peptide level predicts kidney prognosis in patients with predialysis chronic kidney disease. *Nephrol Dial Transplant*. 2012;27(10):3885–3891.
368. Vickery S, Webb MC, Price CP, John RI, Abbas NA, Lamb EJ. Prognostic value of cardiac biomarkers for death in a non-dialysis chronic kidney disease population. *Nephrol Dial Transplant*. 2008;23(11):3546–3553.
369. Astor BC, Yi S, Hiremath L, et al. N-terminal prohormone brain natriuretic peptide as a predictor of cardiovascular disease and mortality in blacks with hypertensive kidney disease: the African American Study of Kidney Disease and Hypertension (AASK). *Circulation*. 2008;117(13):1685–1692.
370. Rinat C, Becker-Cohen R, Nir A, et al. B-type natriuretic peptides are reliable markers of cardiac strain in CKD pediatric patients. *Pediatr Nephrol*. 2012;27(4):617–625.
371. Wahl HG, Graf S, Renz H, Fassbinder W. Elimination of the cardiac natriuretic peptides B-type natriuretic peptide (BNP) and N-terminal proBNP by hemodialysis. *Clin Chem*. 2004;50(6):1071–1074.
372. Mehta RL. Continuous renal replacement therapy in the critically ill patient. *Kidney Int*. 2005;67(2):781–795.
373. Obineche EN, Pathan JY, Fisher S, et al. Natriuretic peptide and adrenomedullin levels in chronic renal failure and effects of peritoneal dialysis. *Kidney Int*. 2006;69(1):152–156.
374. Lee JA, Kim DH, Yoo SJ, Oh DJ, Yu SH, Kang ET. Association between serum n-terminal pro-brain natriuretic peptide concentration and left ventricular dysfunction and extracellular water in continuous ambulatory peritoneal dialysis patients. *Perit Dial Int*. 2006;26(3):360–365.
375. Wang AY, Lam CW, Yu CM, et al. N-terminal pro-brain natriuretic peptide: an independent risk predictor of cardiovascular congestion, mortality, and adverse cardiovascular outcomes in chronic peritoneal dialysis patients. *J Am Soc Nephrol*. 2007;18(1):321–330.
376. Wang AY. Clinical utility of natriuretic peptides in dialysis patients. *Semin Dial*. 2012;25(3):326–333.
377. Papakrivopoulou E, Lillywhite S, Davenport A. Is N-terminal pro-brain-type natriuretic peptide a clinically useful biomarker of volume overload in peritoneal dialysis patients? *Nephrol Dial Transplant*. 2012;27(1):396–401.
378. Zoccali C, Mallamaci F, Benedetto FA, et al. Cardiac natriuretic peptides are related to left ventricular mass and function and predict mortality in dialysis patients. *J Am Soc Nephrol*. 2001;12(7):1508–1515.
379. Guo Q, Barany P, Qureshi AR, et al. N-terminal pro-brain natriuretic peptide independently predicts protein energy wasting and is associated with all-cause mortality in prevalent HD patients. *Am J Nephrol*. 2009;29(6):516–523.
380. Roberts MA, Srivastava PM, Macmillan N, et al. B-type natriuretic peptides strongly predict mortality in patients who are treated with long-term dialysis. *Clin J Am Soc Nephrol*. 2008;3(4):1057–1065.
381. Gutierrez OM, Tamez H, Bhan I, et al. N-terminal pro-B-type natriuretic peptide (NT-proBNP) concentrations in hemodialysis patients: prognostic value of baseline and follow-up measurements. *Clin Chem*. 2008;54(8):1339–1348.
382. Niizuma S, Iwanaga Y, Yahata T, et al. Plasma B-type natriuretic peptide levels reflect the presence and severity of stable coronary artery disease in chronic haemodialysis patients. *Nephrol Dial Transplant*. 2009;24(2):597–603.
383. Breidhardt T, Kalbermatter S, Socrates T, et al. Increasing B-type natriuretic peptide levels predict mortality in unselected haemodialysis patients. *Eur J Heart Fail*. 2011;13(8):860–867.
384. Sangaralingham SJ, McKie PM, Ichiki T, et al. Circulating C-type natriuretic peptide and its relationship to cardiovascular disease in the general population. *Hypertension*. 2015;65(6):1187–1194. <https://doi.org/10.1161/HYPERTENSIONAHA.115.05366>.
385. Suwa M, Seino Y, Nomachi Y, Matsuki S, Funahashi K. Multi-center prospective investigation on efficacy and safety of carperitide for acute heart failure in the ‘real world’ of therapy. *Circ J*. 2005;69(3):283–290.
386. Mitaka C, Kudo T, Jibiki M, et al. Effects of human atrial natriuretic peptide on renal function in patients undergoing abdominal aortic aneurysm repair. *Crit Care Med*. 2008;36(3):745–751.
387. Morikawa S, Sone T, Tsuboi H, et al. Renal protective effects and the prevention of contrast-induced nephropathy by atrial natriuretic peptide. *J Am Coll Cardiol*. 2009;53(12):1040–1046.
388. Nigwekar SU, Navaneethan SD, Parikh CR, Hix JK. Atrial natriuretic peptide for management of acute kidney injury: a systematic review and meta-analysis. *Clin J Am Soc Nephrol*. 2009;4(2):261–272.
389. Mitaka C, Kudo T, Haraguchi G, Tomita M. Cardiovascular and renal effects of carperitide and nesiritide in cardiovascular surgery patients: a systematic review and meta-analysis. *Crit Care*. 2011;15(5):R258.
390. Sezai A, Nakata K, Iida M, et al. Results of low-dose carperitide infusion in high-risk patients undergoing coronary artery bypass grafting. *Ann Thorac Surg*. 2013;96(1):119–126.
391. Sezai A, Hata M, Niino T, et al. Results of low-dose human atrial natriuretic peptide infusion in nondialysis patients with chronic kidney disease undergoing coronary artery bypass grafting: the NU-HIT (Nihon University working group study of low-dose HANP Infusion Therapy during cardiac surgery) trial for CKD. *J Am Coll Cardiol*. 2011;58(9):897–903.
392. Mitaka C, Ohnuma T, Murayama T, et al. Effects of low-dose atrial natriuretic peptide infusion on cardiac surgery-associated acute kidney injury: a multicenter randomized controlled trial. *J Crit Care*. 2017;38:253–258.
393. Elkayam U, Akhter MW, Tummala P, Khan S, Singh H. Nesiritide: a new drug for the treatment of decompensated heart failure. *J Cardiovasc Pharmacol Ther*. 2002;7(3):181–194.
394. Lee CY, Burnett Jr JC. Natriuretic peptides and therapeutic applications. *Heart Fail Rev*. 2007;12(2):131–142.
395. Burger AJ. A review of the renal and neurohormonal effects of B-type natriuretic peptide. *Congest Heart Fail*. 2005;11(1):30–38.
396. Sackner-Bernstein JD, Kowalski M, Fox M, Aaronson K. Short-term risk of death after treatment with nesiritide for decompensated heart failure: a pooled analysis of randomized controlled trials. *JAMA*. 2005;293(15):1900–1905.
397. Sackner-Bernstein JD, Skopicki HA, Aaronson KD. Risk of worsening renal function with nesiritide in patients with acutely decompensated heart failure. *Circulation*. 2005;111(12):1487–1491.
398. O'Connor CM, Starling RC, Hernandez AF, et al. Effect of nesiritide in patients with acute decompensated heart failure. *N Engl J Med*. 2011;365(1):32–43.
399. Mitrovic V, Luss H, Nitsche K, et al. Effects of the renal natriuretic peptide urodilatin (ularitide) in patients with decompensated chronic heart failure: a double-blind, placebo-controlled, ascending-dose trial. *Am Heart J*. 2005;150(6):1239.
400. Mitrovic V, Seferovic PM, Simeunovic D, et al. Haemodynamic and clinical effects of ularitide in decompensated heart failure. *Eur Heart J*. 2006;27(23):2823–2832.
401. Dorner GT, Selenko N, Kral T, Schmetterer L, Eichler HG, Wolzt M. Hemodynamic effects of continuous urodilatin infusion: a dose-finding study. *Clin Pharmacol Ther*. 1998;64(3):322–330.

402. Lüss H, Mitrovic V, Seferovic PM, et al. Renal effects of ularitide in patients with decompensated heart failure. *Am Heart J*. 2008;155(6):1012.e1–1012.e8.
403. Vesely DL. Which of the cardiac natriuretic peptides is most effective for the treatment of congestive heart failure, renal failure and cancer? *Clin Exp Pharmacol Physiol*. 2006;33(3):169–176.
404. Vesely DL. Urodilatin: a better natriuretic peptide? *Curr Heart Fail Rep*. 2007;4(3):147–152.
405. Clark LC, Farghaly H, Saba SR, Vesely DL. Amelioration with vessel dilator of acute tubular necrosis and renal failure established for 2 days. *Am J Physiol Heart Circ Physiol*. 2000;278(5):H1555–H1564.
406. Lisy O, Huntley BK, McCormick DJ, Kurlansky PA, Burnett JC, et al. Design, synthesis, and actions of a novel chimeric natriuretic peptide: CD-NP. *J Am Coll Cardiol*. 2008;52(1):60–68.
407. Lee CY, Chen HH, Lisy O, et al. Pharmacodynamics of a novel designer natriuretic peptide, CD-NP, in a first-in-human clinical trial in healthy subjects. *J Clin Pharmacol*. 2009;49(6):668–673.
408. Lee CY, Huntley BK, McCormick DJ, et al. Cenderitide: structural requirements for the creation of a novel dual particulate guanylyl cyclase receptor agonist with renal-enhancing in vivo and ex vivo actions. *Eur Heart J Cardiovasc Pharmacother*. 2016;2(2):98–105.
409. Helin K, Tikkanen I, Hohenthal U, Fyhrquist F. Inhibition of either angiotensin-converting enzyme or neutral endopeptidase induces both enzymes. *Eur J Pharmacol*. 1994;264(2):135–141. [https://doi.org/10.1016/0014-2999\(94\)00450-1](https://doi.org/10.1016/0014-2999(94)00450-1).
410. Campbell DJ. Long-term neprilysin inhibition—implications for ARNIs. *Nat Rev Cardiol*. 2017;14(3):171–186.
411. Kostis JB, Packer M, Black HR, Schmieder R, Henry D, Levy E. Omapatrilat and enalapril in patients with hypertension: the Omapatrilat Cardiovascular Treatment vs. Enalapril (OCTAVE) trial. *Am J Hypertens*. 2004;17(2):103–111.
412. Packer M, Califf RM, Konstam MA, et al. Comparison of omapatrilat and enalapril in patients with chronic heart failure: the omapatrilat versus enalapril randomized trial of utility in reducing events (OVERTURE). *Circulation*. 2002;106(8):920–926.
413. Hubers SA, Brown NJ. Combined angiotensin receptor antagonism and neprilysin inhibition. *Circulation*. 2016;133(11):1115–1124.
414. Gu J, Noe A, Chandra P, et al. Pharmacokinetics and pharmacodynamics of LCZ696, a novel dual-acting angiotensin receptor-neprilysin inhibitor (ARNi). *J Clin Pharmacol*. 2010;50(4):401–414.
415. Ruilope LM, Dukat A, Bohm M, Lacourciere Y, Gong J, Lefkowitz MP. Blood-pressure reduction with LCZ696, a novel dual-acting inhibitor of the angiotensin II receptor and neprilysin: a randomised, double-blind, placebo-controlled, active comparator study. *Lancet*. 2010;375(9722):1255–1266.
416. Solomon SD, Zile M, Pieske B, et al. The angiotensin receptor neprilysin inhibitor LCZ696 in heart failure with preserved ejection fraction: a phase 2 double-blind randomised controlled trial. *Lancet*. 2012;380(9851):1387–1395.
417. McMurray JJ, Packer M, Desai AS, et al. Angiotensin-neprilysin inhibition versus enalapril in heart failure. *N Engl J Med*. 2014;371(11):993–1004.
418. Solomon SD, McMurray JJV, Anand IS, et al. Angiotensin-neprilysin inhibition in heart failure with preserved ejection fraction. *N Engl J Med*. 2019;381(17):1609–1620.
419. Kurtz TW, Klein U. Next generation multifunctional angiotensin receptor blockers. *Hypertens Res*. 2009;32(10):826–834.
420. Unger T, Paulis L, Sica DA. Therapeutic perspectives in hypertension: novel means for renin-angiotensin-aldosterone system modulation and emerging device-based approaches. *Eur Heart J*. 2011;32(22):2739–2747.
421. Group CTS. Effects of enalapril on mortality in severe congestive heart failure. Results of the Cooperative North Scandinavian enalapril survival study (CONSENSUS). *N Engl J Med*. 1987;316(23):1429–1435.
422. Brum WS, Docherty KF, Ashton NJ, et al. Effect of neprilysin inhibition on Alzheimer disease plasma biomarkers: a secondary analysis of a randomized clinical trial. *JAMA Neurol*. 2024;81(2):197–200.
423. Heidenreich PA, Bozkurt B, Aguilar D, et al. 2022 AHA/ACC/HFSA guideline for the management of heart failure: executive summary: a report of the American College of cardiology/American heart association joint committee on clinical practice guidelines. *Circulation*. 2022;145(18):e876–e894.
424. Voors AA, Gori M, Liu LC, et al. Renal effects of the angiotensin receptor neprilysin inhibitor LCZ696 in patients with heart failure and preserved ejection fraction. *Eur J Heart Fail*. 2015;17(5):510–517.
425. Haynes R, Judge PK, Staplin N, et al. Effects of sacubitril/valsartan versus irbesartan in patients with chronic kidney disease. *Circulation*. 2018;138(15):1505–1514.
426. Nguyen DV, Le TN, Truong BQ, Nguyen HTT. Efficacy and safety of angiotensin receptor-neprilysin inhibition in heart failure patients with end-stage kidney disease on maintenance dialysis: a systematic review and meta-analysis. *Eur J Heart Fail*. 2025;27(1):72–84.
427. Carey RM. Evidence for a splanchnic sodium input monitor regulating renal sodium excretion in man. Lack of dependence upon aldosterone. *Circ Res*. 1978;43(1):19–23.
428. Lennane RJ, Carey RM, Goodwin TJ, Peart WS. A comparison of natriuresis after oral and intravenous sodium loading in sodium-depleted man: evidence for a gastrointestinal or portal monitor of sodium intake. *Clin Sci Mol Med*. 1975;49(5):437–440.
429. Preston RA, Afshartous D, Forte LR, et al. Sodium challenge does not support an acute gastrointestinal-renal natriuretic signaling axis in humans. *Kidney Int*. 2012;82(12):1313–1320.
430. Mueller T, Dieplinger B. The guanylin peptide family and the proposed gastrointestinal-renal natriuretic signaling axis. *Kidney Int*. 2012;82(12):1253–1255.
431. Kitamura K, Kangawa K, Kawamoto M, et al. Adrenomedullin: a novel hypotensive peptide isolated from human pheochromocytoma. *Biochem Biophys Res Commun*. 1993;192(2):553–560.
432. Yanagawa B, Nagaya N. Adrenomedullin: molecular mechanisms and its role in cardiac disease. *Amino Acids*. 2007;32(1):157–164.
433. Velho G, Ragot S, Mohammedi K, et al. Plasma adrenomedullin and allelic variation in the ADM gene and kidney disease in people with type 2 diabetes. *Diabetes*. 2015;64(9):3262–3272.
434. Leeb-Lundberg LM, Marceau F, Muller-Esterl W, Pettibone DJ, Zuraw BL. International union of pharmacology. XLV. Classification of the kinin receptor family: from molecular mechanisms to pathophysiological consequences. *Pharmacol Rev*. 2005;57(1):27–77.
435. Abelous JE, Bardier E. Les substances hypotensives de l'urine humaine normale. *CR Soc Biol*. 1909;66:511–520.
436. Shesely EG, Hu CB, Alhenc-Gelas F, Meneton P, Carretero OA. A second expressed kininogen gene in mice. *Physiol Genomics*. 2006;26(2):152–157.
437. Colman RW, Bagdasarian A, Talamo RC, et al. Williams trait. Human kininogen deficiency with diminished levels of plasminogen proactivator and prekallikrein associated with abnormalities of the Hageman factor-dependent pathways. *J Clin Invest*. 1975;56(6):1650–1662.
438. Campbell DJ. The kallikrein-kinin system in humans. *Clin Exp Pharmacol Physiol*. 2001;28(12):1060–1065. <https://doi.org/10.1046/j.1440-1681.2001.03564.x>.
439. Damas J. The brown Norway rats and the kinin system. *Peptides*. 1996;17(5):859–872.
440. Kraut H, Frey EK, Werle E. Der Nachweis eines Kreislaufhormon in der Pankreasdrüse. *Hoppe-Seyler's Z Physiol Chem*. 1930;189:97–106.
441. Chao J, Yin H, Gao L, et al. Tissue kallikrein elicits cardioprotection by direct kinin b2 receptor activation independent of kinin formation. *Hypertension*. 2008;52(4):715–720.
442. Biyashev D, Tan F, Chen Z, et al. Kallikrein activates bradykinin B2 receptors in absence of kininogen. *Am J Physiol Heart Circ Physiol*. 2006;290(3):H1244–H1250.
443. Krankel N, Madeddu P. Helping the circulatory system heal itself: manipulating kinin signaling to promote neovascularization. *Expert Rev Cardiovasc Ther*. 2009;7(3):215–219.
444. Chao J, Chao L. Functional analysis of human tissue kallikrein in transgenic mouse models. *Hypertension*. 1996;27(3 Pt 2):491–494.
445. Azizi M, Boutouyrie P, Bissery A, et al. Arterial and renal consequences of partial genetic deficiency in tissue kallikrein activity in humans. *J Clin Invest*. 2005;115(3):780–787.
446. Kakoki M, Smithies O. The kallikrein-kinin system in health and in diseases of the kidney. *Kidney Int*. 2009;75(10):1019–1030.
447. Cayla C, Merino VF, Cabrini DA, Silva Jr JA, Pesquero JB, Bader M. Structure of the mammalian kinin receptor gene locus. *Int Immunopharmacol*. 2002;2(13–14):1721–1727.

448. Riad A, Zhuo JL, Schultheiss HP, Tschope C. The role of the renal kallikrein-kinin system in diabetic nephropathy. *Curr Opin Nephrol Hypertens.* 2007;16(1):22–26.
449. Spillmann F, Altmann C, Scheeler M, et al. Regulation of cardiac bradykinin B1- and B2-receptor mRNA in experimental ischemic, diabetic, and pressure-overload-induced cardiomyopathy. *Int Immunopharmacol.* 2002;2(13–14):1823–1832.
450. Griol-Charhbil V, Messadi-Laribi E, Bascands JL, et al. Role of tissue kallikrein in the cardioprotective effects of ischemic and pharmacological preconditioning in myocardial ischemia. *FASEB J.* 2005;19(9):1172–1174.
451. Zhang X, Tan F, Zhang Y, Skidgel RA. Carboxypeptidase M and kinin B1 receptors interact to facilitate efficient b1 signaling from B2 agonists. *J Biol Chem.* 2008;283(12):7994–8004.
452. Ignjatovic T, Stanisavljevic S, Brovkovich V, Skidgel RA, Erdos EG. Kinin B1 receptors stimulate nitric oxide production in endothelial cells: signaling pathways activated by angiotensin I-converting enzyme inhibitors and peptide ligands. *Mol Pharmacol.* 2004;66(5):1310–1316.
453. Savard M, Barbaz D, Belanger S, et al. Expression of endogenous nuclear bradykinin B2 receptors mediating signaling in immediate early gene activation. *J Cell Physiol.* 2008;216(1):234–244.
454. Borkowski JA, Ransom RW, Seabrook GR, et al. Targeted disruption of a B2 bradykinin receptor gene in mice eliminates bradykinin action in smooth muscle and neurons. *J Biol Chem.* 1995;270(23):13706–13710. <https://doi.org/10.1074/jbc.270.23.13706>.
455. Pesquero JB, Araujo RC, Heppenstall PA, et al. Hypoalgesia and altered inflammatory responses in mice lacking kinin B1 receptors. *Proc Natl Acad Sci USA.* 2000;97(14):8140–8145.
456. Cayla C, Todiras M, Ilescu R, et al. Mice deficient for both kinin receptors are normotensive and protected from endotoxin-induced hypotension. *FASEB J.* 2007;21(8):1689–1698.
457. Madeddu P, Varoni MV, Palomba D, et al. Cardiovascular phenotype of a mouse strain with disruption of bradykinin B2-receptor gene. *Circulation.* 1997;96(10):3570–3578.
458. Alfie ME, Yang XP, Hess F, Carretero OA. Salt-sensitive hypertension in bradykinin B2 receptor knockout mice. *Biochem Biophys Res Commun.* 1996;224(3):625–630.
459. Milia AF, Gross V, Plehm R, De Silva Jr JA, Bader M, Luft FC. Normal blood pressure and renal function in mice lacking the bradykinin B(2) receptor. *Hypertension.* 2001;37(6):1473–1479.
460. Kakoki M, McGarrah RW, Kim HS, Smithies O. Bradykinin B1 and B2 receptors both have protective roles in renal ischemia/reperfusion injury. *Proc Natl Acad Sci USA.* 2007;104(18):7576–7581. <https://doi.org/10.1073/pnas.0701617104>.
461. Wang DZ, Chao L, Chao J. Hypotension in transgenic mice overexpressing human bradykinin B2 receptor. *Hypertension.* 1997;29(1 Pt 2):488–493.
462. Ni A, Yin H, Agata J, Yang Z, Chao L, Chao J. Overexpression of kinin B1 receptors induces hypertensive response to des-Arg9-bradykinin and susceptibility to inflammation. *J Biol Chem.* 2003;278(1):219–225.
463. Chao J, Stallone JN, Liang YM, Chen LM, Wang DZ, Chao L. Kallistatin is a potent new vasodilator. *J Clin Invest.* 1997;100(1):11–17.
464. Moreira CR, Schmaier AH, Mahdi F, da Motta G, Nader HB, Shariat-Madar Z. Identification of prolylcarboxypeptidase as the cell matrix-associated prekallikrein activator. *FEBS Lett.* 2002;523(1–3):167–170.
465. el-Dahr S, Yosipiv IV, Muchant DG, Chevalier RL. Salt intake modulates the developmental expression of renal kallikrein and bradykinin B2 receptors. *Am J Physiol.* 1996;270(3 Pt 2):F425–F431.
466. el-Dahr SS, Chao J. Spatial and temporal expression of kallikrein and its mRNA during nephron maturation. *Am J Physiol.* 1992;262(5 Pt 2):F705–F711.
467. el-Dahr SS, Yosipiv I. Developmentally regulated kallikrein enzymatic activity and gene transcription rate in maturing rat kidneys. *Am J Physiol.* 1993;265(1 Pt 2):F146–F150.
468. Madeddu P, Glorioso N, Maioli M, et al. Regulation of rat renal kallikrein expression by estrogen and progesterone. *J Hypertens Suppl.* 1991;9(6):S244–S245.
469. Marin-Castano ME, Schanstra JP, Praddaude F, et al. Differential induction of functional B1-bradykinin receptors along the rat nephron in endotoxin induced inflammation. *Kidney Int.* 1998;54(6):1888–1898.
470. Zeidel ML, Jabs K, Kikeri D, Silva P. Kinins inhibit conductive Na⁺ uptake by rabbit inner medullary collecting duct cells. *Am J Physiol.* 1990;258(6 Pt 2):F1584–F1591.
471. Alfie ME, Alim S, Mehta D, Shesely EG, Carretero OA. An enhanced effect of arginine vasopressin in bradykinin B2 receptor null mutant mice. *Hypertension.* 1999;33(6):1436–1440.
472. Duchene J, Schanstra JP, Pecher C, et al. A novel protein-protein interaction between a G protein-coupled receptor and the phosphatase SHP-2 is involved in bradykinin-induced inhibition of cell proliferation. *J Biol Chem.* 2002;277(43):40375–40383. <https://doi.org/10.1074/jbc.M202744200>.
473. Chambrey R, Picard N. Role of tissue kallikrein in regulation of tubule function. *Curr Opin Nephrol Hypertens.* 2011;20(5):523–528.
474. Picard N, Eladari D, El MS, et al. Defective ENaC processing and function in tissue kallikrein-deficient mice. *J Biol Chem.* 2008;283(8):4602–4611.
475. Levie F, Hubner CA, Houillier P, et al. The Na⁺-dependent chloride-bicarbonate exchanger SLC4A8 mediates an electroneutral Na⁺ reabsorption process in the renal cortical collecting ducts of mice. *J Clin Invest.* 2010;120(5):1627–1635.
476. El MS, Houillier P, Picard N, et al. Tissue kallikrein permits early renal adaptation to potassium load. *Proc Natl Acad Sci USA.* 2010;107(30):13526–13531.
477. Gkika D, Topala CN, Chang Q, et al. Tissue kallikrein stimulates Ca(2+) reabsorption via PKC-dependent plasma membrane accumulation of TRPV5. *EMBO J.* 2006;25(20):4707–4716.
478. Monteiro JS, Blanchard A, Curis E, Chambrey R, Jeunemaitre X, Azizi M. Partial genetic deficiency in tissue kallikrein impairs adaptation to high potassium intake in humans. *Kidney Int.* 2013;84(6):1271–1277.
479. Blanchard A, Azizi M, Peyrard S, et al. Partial human genetic deficiency in tissue kallikrein activity and renal calcium handling. *Clin J Am Soc Nephrol.* 2007;2(2):320–325.
480. Margolius HS, Horwitz D, Geller RG, et al. Urinary kallikrein excretion in normal man. Relationships to sodium intake and sodium-retaining steroids. *Circ Res.* 1974;35(6):812–819.
481. Hirawa N, Uehara Y, Kawabata Y, et al. Mechanistic analysis of renal protection by angiotensin converting enzyme inhibitor in Dahl salt-sensitive rats. *J Hypertens.* 1994;12(8):909–918.
482. Uehara Y, Hirawa N, Kawabata Y, et al. Long-term infusion of kallikrein attenuates renal injury in Dahl salt-sensitive rats. *Hypertension.* 1994;24(6):770–778.
483. Chao J, Li HJ, Yao YY, et al. Kinin infusion prevents renal inflammation, apoptosis, and fibrosis via inhibition of oxidative stress and mitogen-activated protein kinase activity. *Hypertension.* 2007;49(3):490–497.
484. Cervenka L, Vaneckova I, Maly J, Horacek V, El-Dahr SS. Genetic inactivation of the B2 receptor in mice worsens two-kidney, one-clip hypertension: role of NO and the AT2 receptor. *J Hypertens.* 2003;21(8):1531–1538.
485. Griol-Charhbil V, Sabbah L, Colucci J, et al. Tissue kallikrein deficiency and renovascular hypertension in the mouse. *Am J Physiol Regul Integr Comp Physiol.* 2009;296(5):R1385–R1391.
486. Madeddu P, Emanuelli C, El-Dahr S. Mechanisms of disease: the tissue kallikrein-kinin system in hypertension and vascular remodeling. *Nat Clin Pract Nephrol.* 2007;3(4):208–221.
487. Woodley-Miller C, Chao J, Chao L. Restriction fragment length polymorphisms mapped in spontaneously hypertensive rats using kallikrein probes. *J Hypertens.* 1989;7(11):865–871.
488. Yu H, Bowden DW, Spray BJ, Rich SS, Freedman BI. Identification of human plasma kallikrein gene polymorphisms and evaluation of their role in end-stage renal disease. *Hypertension.* 1998;31(4):906–911.
489. Hua H, Zhou S, Liu Y, et al. Relationship between the regulatory region polymorphism of human tissue kallikrein gene and essential hypertension. *J Hum Hypertens.* 2005;19(9):715–721.
490. Slim R, Torremocha F, Moreau T, et al. Loss-of-function polymorphism of the human kallikrein gene with reduced urinary kallikrein activity. *J Am Soc Nephrol.* 2002;13(4):968–976.
491. Rossi GP, Taddei S, Ghiadoni L, et al. Tissue kallikrein gene polymorphisms induce no change in endothelium-dependent or independent vasodilation in hypertensive and normotensive subjects. *J Hypertens.* 2006;24(10):1955–1963. <https://doi.org/10.1097/01.hjh.0000244943.34546.40>.

492. Cui J, Melista E, Chazaro I, et al. Sequence variation of bradykinin receptors B1 and B2 and association with hypertension. *J Hypertens*. 2005;23(1):55–62.
493. Brull D, Dhamrait S, Myerson S, et al. Bradykinin B2BKR receptor polymorphism and left-ventricular growth response. *Lancet*. 2001;358(9288):1155–1156. [https://doi.org/10.1016/s0140-6736\(01\)06273-0](https://doi.org/10.1016/s0140-6736(01)06273-0).
494. Dhamrait SS, Payne JR, Li P, et al. Variation in bradykinin receptor genes increases the cardiovascular risk associated with hypertension. *Eur Heart J*. 2003;24(18):1672–1680.
495. Harvey JN, Jaffa AA, Margolius HS, Mayfield RK. Renal kallikrein and hemodynamic abnormalities of diabetic kidney. *Diabetes*. 1990;39(3):299–304. <https://doi.org/10.2337/diab.39.3.299>.
496. Tschöpe C, Reinecke A, Seidl U, et al. Functional, biochemical, and molecular investigations of renal kallikrein-kinin system in diabetic rats. *Am J Physiol*. 1999;277(6):H2333–H2340.
497. Allen TJ, Cao Z, Youssef S, Hulthén UL, Cooper ME. Role of angiotensin II and bradykinin in experimental diabetic nephropathy. Functional and structural studies. *Diabetes*. 1997;46(10):1612–1618.
498. Buleon M, Allard J, Jaafar A, et al. Pharmacological blockade of B2-kinin receptor reduces renal protective effect of angiotensin-converting enzyme inhibition in db/db mice model. *Am J Physiol Ren Physiol*. 2008;294(5):F1249–F1256.
499. Tschöpe C, Seidl U, Reinecke A, et al. Kinins are involved in the antiproteinuric effect of angiotensin-converting enzyme inhibition in experimental diabetic nephropathy. *Int Immunopharmacol*. 2003;3(3):335–344.
500. Schäfer S, Schmidts HL, Bleich M, Busch AE, Linz W. Nephroprotection in Zucker diabetic fatty rats by vasopeptidase inhibition is partly bradykinin B2 receptor dependent. *Br J Pharmacol*. 2004;143(1):27–32.
501. Kakoki M, Takahashi N, Jennette JC, Smithies O. Diabetic nephropathy is markedly enhanced in mice lacking the bradykinin B2 receptor. *Proc Natl Acad Sci USA*. 2004;101(36):13302–13305.
502. Kakoki M, Kizer CM, Yi X, et al. Senescence-associated phenotypes in Akita diabetic mice are enhanced by absence of bradykinin B2 receptors. *J Clin Invest*. 2006;116(5):1302–1309.
503. Tan Y, Keum JS, Wang B, McHenry MB, Lipsitz SR, Jaffa AA. Targeted deletion of B2-kinin receptors protects against the development of diabetic nephropathy. *Am J Physiol Ren Physiol*. 2007;293(4):F1026–F1035.
504. Kakoki M, Sullivan KA, Backus C, et al. Lack of both bradykinin B1 and B2 receptors enhances nephropathy, neuropathy, and bone mineral loss in Akita diabetic mice. *Proc Natl Acad Sci USA*. 2010;107(22):10190–10195.
505. Bodin S, Chollet C, Gonçalves-Mendes N, et al. Kallikrein protects against microalbuminuria in experimental type I diabetes. *Kidney Int*. 2009;76(4):395–403.
506. Yuan G, Deng J, Wang T, et al. Tissue kallikrein reverses insulin resistance and attenuates nephropathy in diabetic rats by activation of phosphatidylinositol 3-kinase/protein kinase B and adenosine 5'-monophosphate-activated protein kinase signaling pathways. *Endocrinology*. 2007;148(5):2016–2026.
507. Liu W, Yang Y, Liu Y, et al. Exogenous kallikrein protects against diabetic nephropathy. *Kidney Int*. 2016;90(5):1023–1036.
508. Yiu WH, Wong DW, Wu HJ, et al. Kallistatin protects against diabetic nephropathy in db/db mice by suppressing AGE-RAGE-induced oxidative stress. *Kidney Int*. 2016;89(2):386–398.
509. Harvey JN, Edmundson AW, Jaffa AA, Martin LL, Mayfield RK. Renal excretion of kallikrein and eicosanoids in patients with type 1 (insulin-dependent) diabetes mellitus. Relationship to glomerular and tubular function. *Diabetologia*. 1992;35(9):857–862.
510. Maltais I, Bachvarova M, Maheux P, Perron P, Marceau F, Bachvarov D. Bradykinin B2 receptor gene polymorphism is associated with altered urinary albumin/creatinine values in diabetic patients. *Can J Physiol Pharmacol*. 2002;80(4):323–327.
511. Zychma MJ, Gumprecht J, Trautsoit W, Szydłowska I, Grzeszczak W. Polymorphic genes for kinin receptors, nephropathy and blood pressure in type 2 diabetic patients. *Am J Nephrol*. 2003;23(2):112–116.
512. Merchant ML, Niewczas MA, Ficociello LH, et al. Plasma kininogen and kininogen fragments are biomarkers of progressive renal decline in type 1 diabetes. *Kidney Int*. 2013;83(6):1177–1184.
513. Pazoki-Toroudi HR, Hesami A, Vahidi S, Sahebjam F, Seifi B, Djahanguiri B. The preventive effect of captopril or enalapril on reperfusion injury of the kidney of rats is independent of angiotensin II AT1 receptors. *Fundam Clin Pharmacol*. 2003;17(5):595–598.
514. Kitakaze M, Minamino T, Node K, et al. Beneficial effects of inhibition of angiotensin-converting enzyme on ischemic myocardium during coronary hypoperfusion in dogs. *Circulation*. 1995;92(4):950–961.
515. Liu YH, Yang XP, Sharov VG, Sigmon DH, Sabbath HN, Carretero OA. Paracrine systems in the cardioprotective effect of angiotensin-converting enzyme inhibitors on myocardial ischemia/reperfusion injury in rats. *Hypertension*. 1996;27(1):7–13.
516. Park SS, Zhao H, Mueller RA, Xu Z. Bradykinin prevents reperfusion injury by targeting mitochondrial permeability transition pore through glycogen synthase kinase 3beta. *J Mol Cell Cardiol*. 2006;40(5):708–716.
517. Chiang WC, Chien CT, Lin WW, et al. Early activation of bradykinin B2 receptor aggravates reactive oxygen species generation and renal damage in ischemia/reperfusion injury. *Free Radic Biol Med*. 2006;41(8):1304–1314. <https://doi.org/10.1016/j.freeradbiomed.2006.07.011>.
518. Zhou S, Sun Y, Zhuang Y, et al. Effects of kallistatin on oxidative stress and inflammation on renal ischemia-reperfusion injury in mice. *Curr Vasc Pharmacol*. 2015;13(2):265–273. d.
519. Wolf WC, Yoshida H, Agata J, Chao L, Chao J. Human tissue kallikrein gene delivery attenuates hypertension, renal injury, and cardiac remodeling in chronic renal failure. *Kidney Int*. 2000;58(2):730–739.
520. Schanstra JP, Neau E, Drogoz P, et al. In vivo bradykinin B2 receptor activation reduces renal fibrosis. *J Clin Invest*. 2002;110(3):371–379. <https://doi.org/10.1172/JCI15493>.
521. Klein J, Gonzalez J, Duchene J, et al. Delayed blockade of the kinin B1 receptor reduces renal inflammation and fibrosis in obstructive nephropathy. *FASEB J*. 2009;23(1):134–142.
522. Wang PH, Cenedeze MA, Campanholle G, et al. Deletion of bradykinin B1 receptor reduces renal fibrosis. *Int Immunopharmacol*. 2009;9(6):653–657.
523. Pereira RL, Buscariollo BN, Correa-Costa M, et al. Bradykinin receptor 1 activation exacerbates experimental focal and segmental glomerulosclerosis. *Kidney Int*. 2011;79(11):1217–1227.
524. Zychma MJ, Gumprecht J, Zukowska-Szczechowska E, Grzeszczak W. Polymorphisms in the genes encoding for human kinin receptors and the risk of end-stage renal failure: results of transmission/disequilibrium test. The End-Stage Renal Disease Study Group. *J Am Soc Nephrol*. 1999;10(10):2120–2124.
525. Bachvarov DR, Landry M, Pelletier I, et al. Characterization of two polymorphic sites in the human kinin B1 receptor gene: altered frequency of an allele in patients with a history of end-stage renal failure. *J Am Soc Nephrol*. 1998;9(4):598–604.
526. Jozwiak L, Drop A, Buraczynska K, Ksiazek P, Mierzicki P, Buraczynska M. Association of the human bradykinin B2 receptor gene with chronic renal failure. *Mol Diagn*. 2004;8(3):157–161.
527. Liu K, Li QZ, Delgado-Vega AM, et al. Kallikrein genes are associated with lupus and glomerular basement membrane-specific antibody-induced nephritis in mice and humans. *J Clin Invest*. 2009;119(4):911–923.
528. Li QZ, Zhou J, Yang R, et al. The lupus-susceptibility gene kallikrein downmodulates antibody-mediated glomerulonephritis. *Genes Immun*. 2009;10(5):503–508.
529. Kahn R, Hellmark T, Leeb-Lundberg LM, et al. Neutrophil-derived proteinase 3 induces kallikrein-independent release of a novel vasoactive kinin. *J Immunol*. 2009;182(12):7906–7915.
530. Klein J, Gonzalez J, Decramer S, et al. Blockade of the kinin B1 receptor ameliorates glomerulonephritis. *J Am Soc Nephrol*. 2010;21(7):1157–1164.

QUESTIONS

- Acute angiotensin type 1 receptor stimulation by angiotensin II causes all but which one of the following?
 - Elevated intraglomerular pressure (P_{GC})
 - Reduction in the ultrafiltration coefficient (K_f)
 - Reduction in renal blood flow
 - Increased afferent but not efferent arteriolar tone

Answer: d

Rationale: The effects of angiotensin II and the activation of its type 1 receptor on glomerular hemodynamics are pivotal in understanding the role of the RAS in physiology, pathophysiology, and therapeutics. Angiotensin II causes an increase in both afferent and efferent arteriolar tone, but the increase in efferent arteriolar tone is greater than the increase in afferent arteriolar tone, resulting in a rise in glomerular capillary pressure.

- The predominant site for angiotensinogen synthesis is:
 - The juxtaglomerular apparatus
 - Vascular endothelium
 - Pericentral zone hepatocytes
 - Peritubular interstitial fibroblasts

Answer: c

Rationale: The RAS is a multiorgan endocrine system where in specific cell types have key roles. Understanding which cell in which organ does what is an integral part of knowing how the system works.

- Concerning angiotensin-converting enzyme 2, which of the following are true?
 - Shares >85% homology with angiotensin-converting enzyme in terms of its amino acid sequence
 - Cleaves angiotensin II to angiotensin III
 - Is the receptor for the severe adult respiratory syndrome (SARS) coronavirus
 - Within the glomerulus is expressed predominantly in mesangial cells

Answer: c

Rationale: Angiotensin-converting enzyme 2 is a relatively new but important addition to the RAS. Knowledge of its origin and actions is necessary to fully appreciate the complexities of the system.

- B-type natriuretic peptide binds to which of the following receptors?
 - Natriuretic peptide receptor-A
 - Natriuretic peptide receptor-B
 - Natriuretic peptide receptor-B and natriuretic peptide receptor-C
 - Natriuretic peptide receptor-A and natriuretic peptide receptor-C

Answer: d

Rationale: The nomenclature for the natriuretic peptide receptors can be confusing. NPR-A binds ANP and BNP. NPR-B binds CNP, and NPR-C is a clearance receptor for all three peptides.

- Which of the following is not a known substrate of neprilysin?
 - Vascular endothelial growth factor-A
 - Amyloid β -peptide
 - C-type natriuretic peptide
 - Vasopressin

Answer: a

Rationale: Neprilysin is a zinc metalloproteinase that catalyzes the degradation of the natriuretic peptides and has activity against more than 40 other substrates (see [Table 11.2](#)). Vascular endothelial growth factor-A is not a substrate for neprilysin.